MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3420 PRESS. OXDN.- AUTOCL. BLDG.
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
AUTOCL.	:	Autoclave
OXDN.	:	Oxidation
BLDG.	:	Building
PRESS.	:	Pressure

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1. INTENT

This document is to prepare design of foundations for **Pressure Oxidation Autoclave Building**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Doc. No. MD-216-3420-EG-ST-GAD-2230 attached in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 475 Kn/m^2 at 3.0 m below EGL. (Refer Table 5 - BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F4

Size of Columns at node	Along-X	Along-Z
793	0.75m	0.75m

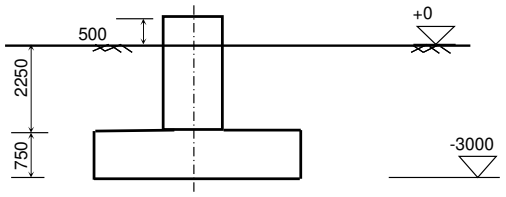
Size of fdn. raft:	3.60m	4.35m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

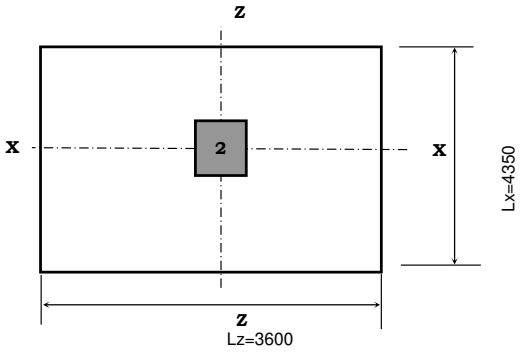
Calculation of Weight of Foundation :

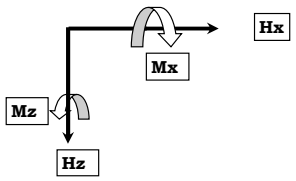
Weight of raft =	(3.6*4.35*0.75*24) =	281.88KN
Weight of pedestal =	0.75x0.75x2.75x24 =	37.13KN
Weight of soil + surcharge=	{(4.35*3.6)-(0.75*0.75)}*2.25*20 =	829.39KN
	Total =	1148.39KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F4

Load at bottom of footing	Node 793				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	149.67	0.00	0.00	0.00	0.00
202 DL+L	149.67	0.00	0.00	0.00	0.00
203 DL+LR	149.67	0.00	0.00	0.00	0.00
204 DL+0.75L+0.75LR	149.67	0.00	0.00	0.00	0.00
205 DL+WL1	149.67	0.00	0.00	320.00	1119.98
206 DL+WL2	149.67	0.00	0.00	-320.00	-1119.98
207 DL+0.75WL1+0.75L+0.75LR	149.67	0.00	0.00	240.00	839.99
208 DL+0.75WL2+0.75L+0.75LR	149.67	0.00	0.00	-240.00	-839.99
209 0.6DL+1.0WL1	89.80	0.00	0.00	320.00	1119.98
210 0.6DL+1.0WL2	89.80	0.00	0.00	-320.00	-1119.98
211 0.6DL+0.7SL1	89.80	0.00	0.00	0.00	0.00
212 0.6DL+0.7SL2	89.80	0.00	0.00	0.00	0.00

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	1298.1	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 DL+L	1298.1	0.00	0.0	0.0	0.0	NA	NA	NA	NA
203 DL+LR	1298.1	0.00	0.0	0.0	0.0	NA	NA	NA	NA
204 DL+0.75L+0.75LR	1298.1	0.00	0.0	0.0	0.0	NA	NA	NA	NA
205 DL+WL1	1298.1	320.00	1120.0	0.0	0.0	NA	2.5	NA	1.6
206 DL+WL2	1298.1	-320.00	-1120.0	0.0	0.0	NA	2.5	NA	1.6
207 DL+0.75WL1+0.75L+0.75LR	1298.1	240.00	840.0	0.0	0.0	NA	3.4	NA	2.2
208 DL+0.75WL2+0.75L+0.75LR	1298.1	-240.00	-840.0	0.0	0.0	NA	3.4	NA	2.2
209 0.6DL+1.0WL1	1238.2	320.00	1120.0	0.0	0.0	NA	2.4	NA	1.5
210 0.6DL+1.0WL2	1238.2	-320.00	-1120.0	0.0	0.0	NA	2.4	NA	1.5
211 0.6DL+0.7SL1	1238.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA
212 0.6DL+0.7SL2	1238.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA

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Base pressure check of footing F4

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	524	201	0	98.36	0
Min Px	524	201	0	98.36	0
Max Py	1516	206	0	194.79	0
Min Py	527	209	0	0.00	0
Max Pz	524	201	0	98.36	0
Min Pz	524	201	0	98.36	0

Gross Maximum Base pressure, p_{max} = 194.8 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.25 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 520 KN/m^2
> 194.8 KN/m^2
Hence OK

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Design of foundation raft F4

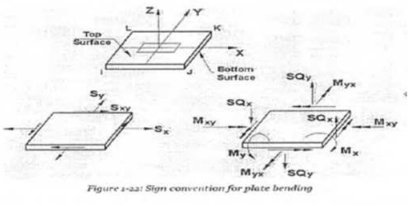
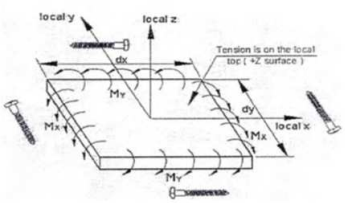


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F4_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1119	0.501	-0.555	0	0	0	-99.678	-272.287	54.714	-154.392	-327.001	105:1.2DL+1.6WL2+1.0L+0.5LR
Min Qx	645	-0.5	-0.101	0	0	0	-72.368	-140.843	60.724	-133.092	-201.567	104:1.2DL+1.6WL1+1.0L+0.5LR
Max Qy	788	-0.206	0.896	0	0	0	-119.11	-382.921	23.84	-142.946	-406.761	108:0.9DL+1.6WL1
Min Qy	783	-0.207	-0.899	0	0	0	-119.08	-383.976	-24.1	-143.177	-408.076	109:0.9DL+1.6WL2
Max Sx	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sx	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Sy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Sxy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sxy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Mx	788	0.191	-0.757	0	0	0	86.955	294.242	-26.313	113.268	320.555	109:0.9DL+1.6WL2
Min Mx	788	-0.209	0.885	0	0	0	-122.84	-361.579	23.369	-146.212	-384.948	104:1.2DL+1.6WL1+1.0L+0.5LR
Max My	783	0.19	0.773	0	0	0	83.389	319.944	27.011	110.4	346.955	104:1.2DL+1.6WL1+1.0L+0.5LR
Min My	783	-0.207	-0.899	0	0	0	-119.08	-383.976	-24.1	-143.177	-408.076	109:0.9DL+1.6WL2
Max Mxy	637	0.173	-0.362	0	0	0	19.703	43.213	87.139	106.842	130.352	104:1.2DL+1.6WL1+1.0L+0.5LR
Min Mxy	646	0.174	0.362	0	0	0	19.788	43.41	-87.141	106.929	130.551	105:1.2DL+1.6WL2+1.0L+0.5LR

PLATE CENTER STRESS : FDN_F4_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1131	0.178	-0.226	0	0	0	-55.263	-94.671	-5.666	-60.929	-100.337	105:1.2DL+1.6WL2+1.0L+0.5LR
Min Qx	641	-0.178	-0.23	0	0	0	-56.301	-95.247	5.7	-62.001	-100.947	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Qy	786	-0.053	0.401	0	0	0	-91.715	-130.408	-1.602	-93.317	-132.01	108:0.9DL+1.6WL1
Min Qy	785	-0.053	-0.402	0	0	0	-91.791	-130.598	1.589	-93.38	-132.187	109:0.9DL+1.6WL2
Max Sx	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sx	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Sy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Sxy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Min Sxy	1491	0.001	0.001	0	0	0	-0.434	-0.16	0.143	-0.577	-0.303	101:1.4DL
Max Mx	785	0.05	0.287	0	0	0	75.165	81.895	-0.383	75.548	82.278	108:0.9DL+1.6WL1
Min Mx	785	-0.054	-0.381	0	0	0	-95.138	-117.318	1.856	-96.994	-119.174	105:1.2DL+1.6WL2+1.0L+0.5LR
Max My	785	0.049	0.316	0	0	0	71.575	96.427	-0.128	71.703	96.555	104:1.2DL+1.6WL1+1.0L+0.5LR
Min My	785	-0.053	-0.402	0	0	0	-91.791	-130.598	1.589	-93.38	-132.187	109:0.9DL+1.6WL2
Max Mxy	591	0.029	-0.168	0	0	0	-1.17	-3.849	55.758	-56.928	-59.607	104:1.2DL+1.6WL1+1.0L+0.5LR
Min Mxy	618	0.029	0.168	0	0	0	-1.169	-3.655	-55.743	-56.912	-59.398	105:1.2DL+1.6WL2+1.0L+0.5LR

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

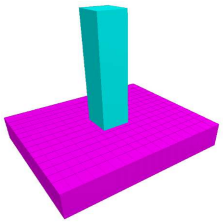
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109:0.9D	113.27	top face	50	692	113.27	0.32	0.0008	522.4	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
104:1.2D	-146.2	bottom face	50	692	146.21	0.41	0.0010	675.4		16	161.48	150	1341	18.91	23.64	0.085	tension control
104:1.2D	106.84	top face	50	692	106.84	0.30	0.0007	492.6		16	161.48	150	1341	18.91	23.64	0.085	tension control
105:1.2D	106.93	top face	50	692	106.93	0.30	0.0007	493.0		16	161.48	150	1341	18.91	23.64	0.085	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN			
Load condition	Max Qx	0.178	1	133.5	60.929	0.002	0.692	689.594	677.825	1187.24	677.83	2535	> 133.5	ok	508.4	φVc>Vu	not required	
	Min Qx	0.178	1	133.5	62.001			688.996			677.83	2535	> 133.5	ok	508.4	φVc>Vu	not required	

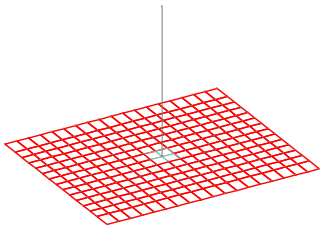
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.2D	346.96	top face	50	692	346.96	0.97	0.0007	465.1	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
109:0.9D	-408.08	bottom face	50	692	408.08	1.14	0.0011	755.9		16	161.48	150	1341	18.91	23.64	0.085	tension control
104:1.2D	130.35	top face	50	692	130.35	0.36	0.0009	601.7		16	161.48	150	1341	18.91	23.64	0.085	tension control
105:1.2D	130.55	top face	50	692	130.55	0.36	0.0009	602.6		16	161.48	150	1341	18.91	23.64	0.085	tension control

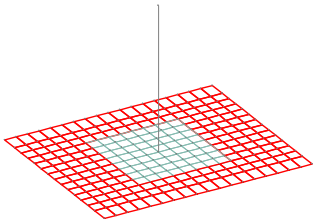
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN			
Load condition	Max Qy	0.401	1	300.75	132.01	0.002	0.692	690.969	677.825	fc'1187	677.83	711	> 300.75	ok	508.4	φVc>Vu	not required	
	Min Qy	0.402	1	301.5	132.19			691.01			677.83	711	> 301.5	ok	508.4	φVc>Vu	not required	



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	150	c/c
At Top	16ϕT	150	c/c
Along Z Direction :			
At Bottom	16ϕT	150	c/c
At Top	16ϕT	150	c/c

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DESIGN CALCULATION OF FOOTING F5

Size of Columns at node	Along-X	Along-Z
2	0.75m	0.75m

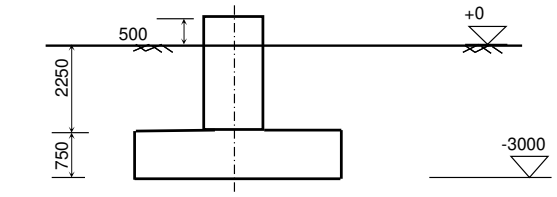
Size of fdn. raft:	3.60m	4.35m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

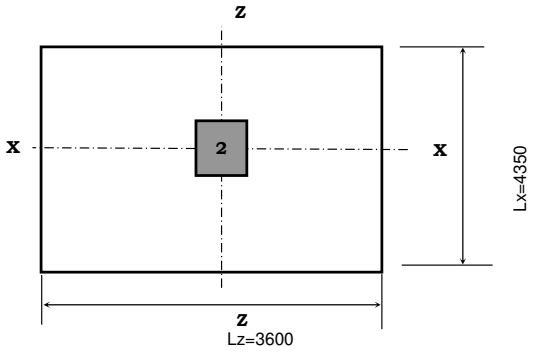
Calculation of Weight of Foundation :

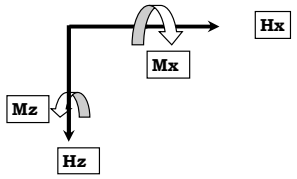
Weight of raft =	(3.6*4.35*0.75*24) =	281.88KN
Weight of pedestal =	0.75x0.75x2.75x24 =	37.13KN
Weight of soil + surcharge=	{(4.35*3.6)-(0.75*0.75)}*2.25*20+150=	829.39KN
	Total =	1148.39KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F5

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	323.27	-1.50	5.25	0.00	0.00
202 DL+L	560.34	-1.50	5.25	-3.85	-13.48
203 DL+LR	519.44	-1.50	5.25	0.00	0.00
204 DL+0.75L+0.75LR	648.20	-1.50	5.25	-2.89	-10.11
205 DL+WL1	-330.73	-1.50	5.25	240.00	839.99
206 DL+WL2	977.27	-1.50	5.25	-240.00	-839.99
207 DL+0.75WL1+0.75L+0.75LR	157.70	-1.50	5.25	177.11	619.88
208 DL+0.75WL2+0.75L+0.75LR	1138.70	-1.50	5.25	-182.89	-640.10
209 0.6DL+1.0WL1	-460.04	-0.90	3.15	240.00	839.99
210 0.6DL+1.0WL2	847.96	-0.90	3.15	-240.00	-839.99
211 0.6DL+0.7SL1	210.07	-0.90	3.15	5.67	19.85
212 0.6DL+0.7SL2	177.86	-0.90	3.15	-5.67	-19.85

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	1471.7	0.00	0.0	-1.5	5.3	504.6	NA	392.4	NA
202 DL+L	1708.7	-3.85	-13.5	-1.5	5.3	585.8	275.8	455.7	177.5
203 DL+LR	1667.8	0.00	0.0	-1.5	5.3	571.8	NA	444.8	NA
204 DL+0.75L+0.75LR	1796.6	-2.89	-10.1	-1.5	5.3	616.0	386.7	479.1	248.8
205 DL+WL1	967.7	240.00	840.0	-1.5	5.3	331.8	2.5	258.0	1.6
206 DL+WL2	2125.7	-240.00	-840.0	-1.5	5.3	728.8	5.5	566.8	3.5
207 DL+0.75WL1+0.75L+0.75LR	1306.1	177.11	619.9	-1.5	5.3	447.8	4.6	348.3	2.9
208 DL+0.75WL2+0.75L+0.75LR	2287.1	-182.89	-640.1	-1.5	5.3	784.1	7.8	609.9	5.0
209 0.6DL+1.0WL1	938.4	240.00	840.0	-0.9	3.2	536.2	2.4	417.0	1.6
210 0.6DL+1.0WL2	1996.4	-240.00	-840.0	-0.9	3.2	1140.8	5.2	887.3	3.3
211 0.6DL+0.7SL1	1358.5	5.67	19.8	-0.9	3.2	776.3	148.9	603.8	95.8
212 0.6DL+0.7SL2	1326.2	-5.67	-19.8	-0.9	3.2	757.9	145.4	589.4	93.6

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Base pressure check of footing F5

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	118.51	0
Min Px	2	201	0	118.51	0
Max Py	1490	209	0	392.84	0
Min Py	2	209	0	0.00	0
Max Pz	2	201	0	118.51	0
Min Pz	2	201	0	118.51	0

Gross Maximum Base pressure, p_{max} = 392.8 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.25 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 520 KN/m^2
> 392.8 KN/m^2
Hence OK

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Design of foundation raft F5

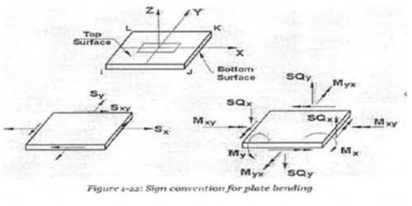
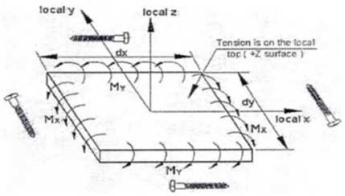


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F5_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	347	1.264	-0.959	0	0	0	-297.6	-461.471	96.199	-393.8	-557.67	105:1.2DL+1.6WL2+1.0L+0.5LR
Min Qx	344	-1.26	-0.952	0	0	0	-295.4	-460.235	-94.754	-390.151	-554.989	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Qy	355	0.194	0.765	0	0	0	162.312	253.439	10.395	172.707	263.834	108:0.9DL+1.6WL1
Min Qy	356	0.256	-1.147	0	0	0	-269.62	-394.315	3.062	-272.686	-397.377	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Sx	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sx	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Sy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Sxy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sxy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Mx	344	0.957	0.577	0	0	0	202.172	261.406	123.292	325.464	384.698	108:0.9DL+1.6WL1
Min Mx	347	1.264	-0.959	0	0	0	-297.6	-461.471	96.199	-393.8	-557.67	105:1.2DL+1.6WL2+1.0L+0.5LR
Max My	344	0.827	0.517	0	0	0	159.086	277.256	139.189	298.275	416.445	104:1.2DL+1.6WL1+1.0L+0.5LR
Min My	347	1.264	-0.959	0	0	0	-297.6	-461.471	96.199	-393.8	-557.67	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Mxy	417	-0.228	0.11	0	0	0	-36.984	31.405	207.151	-244.135	238.556	105:1.2DL+1.6WL2+1.0L+0.5LR
Min Mxy	414	0.232	0.103	0	0	0	-34.788	32.641	-208.594	-243.382	241.235	105:1.2DL+1.6WL2+1.0L+0.5LR

PLATE CENTER STRESS : FDN_F5_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1443	0.632	-0.113	0	0	0	71.462	-450.48	-141.195	212.657	-591.675	108:0.9DL+1.6WL1
Min Qx	1270	-0.267	-0.004	0	0	0	-21.776	-195.835	-9.931	-31.707	-205.766	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Qy	1438	0.584	0.431	0	0	0	91.302	77.883	29.392	120.694	107.275	108:0.9DL+1.6WL1
Min Qy	376	0.067	-0.513	0	0	0	-179.99	-130.115	-9.373	-189.362	-139.488	105:1.2DL+1.6WL2+1.0L+0.5LR
Max Sx	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sx	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Sy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Sxy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Min Sxy	1472	-0.002	-0.002	0	0	0	-0.934	-0.346	0.315	-1.249	-0.661	101:1.4DL
Max Mx	375	0.05	0.301	0	0	0	118.947	69.55	-5.885	124.832	75.435	108:0.9DL+1.6WL1
Min Mx	376	0.067	-0.513	0	0	0	-179.99	-130.115	-9.373	-189.362	-139.488	105:1.2DL+1.6WL2+1.0L+0.5LR
Max My	376	-0.045	0.301	0	0	0	83.276	86.358	2.504	85.78	88.862	104:1.2DL+1.6WL1+1.0L+0.5LR
Min My	1443	0.632	-0.113	0	0	0	71.462	-450.48	-141.195	212.657	-591.675	108:0.9DL+1.6WL1
Max Mxy	420	0.085	0.171	0	0	0	-19.022	-111.54	56.952	-75.974	-168.492	105:1.2DL+1.6WL2+1.0L+0.5LR
Min Mxy	1443	0.632	-0.113	0	0	0	71.462	-450.48	-141.195	212.657	-591.675	108:0.9DL+1.6WL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

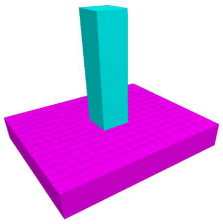
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:0.9D	325.46	top face	50	692	325.46	0.91	0.0005	363.2	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
105:1.2D	-393.8	bottom face	50	690	393.8	1.10	0.0027	1846.7		20	170.19	150	2095.3	29.54	36.93	0.053	tension control
105:1.2D	-244.14	bottom face	50	690	244.14	0.68	0.0016	1136.4		20	252.32	150	2095.3	29.54	36.93	0.053	tension control
105:1.2D	-243.38	bottom face	50	690	243.38	0.68	0.0016	1132.9		20	252.32	150	2095.3	29.54	36.93	0.053	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qx	0.632	1	474	212.66	0.002	0.692	690.191	677.825	1187.24	677.83	2535	> 474	ok	508.4	φVc>Vu	not required
	Min Qx	0.267	1	200.25	31.707			754.661			677.83	2535	> 200.25	ok	508.4	φVc>Vu	not required

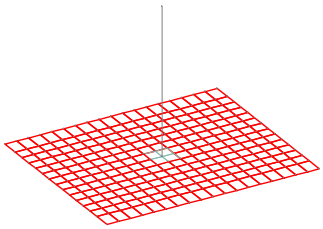
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.2D	416.45	top face	50	692	416.45	1.16	0.0012	795.9	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
105:1.2D	-557.67	bottom face	50	690	557.67	1.56	0.0022	1486.9		20	211.37	150	2095.3	29.54	36.93	0.053	tension control
105:1.2D	238.56	top face	50	690	238.56	0.67	0.0016	1110.2		20	252.32	150	2095.3	29.54	36.93	0.053	tension control
105:1.2D	241.24	top face	50	690	241.24	0.68	0.0016	1122.8		20	252.32	150	2095.3	29.54	36.93	0.053	tension control

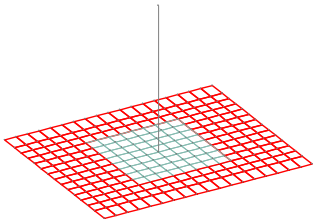
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qy	0.431	1	323.25	107.28	0.002	0.692	702.564	677.825	fc'1187	677.83	711	> 323.25	ok	508.4	φVc>Vu	not required
	Min Qy	0.513	1	384.75	139.49			698.542			677.83	711	> 384.75	ok	508.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **20ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **20ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F6

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m

Size of fdn. raft:	1.50m	1.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

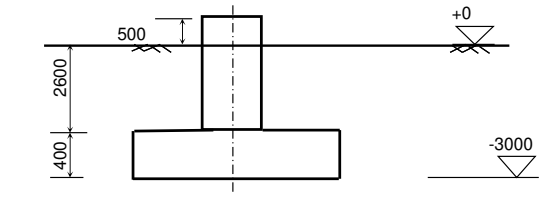
Weight of raft = (1.5*1.5*0.4*24) = 21.60KN

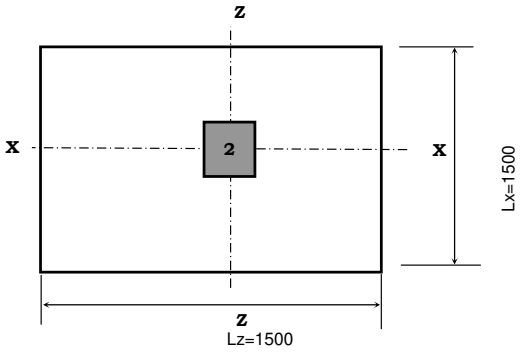
Weight of soil = {(1.5*1.5)-(0.6*0.6)}*2.6*20 = 98.28KN

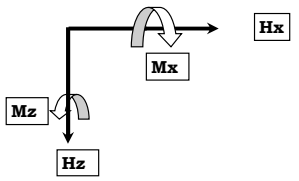
Total = 119.88KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F6

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	169.10	0.00	0.00	0.00	0.00
202 DL+L	476.33	0.00	0.00	0.00	0.00
203 DL+0.75L	399.52	0.00	0.00	0.00	0.00
204 0.6DL+0.7SL1	102.53	0.00	0.00	0.00	0.00
205 0.6DL+0.7SL2	100.39	0.00	0.00	0.00	0.00

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	289.0	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 DL+L	596.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA
203 DL+0.75L	519.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
204 0.6DL+0.7SL1	222.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
205 0.6DL+0.7SL2	220.3	0.00	0.0	0.0	0.0	NA	NA	NA	NA

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Base pressure check of footing F6

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	149.52	0
Min Px	2	201	0	149.52	0
Max Py	2	202	0	290.36	0
Min Py	393	205	0	86.81	0
Max Pz	2	201	0	149.52	0
Min Pz	2	201	0	149.52	0

Gross Maximum Base pressure, p_{max} = 290.4 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.60 m
Unit weight of soil = 20 KN/m^3

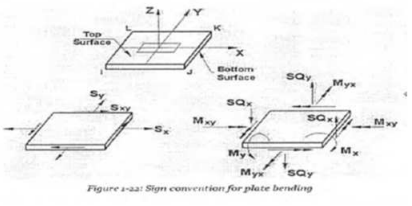
Gross Safe Bearing Capacity = 527 KN/m^2
> 290.4 KN/m^2
Hence OK

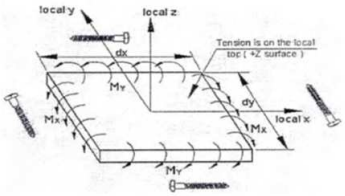
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Design of foundation raft F6





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F6_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.536	-0.096	0	0	0	-24.646	-64.115	-9.516	-34.162	-73.631	102:1.4DL+1.7L
Min Qx	327	-0.536	-0.096	0	0	0	-24.646	-64.115	9.516	-34.162	-73.631	102:1.4DL+1.7L
Max Qy	416	0.096	0.534	0	0	0	-64.035	-24.58	9.52	-73.555	-34.1	102:1.4DL+1.7L
Min Qy	346	0.098	-0.537	0	0	0	-64.192	-24.727	-9.555	-73.747	-34.282	102:1.4DL+1.7L
Max Sx	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sx	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Sy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Sxy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sxy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Mx	1497	-0.029	-0.012	0	0	0	-0.676	-9.049	0.654	-1.33	-9.703	106:0.9DL+1.0SL2
Min Mx	346	0.098	-0.537	0	0	0	-64.192	-24.727	-9.555	-73.747	-34.282	102:1.4DL+1.7L
Max My	406	0.012	0.029	0	0	0	-9.038	-0.675	0.657	-9.695	-1.332	106:0.9DL+1.0SL2
Min My	333	0.536	-0.096	0	0	0	-24.646	-64.115	-9.516	-34.162	-73.631	102:1.4DL+1.7L
Max Mxy	417	0.298	0.299	0	0	0	-26.363	-26.309	16.895	-43.258	-43.204	102:1.4DL+1.7L
Min Mxy	414	-0.298	0.299	0	0	0	-26.363	-26.309	-16.895	-43.258	-43.204	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F6_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	334	0.145	-0.063	0	0	0	-3.408	-45.587	-3.295	-6.703	-48.882	102:1.4DL+1.7L
Min Qx	1497	-0.145	-0.063	0	0	0	-3.408	-45.587	3.295	-6.703	-48.882	102:1.4DL+1.7L
Max Qy	406	0.062	0.144	0	0	0	-45.529	-3.4	3.308	-48.837	-6.708	102:1.4DL+1.7L
Min Qy	356	0.062	-0.145	0	0	0	-45.646	-3.424	-3.318	-48.964	-6.742	102:1.4DL+1.7L
Max Sx	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sx	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Sy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Sxy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Min Sxy	1507	-0.045	0.019	0	0	0	-1.065	-14.219	-1.039	-2.104	-15.258	101:1.4DL
Max Mx	1497	-0.029	-0.012	0	0	0	-0.676	-9.049	0.654	-1.33	-9.703	106:0.9DL+1.0SL2
Min Mx	356	0.062	-0.145	0	0	0	-45.646	-3.424	-3.318	-48.964	-6.742	102:1.4DL+1.7L
Max My	406	0.012	0.029	0	0	0	-9.038	-0.675	0.657	-9.695	-1.332	106:0.9DL+1.0SL2
Min My	1507	-0.145	0.062	0	0	0	-3.415	-45.588	-3.331	-6.746	-48.919	102:1.4DL+1.7L
Max Mxy	418	0.109	0.108	0	0	0	-5.17	-27.349	6.924	-12.094	-34.273	102:1.4DL+1.7L
Min Mxy	1505	-0.109	0.108	0	0	0	-5.17	-27.349	-6.924	-12.094	-34.273	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

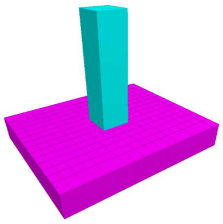
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-1.33	bottom face	50	342	1.33	0.02	0.0000	12.3	615.6	16	326.74	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-73.7	bottom face	50	342	73.747	0.84	0.0020	694.5		16	289.62	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-43.26	bottom face	50	342	43.258	0.49	0.0012	404.9		16	326.74	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-43.26	bottom face	50	342	43.258	0.49	0.0012	404.9		16	326.74	200	1005.8	14.18	17.73	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.145	1	58	6.703	0.003	0.342	374.327	340.826	586.757	340.83	1257	> 58	ok	255.6	φVc>Vu	not required
	Min Qx	0.145	1	58	6.703			374.327			340.83	1257	> 58	ok	255.6	φVc>Vu	not required

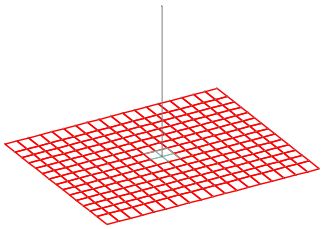
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-1.33	bottom face	50	342	1.332	0.02	0.0000	12.4	615.6	16	326.74	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-73.63	bottom face	50	342	73.631	0.84	0.0020	693.4		16	290.08	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-43.20	bottom face	50	342	43.204	0.49	0.0012	404.4		16	326.74	200	1005.8	14.18	17.73	0.055	tension control
102:1.4D	-43.20	bottom face	50	342	43.204	0.49	0.0012	404.4		16	326.74	200	1005.8	14.18	17.73	0.055	tension control

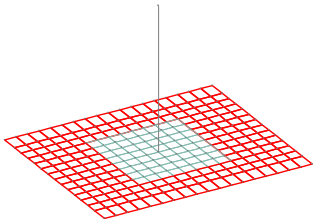
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.144	1	57.6	6.708	0.003	0.342	373.941	340.826	fc'586.8	340.83	356	> 57.6	ok	255.6	φVc>Vu	not required
	Min Qy	0.145	1	58	6.742			374.035			340.83	356	> 58	ok	255.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **200** c/c
At Top **12ϕT** **200** c/c
Along Z Direction :
At Bottom **16ϕT** **200** c/c
At Top **12ϕT** **200** c/c

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DESIGN CALCULATION OF FOOTING F6A

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m
1608	0.60m	0.60m
Size of fdn. raft:	3.70m	2.00m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

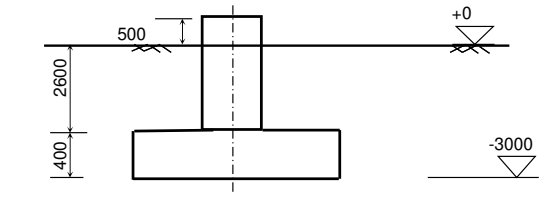
Weight of raft = (3.7*2*0.4*24) = 71.04KN

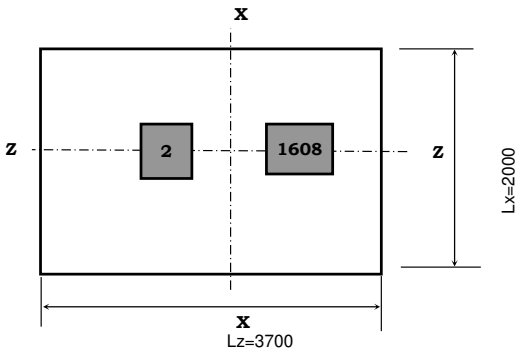
Weight of soil = {(3.7*2)-(0.6*0.6)-(0.6*0.6)}*2.6*20 = 347.36KN

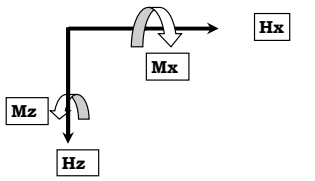
Total = 418.40KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F6A

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	173.60	0.00	0.00	0.00	-69.93
202 DL+L	480.83	0.00	0.00	0.00	-223.54
203 DL+0.75L	404.02	0.00	0.00	0.00	-185.14
204 0.6DL+0.7SL1	105.23	0.00	0.00	0.00	-42.49
205 0.6DL+0.7SL2	103.09	0.00	0.00	0.00	-41.42

Node 1608				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
118.6	-4.6	16.0	0.0	0.0
313.6	-14.6	51.0	0.0	0.0
264.8	-12.1	42.3	0.0	0.0
71.5	-2.9	10.1	0.0	0.0
71.5	-2.9	10.1	0.0	0.0

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	710.6	0.00	-69.9	-4.6	16.0	82.0	10.2	62.1	NA
202 DL+L	1212.8	0.00	-223.5	-14.6	51.0	44.0	5.4	33.3	NA
203 DL+0.75L	1087.3	0.00	-185.1	-12.1	42.3	47.6	5.9	36.0	NA
204 0.6DL+0.7SL1	595.1	0.00	-42.5	-2.9	10.1	108.9	14.0	82.4	NA
205 0.6DL+0.7SL2	593.0	0.00	-41.4	-2.9	10.1	108.5	14.3	82.1	NA

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Base pressure check of footing F6A

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	143.91	0
Min Px	2	201	0	143.91	0
Max Py	325	202	0	341.25	0
Min Py	1663	204	0	50.82	0
Max Pz	2	201	0	143.91	0
Min Pz	2	201	0	143.91	0

Gross Maximum Base pressure, p_{max} = 341.2 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.60 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 527 KN/m^2
> 341.2 KN/m^2
Hence OK

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Design of foundation raft F6A

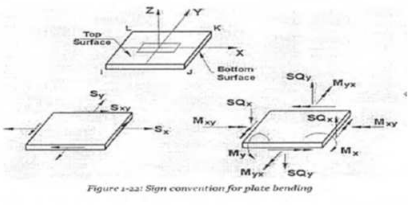
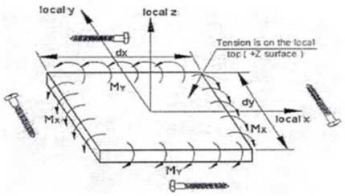


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F6A_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	347	1.389	-0.134	0	0	0	-95.424	-8.797	25.281	-120.705	-34.078	102:1.4DL+1.7L
Min Qx	344	-1.321	-0.14	0	0	0	-97.638	-9.132	-21.496	-119.134	-30.628	102:1.4DL+1.7L
Max Qy	427	0.412	0.984	0	0	0	-45.778	71.275	43.26	-89.038	114.535	102:1.4DL+1.7L
Min Qy	1551	0.038	-0.801	0	0	0	-66.345	46.194	6.684	-73.029	52.878	102:1.4DL+1.7L
Max Sx	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sx	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Sy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Sxy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sxy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Mx	1598	-0.032	-0.102	0	0	0	0.017	14.733	4.181	4.198	18.914	102:1.4DL+1.7L
Min Mx	345	-0.584	-0.731	0	0	0	-176.45	-33.378	-8.731	-185.177	-42.109	102:1.4DL+1.7L
Max My	416	-0.176	0.086	0	0	0	-52.421	232.67	27.62	-80.041	260.29	102:1.4DL+1.7L
Min My	345	-0.584	-0.731	0	0	0	-176.45	-33.378	-8.731	-185.177	-42.109	102:1.4DL+1.7L
Max Mxy	417	-0.223	0.311	0	0	0	-23.382	192.403	49.674	-73.056	242.077	102:1.4DL+1.7L
Min Mxy	414	0.25	0.307	0	0	0	-25.047	188.557	-39.817	-64.864	228.374	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F6A_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1570	0.207	0.119	0	0	0	-0.908	9.308	9.175	-10.083	18.483	102:1.4DL+1.7L
Min Qx	1585	-0.199	0.119	0	0	0	-1.529	8.419	-7.877	-9.406	16.296	102:1.4DL+1.7L
Max Qy	1588	-0.067	0.229	0	0	0	-4.012	101.135	-17.632	-21.644	118.767	102:1.4DL+1.7L
Min Qy	1534	0.015	-0.524	0	0	0	-43.054	96.257	9.849	-52.903	106.106	102:1.4DL+1.7L
Max Sx	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sx	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Sy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Sxy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Min Sxy	1598	-0.013	-0.031	0	0	0	-0.104	4.311	1.255	-1.359	5.566	101:1.4DL
Max Mx	1598	-0.032	-0.102	0	0	0	0.017	14.733	4.181	4.198	18.914	102:1.4DL+1.7L
Min Mx	1534	0.015	-0.524	0	0	0	-43.054	96.257	9.849	-52.903	106.106	102:1.4DL+1.7L
Max My	406	-0.003	0.072	0	0	0	-23.698	217.855	12.348	-36.046	230.203	102:1.4DL+1.7L
Min My	1582	0.09	-0.043	0	0	0	-4.741	-0.806	1.445	-6.186	-2.251	102:1.4DL+1.7L
Max Mxy	408	-0.028	0.169	0	0	0	-11.418	196.149	26.416	-37.834	222.565	102:1.4DL+1.7L
Min Mxy	1588	-0.067	0.229	0	0	0	-4.012	101.135	-17.632	-21.644	118.767	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

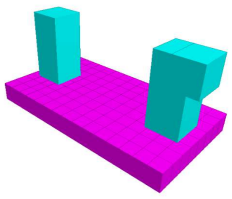
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.4D	4.20	top face	50	342	4.198	0.05	0.0001	39.0	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-185.2	bottom face	50	342	185.18	2.11	0.0036	1214.6		16	165.6	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-73.06	bottom face	50	342	73.056	0.83	0.0020	687.9		16	292.4	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-64.86	bottom face	50	342	64.864	0.74	0.0018	609.8		16	326.74	150	1341	18.91	23.64	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.207	1	82.8	10.083	0.004	0.342	387.752	346.525	586.757	346.52	1261	> 82.8	ok	259.9	φVc>Vu	not required
	Min Qx	0.199	1	79.6	9.406			389.708			346.52	1261	> 79.6	ok	259.9	φVc>Vu	not required

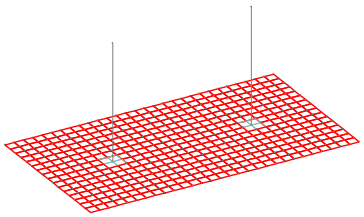
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.4D	260.29	top face	50	340	260.29	3.00	0.0059	2000.5	612	20	157.1	150	2095.3	29.54	36.93	0.025	tension control
102:1.4D	-42.11	bottom face	50	342	42.109	0.48	0.0012	394.1		16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	242.08	top face	50	340	242.08	2.79	0.0053	1811.0		20	173.54	150	2095.3	29.54	36.93	0.025	tension control
102:1.4D	228.37	top face	50	340	228.37	2.63	0.0049	1669.5		20	188.25	150	2095.3	29.54	36.93	0.025	tension control

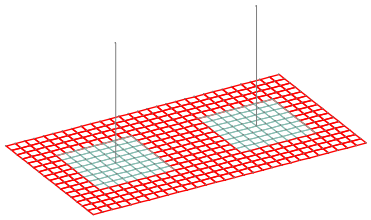
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.229	1	91.6	118.77	0.006	0.34	329.741	346.525	fc'586.8	329.74	347	> 91.6	ok	247.3	φVc>Vu	not required
	Min Qy	0.524	1	209.6	106.11			339.129			339.13	355	> 209.6	ok	254.3	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **20ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F7

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m

Size of fdn. raft:	1.85m	3.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

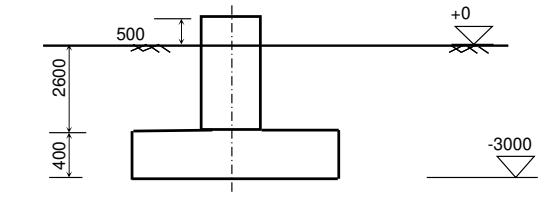
Weight of raft = (1.85*3.5*0.4*24) = 62.16KN

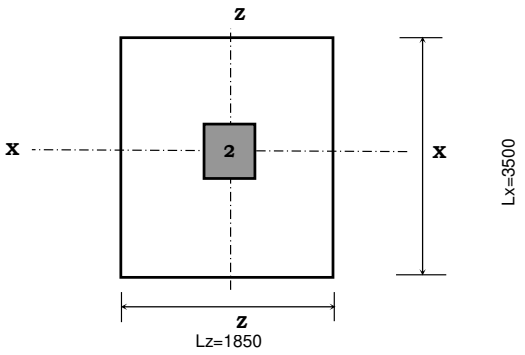
Weight of soil = {(1.85*3.5)-(0.6*0.6)}*2.6*20 = 317.98KN

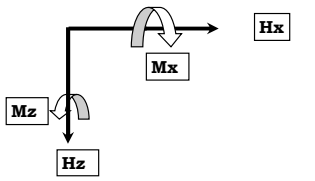
Total = 380.14KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F7

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	230.50	0.00	0.00	-5.25	-18.38
202 DL+L	706.75	0.00	0.00	-17.75	-62.12
203 DL+0.75L	587.69	0.00	0.00	-14.63	-51.19
204 0.6DL+0.7SL1	144.96	0.00	0.00	-3.15	-11.03
205 0.6DL+0.7SL2	131.64	0.00	0.00	-3.15	-11.03

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	610.6	-5.25	-18.4	0.0	0.0	NA	58.2	NA	46.5
202 DL+L	1086.9	-17.75	-62.1	0.0	0.0	NA	30.6	NA	24.5
203 DL+0.75L	967.8	-14.63	-51.2	0.0	0.0	NA	33.1	NA	26.5
204 0.6DL+0.7SL1	525.1	-3.15	-11.0	0.0	0.0	NA	83.3	NA	66.7
205 0.6DL+0.7SL2	511.8	-3.15	-11.0	0.0	0.0	NA	81.2	NA	65.0

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Base pressure check of footing F7

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	117.47	0
Min Px	2	201	0	117.47	0
Max Py	2	202	0	211.42	0
Min Py	1631	205	0	54.74	0
Max Pz	2	201	0	117.47	0
Min Pz	2	201	0	117.47	0

Gross Maximum Base pressure, p_{max} = 211.4 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.60 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 527 KN/m^2
> 211.4 KN/m^2
Hence OK

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Design of foundation raft F7

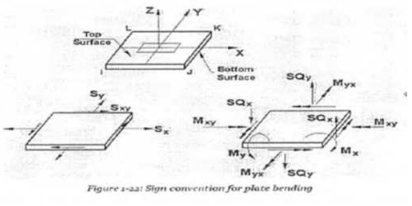
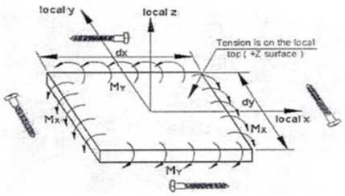


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F7_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	334	0.39	-0.084	0	0	0	-9.241	-183.61	-4.054	-13.295	-187.664	102:1.4DL+1.7L
Min Qx	1497	-0.39	-0.084	0	0	0	-9.241	-183.61	4.054	-13.295	-187.664	102:1.4DL+1.7L
Max Qy	406	0.071	0.537	0	0	0	-51.196	-72.892	8.324	-59.52	-81.216	102:1.4DL+1.7L
Min Qy	356	0.122	-0.743	0	0	0	-67.222	-104.168	-9.471	-76.693	-113.639	102:1.4DL+1.7L
Max Sx	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sx	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Sy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Sxy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sxy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Mx	1570	-0.015	-0.006	0	0	0	-0.026	-29.652	0.315	-0.341	-29.967	106:0.9DL+1.0SL2
Min Mx	356	0.122	-0.743	0	0	0	-67.222	-104.168	-9.471	-76.693	-113.639	102:1.4DL+1.7L
Max My	1552	0	0.006	0	0	0	-1.753	-0.179	0.107	-1.86	-0.286	106:0.9DL+1.0SL2
Min My	1497	-0.39	-0.084	0	0	0	-9.241	-183.61	4.054	-13.295	-187.664	102:1.4DL+1.7L
Max Mxy	354	-0.224	-0.569	0	0	0	-42.759	-106.801	21.485	-64.244	-128.286	102:1.4DL+1.7L
Min Mxy	357	0.224	-0.569	0	0	0	-42.759	-106.801	-21.485	-64.244	-128.286	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F7_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1511	0.1	-0.42	0	0	0	-28.893	-62.24	-14.073	-42.966	-76.313	102:1.4DL+1.7L
Min Qx	1508	-0.1	-0.42	0	0	0	-28.893	-62.24	14.073	-42.966	-76.313	102:1.4DL+1.7L
Max Qy	1538	0.023	0.355	0	0	0	-31.55	-41.38	4.781	-36.331	-46.161	102:1.4DL+1.7L
Min Qy	1510	0.043	-0.484	0	0	0	-42.004	-58.637	-5.925	-47.929	-64.562	102:1.4DL+1.7L
Max Sx	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sx	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Sy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Sxy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Min Sxy	1583	-0.002	0.007	0	0	0	-0.25	-0.613	-0.168	-0.418	-0.781	101:1.4DL
Max Mx	1570	-0.015	-0.006	0	0	0	-0.026	-29.652	0.315	-0.341	-29.967	106:0.9DL+1.0SL2
Min Mx	1510	0.043	-0.484	0	0	0	-42.004	-58.637	-5.925	-47.929	-64.562	102:1.4DL+1.7L
Max My	1552	0	0.006	0	0	0	-1.753	-0.179	0.107	-1.86	-0.286	106:0.9DL+1.0SL2
Min My	1570	-0.085	-0.032	0	0	0	-0.145	-169.827	1.707	-1.852	-171.534	102:1.4DL+1.7L
Max Mxy	1523	-0.089	-0.329	0	0	0	-12.492	-66.788	14.187	-26.679	-80.975	102:1.4DL+1.7L
Min Mxy	1512	0.089	-0.329	0	0	0	-12.492	-66.788	-14.187	-26.679	-80.975	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

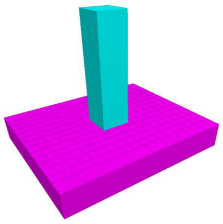
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-0.34	bottom face	50	342	0.341	0.00	0.0000	3.2	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-76.7	bottom face	50	342	76.693	0.87	0.0021	722.7		16	278.33	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-64.24	bottom face	50	342	64.244	0.73	0.0018	603.9		16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-64.24	bottom face	50	342	64.244	0.73	0.0018	603.9		16	326.74	150	1341	18.91	23.64	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.1	1	40	42.966	0.004	0.342	330.986	346.525	586.757	330.99	1250	> 40	ok	248.2	φVc>Vu	not required
	Min Qx	0.1	1	40	42.966			330.986			330.99	1250	> 40	ok	248.2	φVc>Vu	not required

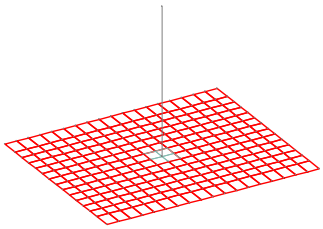
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-0.29	bottom face	50	342	0.286	0.00	0.0000	2.7	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-187.66	bottom face	50	342	187.66	2.14	0.0036	1239.6		16	162.27	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-128.29	bottom face	50	342	128.29	1.46	0.0036	1221.6		16	164.65	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-128.29	bottom face	50	342	128.29	1.46	0.0036	1221.6		16	164.65	150	1341	18.91	23.64	0.04	tension control

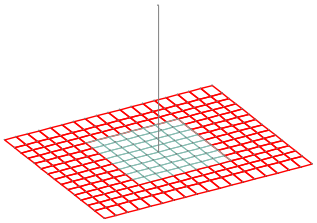
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.355	1	142	46.161	0.004	0.342	347.712	346.525	fc'586.8	346.52	360	> 142	ok	259.9	φVc>Vu	not required
	Min Qy	0.484	1	193.6	64.562			347.107			346.52	360	> 193.6	ok	259.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
- At Bottom **16ϕT** **150** c/c
- At Top **12ϕT** **150** c/c

Along Z Direction :
- At Bottom **16ϕT** **150** c/c
- At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F8

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m
1575	0.60m	0.60m
Size of fdn. raft:	1.85m	3.75m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-2.00	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

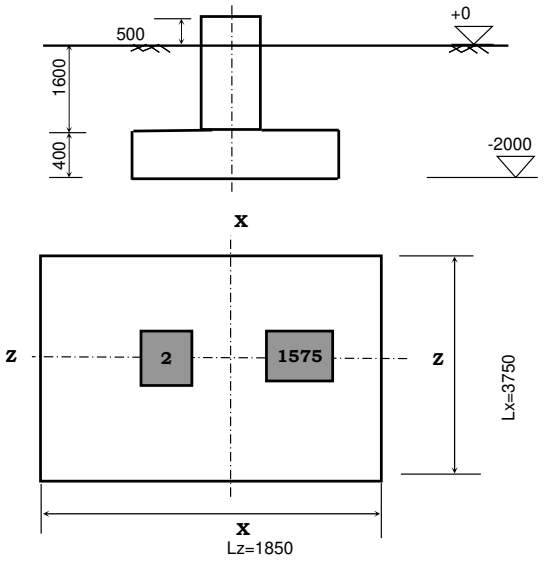
Weight of raft = (1.85*3.75*0.4*24) = 66.60KN

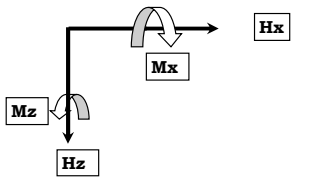
Weight of soil = {(3.75*1.85)-(0.6*0.6)-(0.6*0.6)}*1.6*20 = 198.96KN

Total = 265.56KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F8

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	221.50	0.00	0.00	-5.25	-13.13
202 DL+L	697.75	0.00	0.00	-17.75	-44.38
203 DL+0.75L	578.69	0.00	0.00	-14.63	-36.56
204 0.6DL+0.7SL1	139.56	0.00	0.00	-3.15	-7.88
205 0.6DL+0.7SL2	126.24	0.00	0.00	-3.15	-7.88

Node 1575				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
41.8	0.0	0.0	-0.5	-1.4
83.8	0.0	0.0	-2.1	-5.2
73.3	0.0	0.0	-1.7	-4.3
25.7	0.0	0.0	-0.3	-0.8
25.7	0.0	0.0	-0.3	-0.8

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	528.9	-5.79	-14.5	0.0	0.0	NA	68.5	NA	36.5
202 DL+L	1047.1	-19.84	-49.6	0.0	0.0	NA	39.6	NA	21.1
203 DL+0.75L	917.6	-16.33	-40.8	0.0	0.0	NA	42.1	NA	22.5
204 0.6DL+0.7SL1	430.8	-3.47	-8.7	0.0	0.0	NA	93.0	NA	49.6
205 0.6DL+0.7SL2	417.5	-3.47	-8.7	0.0	0.0	NA	90.1	NA	48.1

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Base pressure check of footing F8

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	108.02	0
Min Px	2	201	0	108.02	0
Max Py	2	202	0	207.02	0
Min Py	1664	205	0	50.14	0
Max Pz	2	201	0	108.02	0
Min Pz	2	201	0	108.02	0

Gross Maximum Base pressure, p_{max} = 207.0 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 1.60 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 507 KN/m^2
> 207.0 KN/m^2
Hence OK

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Design of foundation raft F8

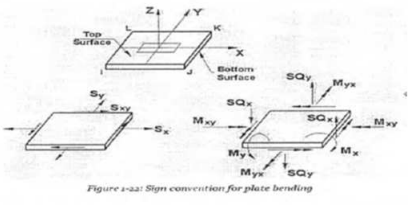
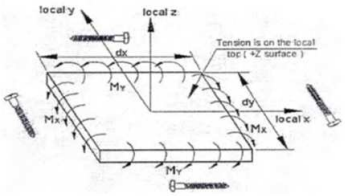


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.906	-0.028	0	0	0	-48.175	-205.664	-13.284	-61.459	-218.948	102:1.4DL+1.7L
Min Qx	327	-0.906	-0.028	0	0	0	-48.175	-205.664	13.284	-61.459	-218.948	102:1.4DL+1.7L
Max Qy	416	0.084	0.894	0	0	0	-93.129	-110.977	17.734	-110.863	-128.711	102:1.4DL+1.7L
Min Qy	346	0.208	-1.115	0	0	0	-108.67	-164.801	-14.437	-123.109	-179.238	102:1.4DL+1.7L
Max Sx	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sx	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Sy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Sxy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sxy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Mx	1570	-0.014	-0.001	0	0	0	-0.063	-26.538	0.32	-0.383	-26.858	106:0.9DL+1.0SL2
Min Mx	346	0.208	-1.115	0	0	0	-108.67	-164.801	-14.437	-123.109	-179.238	102:1.4DL+1.7L
Max My	1595	0.004	-0.01	0	0	0	-4.002	-0.237	-0.211	-4.213	-0.448	106:0.9DL+1.0SL2
Min My	333	0.906	-0.028	0	0	0	-48.175	-205.664	-13.284	-61.459	-218.948	102:1.4DL+1.7L
Max Mxy	417	0.353	0.601	0	0	0	-40.667	-119.13	33.281	-73.948	-152.411	102:1.4DL+1.7L
Min Mxy	414	-0.353	0.601	0	0	0	-40.667	-119.13	-33.281	-73.948	-152.411	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F8_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	334	0.381	-0.064	0	0	0	-9.641	-173.635	-5.3	-14.941	-178.935	102:1.4DL+1.7L
Min Qx	1497	-0.381	-0.064	0	0	0	-9.641	-173.635	5.3	-14.941	-178.935	102:1.4DL+1.7L
Max Qy	406	0.078	0.56	0	0	0	-53.461	-62.871	8.238	-61.699	-71.109	102:1.4DL+1.7L
Min Qy	1510	0.048	-0.393	0	0	0	-46.153	-62.594	-4.855	-51.008	-67.449	102:1.4DL+1.7L
Max Sx	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sx	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Sy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Sxy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Min Sxy	1632	-0.003	0.011	0	0	0	-0.406	-0.963	-0.284	-0.69	-1.247	101:1.4DL
Max Mx	1570	-0.014	-0.001	0	0	0	-0.063	-26.538	0.32	-0.383	-26.858	106:0.9DL+1.0SL2
Min Mx	406	0.078	0.56	0	0	0	-53.461	-62.871	8.238	-61.699	-71.109	102:1.4DL+1.7L
Max My	1595	0.004	-0.01	0	0	0	-4.002	-0.237	-0.211	-4.213	-0.448	106:0.9DL+1.0SL2
Min My	1497	-0.381	-0.064	0	0	0	-9.641	-173.635	5.3	-14.941	-178.935	102:1.4DL+1.7L
Max Mxy	418	0.221	0.365	0	0	0	-11.856	-122.871	19.654	-31.51	-142.525	102:1.4DL+1.7L
Min Mxy	1505	-0.221	0.365	0	0	0	-11.856	-122.871	-19.654	-31.51	-142.525	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

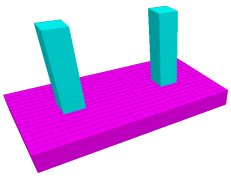
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-0.38	bottom face	50	342	0.383	0.00	0.0000	3.6	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-123.1	bottom face	50	342	123.11	1.40	0.0034	1171.1		16	171.76	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-73.95	bottom face	50	342	73.948	0.84	0.0020	696.4		16	288.82	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-73.95	bottom face	50	342	73.948	0.84	0.0020	696.4		16	288.82	150	1341	18.91	23.64	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.381	1	152.4	14.941	0.004	0.342	403.254	346.525	586.757	346.52	1261	> 152.4	ok	259.9	φVc>Vu	not required
	Min Qx	0.381	1	152.4	14.941			403.254			346.52	1261	> 152.4	ok	259.9	φVc>Vu	not required

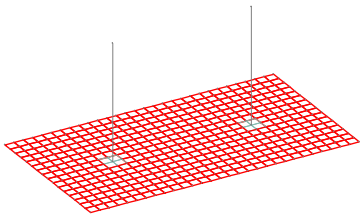
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:0.9D	-0.45	bottom face	50	342	0.448	0.01	0.0000	4.2	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-218.95	bottom face	50	342	218.95	2.50	0.0037	1270.6		16	158.3	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-152.41	bottom face	50	342	152.41	1.74	0.0026	888.7		16	226.34	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-152.41	bottom face	50	342	152.41	1.74	0.0026	888.7		16	226.34	150	1341	18.91	23.64	0.04	tension control

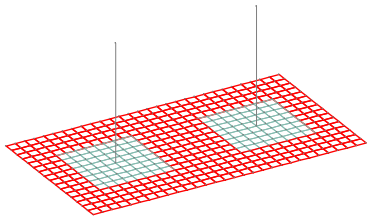
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qy	0.56	1	224	71.109	0.004	0.342	348.288	346.525	fc'586.8	346.52	360	> 224	ok	259.9	φVc>Vu	not required
	Min Qy	0.393	1	157.2	67.449			341.899			341.9	357	> 157.2	ok	256.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F9

Size of Columns at node	Along-X	Along-Z
2	0.60m	0.60m
1753	0.60m	0.60m
Size of fdn. raft:	2.95m	2.25m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.50	m
U/S level of fdn:	-2.30	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

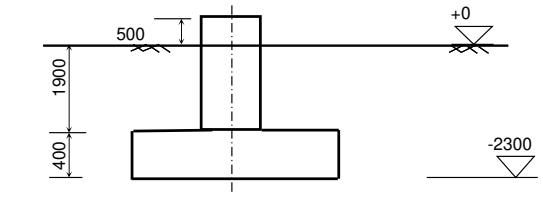
Weight of raft = (2.95*2.25*0.4*24) = 63.72KN

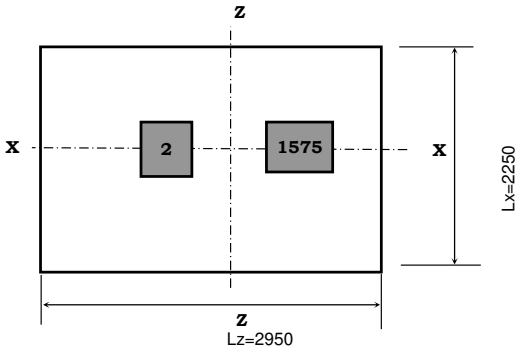
Weight of soil = {(2.95*2.25)-(0.6*0.6)-(0.6*0.6)}*1.9*20 = 224.87KN

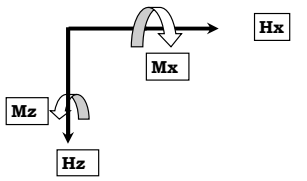
Total = 288.59KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F9

Load at bottom of footing	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL	224.20	0.00	0.00	-5.25	-14.70
202 DL+L	700.45	0.00	0.00	-17.75	-49.70
203 DL+0.75L	581.39	0.00	0.00	-14.63	-40.95
204 0.6DL+0.7SL1	141.18	0.00	0.00	-3.15	-8.82
205 0.6DL+0.7SL2	127.86	0.00	0.00	-3.15	-8.82

Node 1753				
V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
224.2	0.0	0.0	-5.3	-14.7
700.5	0.0	0.0	-17.8	-49.7
581.4	0.0	0.0	-14.6	-40.9
141.2	0.0	0.0	-3.2	-8.8
127.9	0.0	0.0	-3.2	-8.8

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL	737.0	-10.50	-29.4	0.0	0.0	NA	28.2	NA	28.1
202 DL+L	1689.5	-35.50	-99.4	0.0	0.0	NA	19.1	NA	19.0
203 DL+0.75L	1451.4	-29.25	-81.9	0.0	0.0	NA	19.9	NA	19.8
204 0.6DL+0.7SL1	570.9	-6.30	-17.6	0.0	0.0	NA	36.4	NA	36.3
205 0.6DL+0.7SL2	544.3	-6.30	-17.6	0.0	0.0	NA	34.7	NA	34.6

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Base pressure check of footing F9

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	146.13	0
Min Px	2	201	0	146.13	0
Max Py	1846	202	0	399.13	0
Min Py	1758	205	0	54.97	0
Max Pz	2	201	0	146.13	0
Min Pz	2	201	0	146.13	0

Gross Maximum Base pressure, p_{max} = 399.1 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2
Depth of soil above footing = 2.60 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 527 KN/m^2
> 399.1 KN/m^2
Hence OK

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Design of foundation raft F9

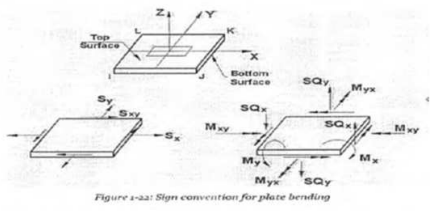
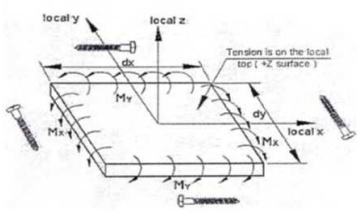


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	1.12	-0.108	0	0	0	60.866	-145.046	-8.45	69.316	-153.496	102:1.4DL+1.7L
Min Qx	1660	-1.12	-0.107	0	0	0	55.465	-142.546	9.633	65.098	-152.179	102:1.4DL+1.7L
Max Qy	415	0.183	0.834	0	0	0	-31.323	-79.913	7.764	-39.087	-87.677	102:1.4DL+1.7L
Min Qy	1602	0.104	-1.23	0	0	0	-39.664	-119.061	5.472	-45.136	-124.533	102:1.4DL+1.7L
Max Sx	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sx	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Sy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Sxy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sxy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Mx	1556	0.161	-0.118	0	0	0	137.597	-87.334	0.106	137.703	-87.44	102:1.4DL+1.7L
Min Mx	1738	-0.075	-1.212	0	0	0	-45.602	-113.831	-7.928	-53.53	-121.759	102:1.4DL+1.7L
Max My	1646	0.032	0.078	0	0	0	2.198	0.107	3.092	5.29	3.199	102:1.4DL+1.7L
Min My	1642	-0.397	-0.077	0	0	0	-7.924	-201.076	11.846	-19.77	-212.922	102:1.4DL+1.7L
Max Mxy	417	0.506	0.433	0	0	0	57.273	-76.432	45.831	103.104	-122.263	102:1.4DL+1.7L
Min Mxy	1666	-0.51	0.43	0	0	0	52.747	-75.157	-42.061	94.808	-117.218	102:1.4DL+1.7L

PLATE CENTER STRESS : FDN_F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	334	0.55	-0.158	0	0	0	115.145	-106.697	-1.058	116.203	-107.755	102:1.4DL+1.7L
Min Qx	1659	-0.56	-0.158	0	0	0	111.613	-105.458	2.242	113.855	-107.7	102:1.4DL+1.7L
Max Qy	1643	0.016	0.562	0	0	0	-4.56	-34.42	7.871	-12.431	-42.291	102:1.4DL+1.7L
Min Qy	1740	-0.103	-0.628	0	0	0	-9.655	-40.246	1.017	-10.672	-41.263	102:1.4DL+1.7L
Max Sx	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sx	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Sy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Sxy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Min Sxy	1766	-0.006	-0.05	0	0	0	-0.653	-2.237	0.087	-0.74	-2.324	101:1.4DL
Max Mx	1556	0.161	-0.118	0	0	0	137.597	-87.334	0.106	137.703	-87.44	102:1.4DL+1.7L
Min Mx	1740	-0.103	-0.628	0	0	0	-9.655	-40.246	1.017	-10.672	-41.263	102:1.4DL+1.7L
Max My	1646	0.032	0.078	0	0	0	2.198	0.107	3.092	5.29	3.199	102:1.4DL+1.7L
Min My	334	0.55	-0.158	0	0	0	115.145	-106.697	-1.058	116.203	-107.755	102:1.4DL+1.7L
Max Mxy	407	0.265	0.285	0	0	0	52.708	-36.807	31.863	84.571	-68.67	102:1.4DL+1.7L
Min Mxy	1663	-0.272	0.284	0	0	0	48.847	-36.468	-27.7	76.547	-64.168	102:1.4DL+1.7L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

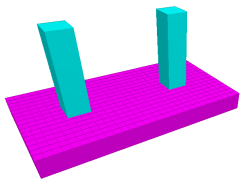
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.4D	137.70	top face	50	342	137.7	1.57	0.0038	1313.9	615.6	16	153.09	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-53.5	bottom face	50	342	53.53	0.61	0.0015	502.1		16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	103.10	top face	50	342	103.1	1.18	0.0029	976.8		16	205.93	150	1341	18.91	23.64	0.04	tension control
102:1.4D	94.81	top face	50	342	94.808	1.08	0.0026	896.6		16	224.33	150	1341	18.91	23.64	0.04	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	Max Qx	0.55	1	220	116.2	0.004	0.342	338.489	346.525	586.757	338.49	1255 > 220	ok	253.9	φVc>Vu	not required
	Min Qx	0.56	1	224	113.86			339.067			339.07	1256 > 224	ok	254.3	φVc>Vu	not required

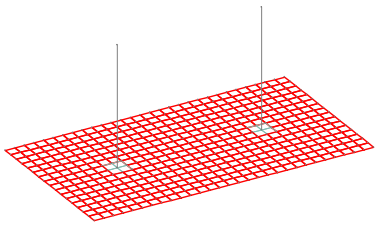
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102:1.4D	3.20	top face	50	342	3.199	0.04	0.0001	29.7	615.6	16	326.74	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-212.92	bottom face	50	342	212.92	2.43	0.0035	1209.4		16	166.32	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-122.26	bottom face	50	342	122.26	1.39	0.0034	1162.8		16	172.98	150	1341	18.91	23.64	0.04	tension control
102:1.4D	-117.22	bottom face	50	342	117.22	1.34	0.0033	1113.7		16	180.61	150	1341	18.91	23.64	0.04	tension control

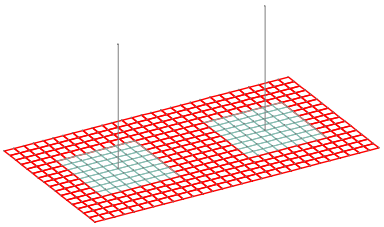
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	Max Qy	0.562	1	224.8	42.291	0.004	0.342	365.171	346.525	fc'586.8	346.52	360 > 224.8	ok	259.9	φVc>Vu	not required
	Min Qy	0.645	1	258	41.263			372.477			346.52	360 > 258	ok	259.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**150**	c/c
At Top	**16ϕT**	**150**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**150**	c/c
At Top	**16ϕT**	**150**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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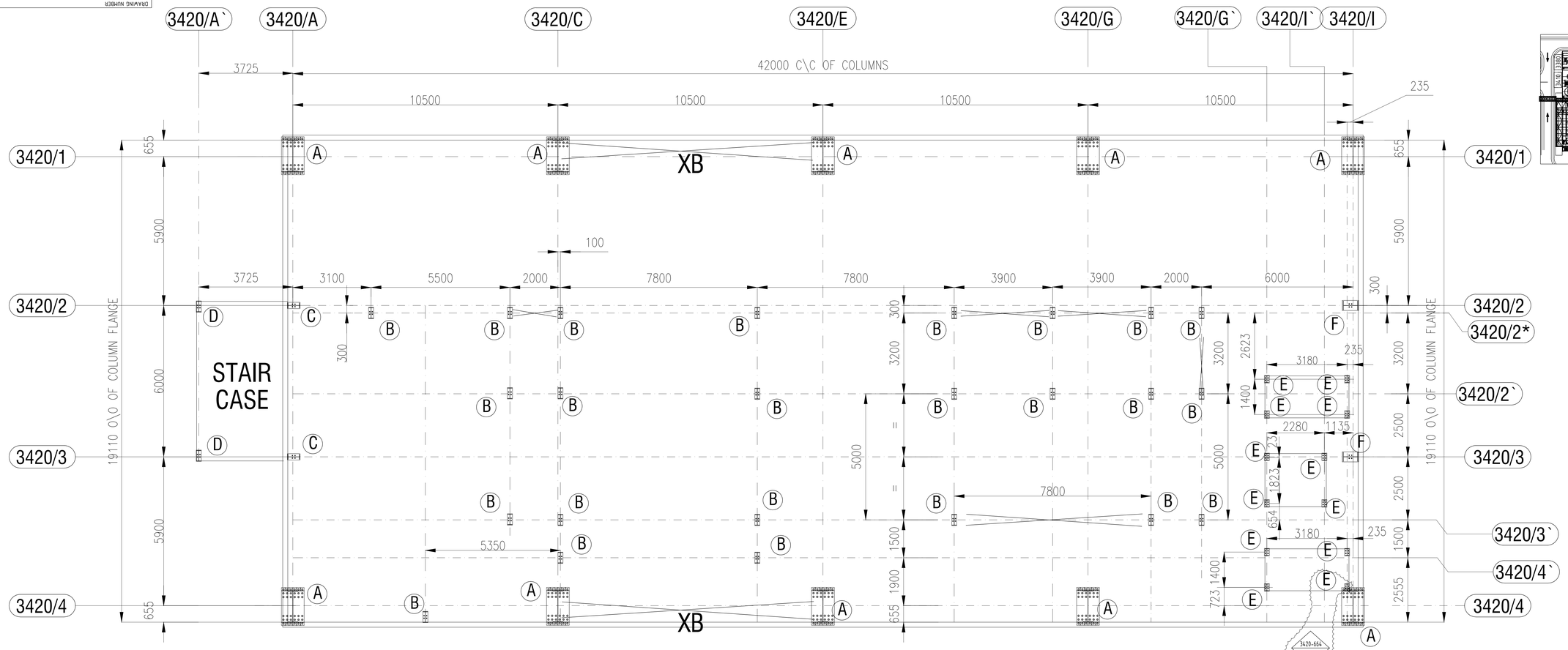
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ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER MD-216-3420-EG-ST-GAD-2229 REV D01

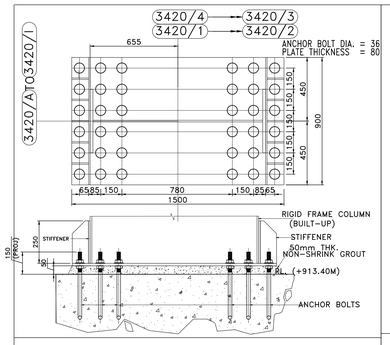


ANCHOR BOLT PLAN

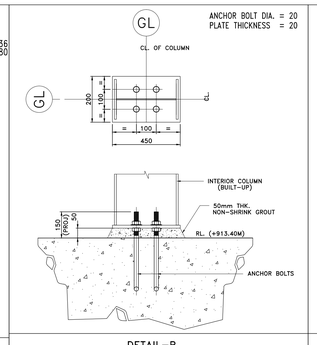
FFL IS AT LEVEL +913.40

I DENOTES TO PLATFORM COLUMNS

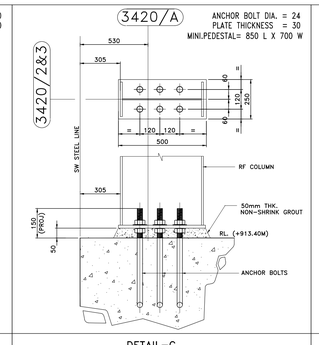
XB DENOTE TO X-BRACING.



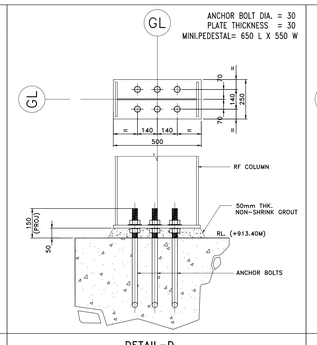
DETAIL–A



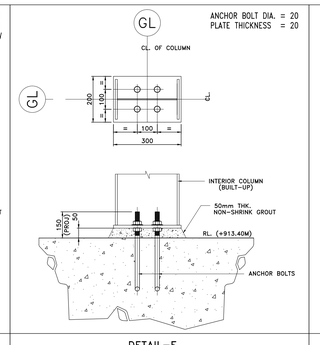
DETAIL–B



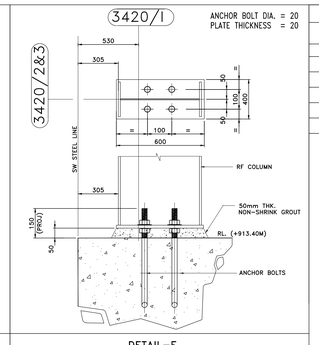
DETAIL–C



DETAIL–D



DETAIL–E



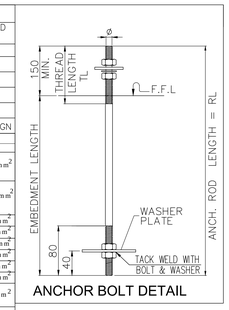
DETAIL–F

NOTE: ANCHOR BOLT SPECIFICATION (AS PER GR. 8.8)

QTY.	DIA. OF BOLT	DIA. OF HOLE	THREAD LENGTH	ANCHOR ROD LENGTH
	ø	D	TL	RL
--	M20	26	150	450
--	M24	30	150	500
--	M30	36	200	700
--	M36	42	200	1000
	>M36	ø + 8	200	AS PER DESIGN

MATERIAL SPECIFICATIONS

NO.	MATERIALS		GRADE & SPECIFICATION	STRENGTH
01	ANCHOR BOLTS	J-type	ASTM F1554 Grade A36M, Hot dip galvanized to ASTM A153M, Heavy Hex Nuts ASESM-10S Washers-ASTMF436	Fu = 400 N/mm²
		STRAIGHT -type	BS EN ISO 898-1 GRDE 8.8, Hot dip galvanized to BS EN ISO1461 BS EN ISO 898-2 10 STYLE2 WASHER-BS4320	Fu = 800 N/mm²
02	BUILT-UP SECTIONS		ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
03	HOT ROLLED SECTIONS	Tubes	E10025 S355HR or equivalent	Fy = 345 N/mm²
		I-Sections	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
		Channels	E10025 S355JR or equivalent	Fy = 355 N/mm²
		Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
		Pipes	ASTM A53 Grade B or equivalent	Fy = 240 N/mm²
04	Primary Connection bolts		ASTM A325M TYPE-1; Fully threaded hot dip galvanized	Ft = 827 N/mm²
05	DIAGONAL "X" BRACING	HR Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
06	Cold formed secondary members		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm²
07	Secondary Connection bolts (for purlins,girts and flange braces)		Grade 8.8 Fully threaded hot dip galvanized	Ft = 400 N/mm²
08	Sag Rods		ASTM Grade A36M Electro galvanized, Fully threaded with 4 nuts Coating Z275 (Grade 90)	Fy = 245 N/mm²
09	Mezzanine Deck Panels		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm²



ANCHOR BOLT DETAIL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3420-EG-ME-GAD-1015	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL929.300								
MD-216-3420-EG-ME-GAD-1014	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL920.100								
MD-216-3420-EG-ME-GAD-1013	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1012	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL929.30	D01	20/11/2023	AS BUILT	--	SYC	--	PRI	
MD-216-3420-EG-ME-GAD-1011	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL912.900	C01	11/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	--	SYC	--	PRI	
MD-216-3420-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING PLAN	A01	02/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. +966 2 691 0999
Fax +966 2 691 0099
E-mail asc@ascsteel.com
Site www.ascsteel.com

DRAWING NUMBER		REVISION
MD-P20131-B03-A001		03
DATE 29/06/2020	DATE 29/06/2020	DATE 29/06/2020
DRAWN 7014	CHECKED 7013	APPROVED 7001

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_S_G_26043_001	2
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 _ GAU01 _ HH12 _ MTL01_ 34211	OU100668674 / 2

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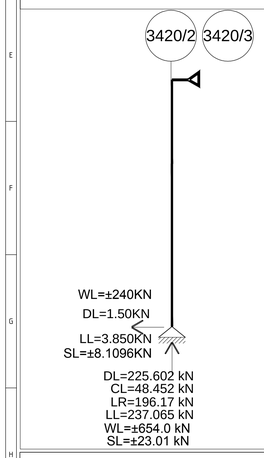
DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	02/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	PRESSURE OXIDATION-AUTOCLAVE BUILDING
APPROVED	PRI	02/07/2020	ANCHOR BOLT PLAN AND DETAILS
EMPLOYER			
SCALE NTS	A0	DRAWING NUMBER MD-216-3420-EG-ST-GAD-2229	REV D01

DRAWING NUMBER MD-216-3420-EG-ST-GAD-2230 REV D01

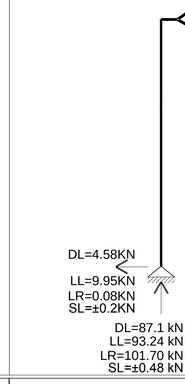
LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal pressure
Wind_Left3	Denotes wind load from left to right with internal suction
Wind_Right4	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2.
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

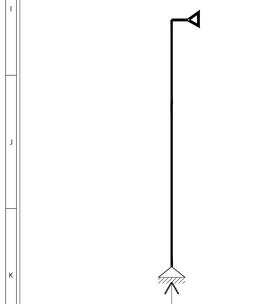
ENDWALL COLUMN REACTION AT GL I



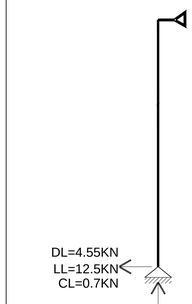
PLATFORM COLUMN REACTION AT GL 3420/2



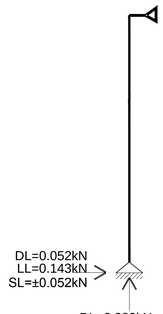
PLATFORM COLUMN REACTION AT GL 3420/2`

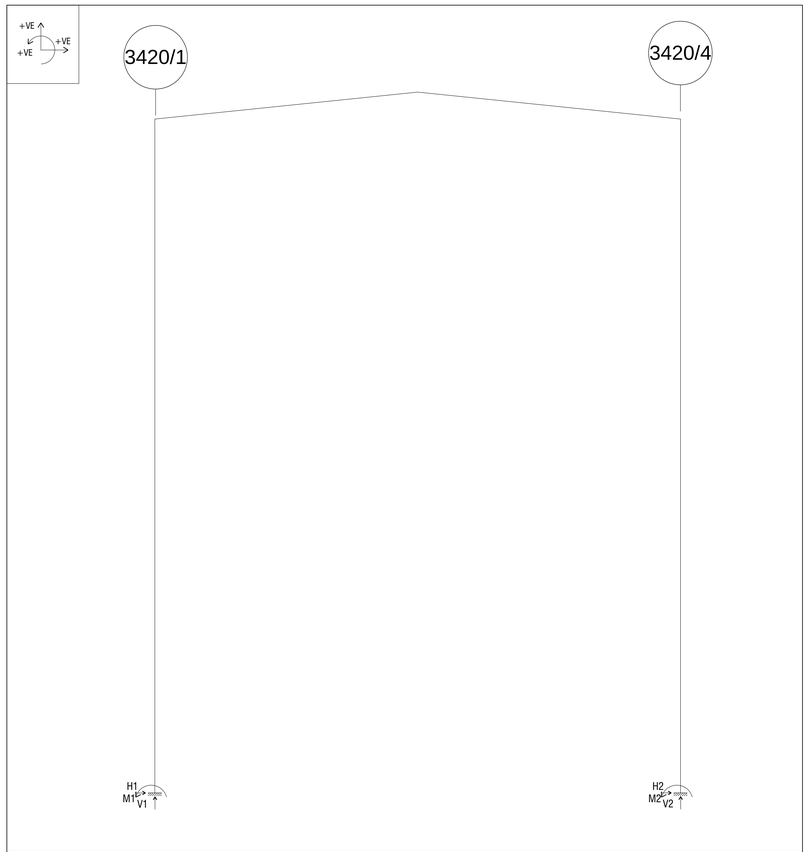


PLATFORM COLUMN REACTION AT GL 3420/3`



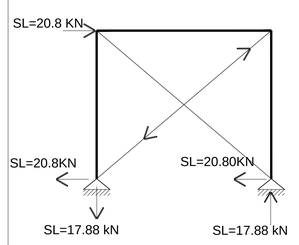
TYP.PLATFORM COLUMN REACTION AT GL 3420/G`&I` (separate platform)



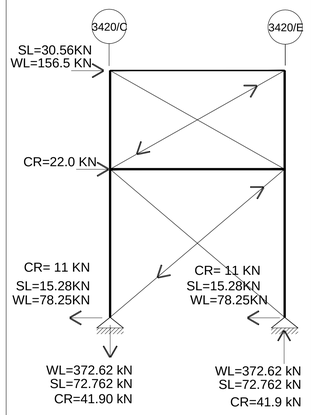


TYP. FRAME @GL 3420/A TO GL 3420/I							
CATEGORY	LOAD CASE	LEFT COL @GL 3420/1			RIGHT COL @GL 3420/4		
		H1 (KN)	V1 (KN)	M1(KN.M)	H2 (KN)	V2 (KN)	M2 (KN.M)
Dead	Dead	4.9	94.8	-55.6	-12.1	224.4	108.3
Collateral	Collateral	-0.2	24.8	-14.4	0.2	167.3	-19.7
Roof Live	Live	13.5	72.6	-160.4	-13.5	144.5	162.0
FLOOR Live	Floor Live	-1.0	52.2	19.8	-18.2	325.7	120.0
Crane	Crane1	-8.0	29.8	148.2	-14.7	140.5	214.9
	Crane2	-7.9	131.7	184.6	-14.8	38.6	254.2
	Crane3	14.3	140.5	-205.7	8.4	29.8	-157.2
	Crane4	14.3	38.6	-242.2	8.4	131.7	-196.3
Wind	Wind_Left1	-338.8	-281.6	3672.1	-313.7	-142.4	3180.7
	Wind_Left2	-473.9	-196.1	4432.0	-188.5	-57.5	2560.2
	Wind_Left3	-324.9	-129.6	3543.1	-344.4	-21.3	3618.7
	Wind_Left4	-452.6	-48.7	4237.0	-216.7	60.9	2951.2
	Wind_Right1	309.8	-140.0	-3077.7	352.7	-276.8	-3895.1
	Wind_Right2	182.0	-58.0	-2410.4	480.5	-195.6	-4591.3
	Wind_Right3	337.6	-21.3	-3460.1	331.7	-129.5	-3703.0
	Wind_Right4	209.8	60.6	-2794.0	459.4	-48.4	-4399.4
	Wind_Long1	280.3	-145.4	-1523.8	-280.3	-145.9	1528.9
	Wind_Long2	152.6	-64.1	-847.1	-152.6	-64.3	850.0
	Wind_Long3	293.1	-75.1	-1678.0	-293.1	-75.7	1683.9
	Wind_Long4	165.6	6.3	-1001.7	-165.6	5.9	1005.4
Seismic	Seismic_L	-5.1	-2.8	98.3	-24.9	2.8	217.4
	Seismic_R	5.1	2.8	-98.3	24.9	-2.8	-217.4
Temperature	25Degree	1.1	0.0	-23.7	-1.1	0.0	23.9
	-25Degree	-1.1	0.0	23.7	1.1	0.0	-23.9

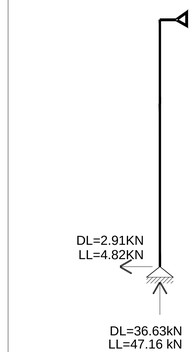
SIDEWALL (XB) BRACING REACTION AT GL 3420/2&3420/3`



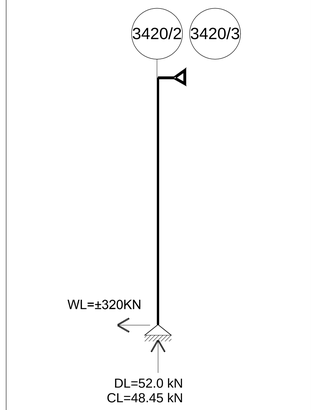
SIDEWALL (XB) BRACING REACTION AT GL 3420/1& 3420/4



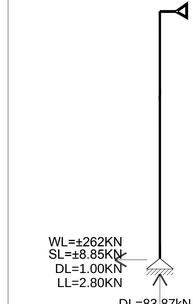
PLATFORM COLUMN REACTION AT GL 3420/4



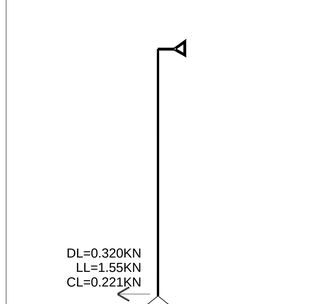
ENDWALL COLUMN REACTION AT GL 3240/A



STAIR COLUMN REACTION



PLATFORM COLUMN REACTION AT GL 3420/4`



WIND IS CALCULATED BASED ON SBC-301(2018)
ALL HORIZONTAL REACTION UNDER PLATFORM COLUMNS ARE IN TRANSVERSE DIRECTION.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3420-EG-ME-GAD-1015	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL929.300								
MD-216-3420-EG-ME-GAD-1014	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL920.100								
MD-216-3420-EG-ME-GAD-1013	LOAD PLAN, AUTOCLAV ,PLAN AT LEVEL912.900								
MD-216-3420-EG-ME-GAD-1012	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL929.30	D01	20/11/2023	AS BUILT	--	SYC	--	PRI	
MD-216-3420-EG-ME-GAD-1011	LOAD PLAN, FEED STORAGE,PLAN AT LEVEL912.900	C01	11/08/2020	ISSUED FOR CONSTRUCTION	--	SYC	--	PRI	
MD-216-3420-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING PLAN	A01	02/07/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

P.O.Box : 41017 Jeddah 21521 K.S.A
Tel. +966 2 591 0999
Fax +966 2 591 0099
E-mail aic@aicsteel.com
Site www.aicsteel.com

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DRAWING NUMBER	MD-P20131-B03-A002
REVISION	03
DATE / DRAWN	29/06/2020 / 7014
DATE / CHECKED	29/06/2020 / 7013
DATE / APPROVED	29/06/2020 / 7001
LARSEN & TOUBRO DRAWING NUMBER	019045_450_S_G_26043_002
REVISION	2
OUTOTEC DRAWING NUMBER	900707 __ GAU01 __ HH12 __ MTL01 __ 34212
DOCUMENT ID	OU100668475
REVISION	2
DRAWN	-- / --
CHECKED	SYC / 02/07/2020
ENGINEER	-- / --
APPROVED	PRI / 02/07/2020
EMPLOYER	
SCALE	NTS / A0
PROJECT	MANSOURAH - MASSARAH GOLD PROJECT
	GENERAL ARRANGEMENT DRAWING
	PRESSURE OXIDATION-AUTOCLAVE BUILDING
	COLUMN REACTIONS
DRAWING NUMBER	MD-216-3420-EG-ST-GAD-2230
REV	D01

C01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F4

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F4
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
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ELEMENT INCIDENCES SHELL
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1098 798 1122 1123 799; 1099 800 1124 1122 798; 1100 801 1125 1124 800;
1101 802 1126 1125 801; 1102 793 1117 1126 802; 1117 1116 1141 1142 1117;
1118 1118 1143 1141 1116; 1119 1119 1144 1143 1118; 1120 1120 1145 1144 1119;
1121 1121 1146 1145 1120; 1122 1122 1147 1148 1123; 1123 1124 1149 1147 1122;
1124 1125 1150 1149 1124; 1125 1126 1151 1150 1125; 1126 1117 1142 1151 1126;
1127 1141 1152 1153 1142; 1128 1143 1154 1152 1141; 1129 1144 1155 1154 1143;
1130 1145 1156 1155 1144; 1131 1146 1157 1156 1145; 1132 1147 1158 1159 1148;
1133 1149 1160 1158 1147; 1134 1150 1161 1160 1149; 1135 1151 1162 1161 1150;
1136 1142 1153 1162 1151; 1157 1184 1185 603 529; 1158 1185 1186 604 603;
1159 1186 1187 605 604; 1160 1187 1188 626 605; 1161 1188 1189 645 626;
1162 1189 1190 668 645; 1163 1190 1191 797 668; 1164 1191 1192 1121 797;
1165 1192 1193 1146 1121; 1166 1193 1194 1157 1146; 1178 531 607 1207 1206;
1179 607 609 1208 1207; 1180 609 611 1209 1208; 1181 611 628 1210 1209;
1182 628 647 1211 1210; 1183 647 670 1212 1211; 1184 670 799 1213 1212;
1185 799 1123 1214 1213; 1186 1123 1148 1215 1214; 1187 1148 1159 1216 1215;
1231 1152 1262 1263 1153; 1232 1154 1264 1262 1152; 1233 1155 1265 1264 1154;
1234 1156 1266 1265 1155; 1235 1157 1267 1266 1156; 1236 1158 1268 1269 1159;
1237 1160 1270 1268 1158; 1238 1161 1271 1270 1160; 1239 1162 1272 1271 1161;
1240 1153 1263 1272 1162; 1251 1194 1284 1267 1157; 1253 1159 1269 1286 1216;
```

```
1240 1153 1263 1272 1162; 1251 1194 1284 1267 1157; 1253 1159 1269 1286 1216;
1282 1262 1317 1318 1263; 1283 1264 1319 1317 1262; 1284 1265 1320 1319 1264;
1285 1266 1321 1320 1265; 1286 1267 1322 1321 1266; 1287 1268 1323 1324 1269;
1288 1270 1325 1323 1268; 1289 1271 1326 1325 1270; 1290 1272 1327 1326 1271;
1291 1263 1318 1327 1272; 1292 1317 1328 1329 1318; 1293 1319 1330 1328 1317;
1294 1320 1331 1330 1319; 1295 1321 1332 1331 1320; 1296 1322 1333 1332 1321;
1297 1323 1334 1335 1324; 1298 1325 1336 1334 1323; 1299 1326 1337 1336 1325;
1300 1327 1338 1337 1326; 1301 1318 1329 1338 1327; 1302 1284 1339 1322 1267;
1303 1339 1340 1333 1322; 1304 1269 1324 1341 1286; 1305 1324 1335 1342 1341;
1306 1328 1343 1344 1329; 1307 1330 1345 1343 1328; 1308 1331 1346 1345 1330;
1309 1332 1347 1346 1331; 1310 1333 1348 1347 1332; 1311 1334 1349 1350 1335;
1312 1336 1351 1349 1334; 1313 1337 1352 1351 1336; 1314 1338 1353 1352 1337;
1315 1329 1344 1353 1338; 1316 1340 1354 1348 1333; 1317 1335 1350 1355 1342;
1364 1206 1207 1406 1405; 1365 1207 1208 1407 1406; 1366 1208 1209 1408 1407;
1367 1209 1210 1409 1408; 1368 1210 1211 1410 1409; 1369 1211 1212 1411 1410;
1370 1212 1213 1412 1411; 1371 1213 1214 1413 1412; 1372 1214 1215 1414 1413;
1373 1215 1216 1415 1414; 1385 1405 1406 1428 1427; 1386 1406 1407 1429 1428;
1387 1407 1408 1430 1429; 1388 1408 1409 1431 1430; 1389 1409 1410 1432 1431;
1390 1410 1411 1433 1432; 1391 1411 1412 1434 1433; 1392 1412 1413 1435 1434;
1393 1413 1414 1436 1435; 1394 1414 1415 1437 1436; 1396 1216 1286 1439 1415;
1397 1415 1439 1440 1437; 1400 1286 1341 1443 1439; 1401 1341 1342 1444 1443;
1402 1439 1443 1445 1440; 1403 1443 1444 1446 1445; 1404 1342 1355 1447 1444;
1405 1444 1447 1448 1446; 1422 1465 1466 1185 1184; 1423 1466 1467 1186 1185;
1424 1467 1468 1187 1186; 1425 1468 1469 1188 1187; 1426 1469 1470 1189 1188;
1427 1470 1471 1190 1189; 1428 1471 1472 1191 1190; 1429 1472 1473 1192 1191;
1430 1473 1474 1193 1192; 1431 1474 1475 1194 1193; 1443 1487 1488 1466 1465;
1444 1488 1489 1467 1466; 1445 1489 1490 1468 1467; 1446 1490 1491 1469 1468;
1447 1491 1492 1470 1469; 1448 1492 1493 1471 1470; 1449 1493 1494 1472 1471;
1450 1494 1495 1473 1472; 1451 1495 1496 1474 1473; 1452 1496 1497 1475 1474;
1454 1475 1499 1284 1194; 1455 1497 1500 1499 1475; 1458 1499 1503 1339 1284;
1459 1503 1504 1340 1339; 1460 1500 1505 1503 1499; 1461 1505 1506 1504 1503;
1462 1504 1507 1354 1340; 1463 1506 1508 1507 1504; 1464 1509 1510 1488 1487;
1465 1510 1511 1489 1488; 1466 1511 1512 1490 1489; 1467 1512 1513 1491 1490;
1468 1513 1514 1492 1491; 1469 1514 1515 1493 1492; 1470 1515 1516 1494 1493;
1471 1516 1517 1495 1494; 1472 1517 1518 1496 1495; 1473 1518 1519 1497 1496;
1474 1519 1520 1500 1497; 1475 1520 1521 1505 1500; 1476 1521 1522 1506 1505;
1477 1522 1523 1508 1506; 1478 1427 1428 1524 1525; 1479 1428 1429 1526 1524;
1480 1429 1430 1527 1526; 1481 1430 1431 1528 1527; 1482 1431 1432 1529 1528;
1483 1432 1433 1530 1529; 1484 1433 1434 1531 1530; 1485 1434 1435 1532 1531;
1486 1435 1436 1533 1532; 1487 1436 1437 1534 1533; 1488 1437 1440 1535 1534;
1489 1440 1445 1536 1535; 1490 1445 1446 1537 1536; 1491 1446 1448 1538 1537;
START GROUP DEFINITION
ELEMENT
_RAFT 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 -
1157 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491
_RAFT_MOM 589 TO 628 637 TO 646 661 TO 666 783 TO 788 1095 TO 1100 -
1119 TO 1124 1127 TO 1136 1157 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 -
1282 TO 1317 1364 TO 1373 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 -
1443 TO 1452 1454 1455 1458 TO 1491
_RAFT_SHEAR 589 TO 618 623 624 641 642 663 664 785 786 1097 1098 1121 1122 -
1131 1132 1157 TO 1166 1178 TO 1187 1235 1236 1251 1253 1282 TO 1317 1364 -
1365 TO 1373 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 -
1454 1455 1458 TO 1491
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491 THICKNESS 0.75
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1092 PRIS YD 0.75 ZD 0.75
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
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1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491 PR GY -50
* LOAD DUE TO SURCHARGE
589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491 PR GY -20
JOINT LOAD
1115 FY -100.45
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
1115 FZ -320
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
1115 FZ 320
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1115 FY 0.0001
PERFORM ANALYSIS
CHANGE
SUPPORTS
589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
1459 TO 1491 PLATE MAT DIRECT Y SUBGRADE 57000 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+WL1
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL2
REPEAT LOAD
```

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REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 0.6DL+WL1
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 0.6DL+WL2
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 6 0.7
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.7L+0.5LR
REPEAT LOAD
1 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 5 1.6
```



```
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 111
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 1092
CLB 0.05 MEMB 1092
CLS 0.05 ALL
CLT 0.05 MEMB 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1157 TO 1163 1178 -
1179 TO 1184 1364 TO 1370 1385 TO 1391 1422 TO 1428 1443 TO 1449 1464 TO 1470 -
1478 TO 1484
MAXMAIN 32 ALL
MINMAIN 32 ALL
MINSEC 10 ALL
DESIGN COLUMN 1092
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F4_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
793 3.74 909.9 3.175; 1115 3.74 913.4 3.175;
MEMBER INCIDENCES
1092 793 1115;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1092 PRIS YD 0.75 ZD 0.75
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
793 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
1115 FY -100.45
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
1115 FZ -320
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
1115 FZ 320
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
1115 FY -0.0001
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1115 FY 0.0001
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 3 1.0
LOAD COMB 203 DL+LR
1 1.0 2 1.0
LOAD COMB 204 DL+0.75L+0.75LR
1 1.0 2 0.75 3 0.75
LOAD COMB 205 DL+WL1
1 1.0 4 1.0
LOAD COMB 206 DL+WL2
1 1.0 5 1.0
LOAD COMB 207 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 208 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 209 0.6DL+1.0WL1
```



```
LOAD COMB 209 0.6DL+1.0WL1
1 0.6 4 1.0
LOAD COMB 210 0.6DL+1.0WL2
1 0.6 5 1.0
LOAD COMB 211 0.6DL+0.7SL1
1 0.6 6 0.7
LOAD COMB 212 0.6DL+0.7SL2
1 0.6 7 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 3 STAAD INPUT – FOUNDATION F5

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F5
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175;
41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175;
302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438;
306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438;
309 1 909.9 3.175; 311 1.25 909.9 3.175; 313 1 909.9 3.42438;
314 1.25 909.9 3.42438; 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438;
324 0 909.9 3.67438; 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438;
327 0.75 909.9 3.67438; 328 1 909.9 3.67438; 329 1.25 909.9 3.67438;
333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438;
336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438;
339 1 909.9 3.92438; 340 1.25 909.9 3.92438; 344 -0.5 909.9 4.17438;
345 -0.25 909.9 4.17438; 346 0 909.9 4.17438; 347 0.25 909.9 4.17438;
348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438; 350 1 909.9 4.17438;
351 1.25 909.9 4.17438; 355 -0.5 909.9 4.42438; 356 -0.25 909.9 4.42438;
357 0 909.9 4.42438; 358 0.25 909.9 4.42438; 359 0.5 909.9 4.42438;
360 0.75 909.9 4.42438; 361 1 909.9 4.42438; 362 1.25 909.9 4.42438;
366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925; 369 -0.25 909.9 2.17438;
370 -0.25 909.9 1.925; 371 0 909.9 2.17438; 372 0 909.9 1.925;
373 0.25 909.9 2.17438; 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438;
376 0.5 909.9 1.925; 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925;
379 1 909.9 2.17438; 380 1 909.9 1.925; 381 1.25 909.9 2.17438;
382 1.25 909.9 1.925; 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438;
390 0 909.9 2.42438; 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438;
393 0.75 909.9 2.42438; 394 1 909.9 2.42438; 395 1.25 909.9 2.42438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
405 1 909.9 2.67438; 406 1.25 909.9 2.67438; 410 -0.5 909.9 2.92438;
411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438;
414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438;
417 1.25 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175;
422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438;
425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925;
428 1.5 909.9 2.42438; 429 1.5 909.9 2.67438; 430 1.5 909.9 2.92438;
1163 -0.75 909.9 3.42438; 1164 -0.75 909.9 3.175; 1165 -0.75 909.9 3.67438;
1166 -0.75 909.9 3.92438; 1167 -0.75 909.9 4.17438; 1168 -0.75 909.9 4.42438;
1169 -0.75 909.9 2.17438; 1170 -0.75 909.9 1.925; 1171 -0.75 909.9 2.42438;
1172 -0.75 909.9 2.67438; 1173 -0.75 909.9 2.92438; 1174 -0.5 909.9 4.67438;
1175 -0.25 909.9 4.67438; 1176 0 909.9 4.67438; 1177 0.25 909.9 4.67438;
1178 0.5 909.9 4.67438; 1179 0.75 909.9 4.67438; 1180 1 909.9 4.67438;
1181 1.25 909.9 4.67438; 1182 1.5 909.9 4.67438; 1195 -0.75 909.9 4.67438;
1196 -0.25 909.9 1.675; 1197 -0.5 909.9 1.675; 1198 0 909.9 1.675;
1199 0.25 909.9 1.675; 1200 0.5 909.9 1.675; 1201 0.75 909.9 1.675;
1202 1 909.9 1.675; 1203 1.25 909.9 1.675; 1204 1.5 909.9 1.675;
1217 -0.75 909.9 1.675; 1292 -1 909.9 3.42438; 1293 -1 909.9 3.175;
1294 -1 909.9 3.67438; 1295 -1 909.9 3.92438; 1296 -1 909.9 4.17438;
1297 -1 909.9 4.42438; 1298 -1 909.9 2.17438; 1299 -1 909.9 1.925;
1300 -1 909.9 2.42438; 1301 -1 909.9 2.67438; 1302 -1 909.9 2.92438;
1303 -1.25 909.9 3.42438; 1304 -1.25 909.9 3.175; 1305 -1.25 909.9 3.67438;
1306 -1.25 909.9 3.92438; 1307 -1.25 909.9 4.17438; 1308 -1.25 909.9 4.42438;
1309 -1.25 909.9 2.17438; 1310 -1.25 909.9 1.925; 1311 -1.25 909.9 2.42438;
1312 -1.25 909.9 2.67438; 1313 -1.25 909.9 2.92438; 1314 -1 909.9 4.67438;
1315 -1.25 909.9 4.67438; 1316 -1 909.9 1.675; 1317 -1.25 909.9 1.675;
1337 -1.5 909.9 3.42438; 1338 -1.5 909.9 3.175; 1339 -1.5 909.9 3.67438;
1340 -1.5 909.9 3.92438; 1341 -1.5 909.9 4.17438; 1342 -1.5 909.9 4.42438;
1343 -1.5 909.9 2.17438; 1344 -1.5 909.9 1.925; 1345 -1.5 909.9 2.42438;
1346 -1.5 909.9 2.67438; 1347 -1.5 909.9 2.92438; 1348 -1.5 909.9 4.67438;
1349 -1.5 909.9 1.675; 1350 1.8 909.9 3.42438; 1351 1.8 909.9 3.175;
1352 1.8 909.9 3.67438; 1353 1.8 909.9 3.92438; 1354 1.8 909.9 4.17438;
1355 1.8 909.9 4.42438; 1356 1.8 909.9 2.17438; 1357 1.8 909.9 1.925;
1358 1.8 909.9 2.42438; 1359 1.8 909.9 2.67438; 1360 1.8 909.9 2.92438;
1361 1.8 909.9 4.67438; 1362 1.8 909.9 1.675; 1363 -1.8 909.9 3.42438;
1364 -1.8 909.9 3.175; 1365 -1.8 909.9 3.67438; 1366 -1.8 909.9 3.92438;
1367 -1.8 909.9 4.17438; 1368 -1.8 909.9 4.42438; 1369 -1.8 909.9 2.17438;
1370 -1.8 909.9 1.925; 1371 -1.8 909.9 2.42438; 1372 -1.8 909.9 2.67438;
```



```
1370 -1.8 909.9 1.925; 1371 -1.8 909.9 2.42438; 1372 -1.8 909.9 2.67438;
1373 -1.8 909.9 2.92438; 1374 -1.8 909.9 4.67438; 1375 -1.8 909.9 1.675;
1428 -0.25 909.9 1.425; 1429 -0.5 909.9 1.425; 1430 0 909.9 1.425;
1431 0.25 909.9 1.425; 1432 0.5 909.9 1.425; 1433 0.75 909.9 1.425;
1434 1 909.9 1.425; 1435 1.25 909.9 1.425; 1436 1.5 909.9 1.425;
1437 -0.75 909.9 1.425; 1438 -1 909.9 1.425; 1439 -1.25 909.9 1.425;
1440 -1.5 909.9 1.425; 1441 1.8 909.9 1.425; 1442 -1.8 909.9 1.425;
1447 -0.5 909.9 4.92438; 1448 -0.25 909.9 4.92438; 1449 0 909.9 4.92438;
1450 0.25 909.9 4.92438; 1451 0.5 909.9 4.92438; 1452 0.75 909.9 4.92438;
1453 1 909.9 4.92438; 1454 1.25 909.9 4.92438; 1455 1.5 909.9 4.92438;
1456 -0.75 909.9 4.92438; 1457 -1 909.9 4.92438; 1458 -1.25 909.9 4.92438;
1459 -1.5 909.9 4.92438; 1460 1.8 909.9 4.92438; 1461 -1.8 909.9 4.92438;
1462 -0.25 909.9 1.175; 1463 -0.5 909.9 1.175; 1464 0 909.9 1.175;
1465 0.25 909.9 1.175; 1466 0.5 909.9 1.175; 1467 0.75 909.9 1.175;
1468 1 909.9 1.175; 1469 1.25 909.9 1.175; 1470 1.5 909.9 1.175;
1471 -0.75 909.9 1.175; 1472 -1 909.9 1.175; 1473 -1.25 909.9 1.175;
1474 -1.5 909.9 1.175; 1475 1.8 909.9 1.175; 1476 -1.8 909.9 1.175;
1477 -0.25 909.9 1; 1478 -0.5 909.9 1; 1479 0 909.9 1; 1480 0.25 909.9 1;
1481 0.5 909.9 1; 1482 0.75 909.9 1; 1483 1 909.9 1; 1484 1.25 909.9 1;
1485 1.5 909.9 1; 1486 -0.75 909.9 1; 1487 -1 909.9 1; 1488 -1.25 909.9 1;
1489 -1.5 909.9 1; 1490 1.8 909.9 1; 1491 -1.8 909.9 1;
1492 -0.5 909.9 5.17438; 1493 -0.25 909.9 5.17438; 1494 0 909.9 5.17438;
1495 0.25 909.9 5.17438; 1496 0.5 909.9 5.17438; 1497 0.75 909.9 5.17438;
1498 1 909.9 5.17438; 1499 1.25 909.9 5.17438; 1500 1.5 909.9 5.17438;
1501 -0.75 909.9 5.17438; 1502 -1 909.9 5.17438; 1503 -1.25 909.9 5.17438;
1504 -1.5 909.9 5.17438; 1505 1.8 909.9 5.17438; 1506 -1.8 909.9 5.17438;
1507 -0.5 909.9 5.34938; 1508 -0.25 909.9 5.34938; 1509 0 909.9 5.34938;
1510 0.25 909.9 5.34938; 1511 0.5 909.9 5.34938; 1512 0.75 909.9 5.34938;
1513 1 909.9 5.34938; 1514 1.25 909.9 5.34938; 1515 1.5 909.9 5.34938;
1516 -0.75 909.9 5.34938; 1517 -1 909.9 5.34938; 1518 -1.25 909.9 5.34938;
1519 -1.5 909.9 5.34938; 1520 1.8 909.9 5.34938; 1521 -1.8 909.9 5.34938;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309; 344 322 323 304 302;
345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306;
348 326 327 308 307; 349 327 328 313 308; 350 328 329 314 313;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327;
360 339 340 329 328; 364 344 345 334 333; 365 345 346 335 334;
366 346 347 336 335; 367 347 348 337 336; 368 348 349 338 337;
369 349 350 339 338; 370 350 351 340 339; 374 355 356 345 344;
375 356 357 346 345; 376 357 358 347 346; 377 358 359 348 347;
378 359 360 349 348; 379 360 361 350 349; 380 361 362 351 350;
384 366 369 370 367; 385 369 371 372 370; 386 371 373 374 372;
387 373 375 376 374; 388 375 377 378 376; 389 377 379 380 378;
390 379 381 382 380; 394 388 389 369 366; 395 389 390 371 369;
396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375;
399 393 394 379 377; 400 394 395 381 379; 404 399 400 389 388;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 409 404 405 394 393; 410 405 406 395 394;
414 410 411 400 399; 415 411 412 401 400; 416 412 413 402 401;
417 413 414 403 402; 418 414 415 404 403; 419 415 416 405 404;
420 416 417 406 405; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 430 309 311 417 416;
433 314 420 421 311; 434 329 422 420 314; 435 340 423 422 329;
436 351 424 423 340; 437 362 425 424 351; 438 381 426 427 382;
439 395 428 426 381; 440 406 429 428 395; 441 417 430 429 406;
442 311 421 430 417; 1137 1163 302 34 1164; 1138 1165 322 302 1163;
1139 1166 333 322 1165; 1140 1167 344 333 1166; 1141 1168 355 344 1167;
1142 1169 366 367 1170; 1143 1171 388 366 1169; 1144 1172 399 388 1171;
1145 1173 410 399 1172; 1146 1164 34 410 1173; 1147 1174 1175 356 355;
1148 1175 1176 357 356; 1149 1176 1177 358 357; 1150 1177 1178 359 358;
1151 1178 1179 360 359; 1152 1179 1180 361 360; 1153 1180 1181 362 361;
1154 1181 1182 425 362; 1167 1195 1174 355 1168; 1168 367 370 1196 1197;
1169 370 372 1198 1196; 1170 372 374 1199 1198; 1171 374 376 1200 1199;
1172 376 378 1201 1200; 1173 378 380 1202 1201; 1174 380 382 1203 1202;
1175 382 427 1204 1203; 1188 1170 367 1197 1217; 1259 1292 1163 1164 1293;
1260 1294 1165 1163 1292; 1261 1295 1166 1165 1294; 1262 1296 1167 1166 1295;
1263 1297 1168 1167 1296; 1264 1298 1169 1170 1299; 1265 1300 1171 1169 1298;
1266 1301 1172 1171 1300; 1267 1302 1173 1172 1301; 1268 1293 1164 1173 1302;
1269 1303 1292 1293 1304; 1270 1305 1294 1292 1303; 1271 1306 1295 1294 1305;
1272 1307 1296 1295 1306; 1273 1308 1297 1296 1307; 1274 1309 1298 1299 1310;
1275 1311 1300 1298 1309; 1276 1312 1301 1300 1311; 1277 1313 1302 1301 1312;
1278 1304 1293 1302 1313; 1279 1314 1195 1168 1297; 1280 1315 1314 1297 1308;
1281 1299 1170 1217 1316; 1282 1310 1299 1316 1317; 1301 1337 1303 1304 1338;
1302 1339 1305 1303 1337; 1303 1340 1306 1305 1339; 1304 1341 1307 1306 1340;
1305 1342 1308 1307 1341; 1306 1343 1309 1310 1344; 1307 1345 1311 1309 1343;
```

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1305 1342 1308 1307 1341; 1306 1343 1309 1310 1344; 1307 1345 1311 1309 1343;
1308 1346 1312 1311 1345; 1309 1347 1313 1312 1346; 1310 1338 1304 1313 1347;
1311 1348 1315 1308 1342; 1312 1344 1310 1317 1349; 1313 420 1350 1351 421;
1314 422 1352 1350 420; 1315 423 1353 1352 422; 1316 424 1354 1353 423;
1317 425 1355 1354 424; 1318 426 1356 1357 427; 1319 428 1358 1356 426;
1320 429 1359 1358 428; 1321 430 1360 1359 429; 1322 421 1351 1360 430;
1323 1182 1361 1355 425; 1324 427 1357 1362 1204; 1325 1363 1337 1338 1364;
1326 1365 1339 1337 1363; 1327 1366 1340 1339 1365; 1328 1367 1341 1340 1366;
1329 1368 1342 1341 1367; 1330 1369 1343 1344 1370; 1331 1371 1345 1343 1369;
1332 1372 1346 1345 1371; 1333 1373 1347 1346 1372; 1334 1364 1338 1347 1373;
1335 1374 1348 1342 1368; 1336 1370 1344 1349 1375; 1385 1197 1196 1428 1429;
1386 1196 1198 1430 1428; 1387 1198 1199 1431 1430; 1388 1199 1200 1432 1431;
1389 1200 1201 1433 1432; 1390 1201 1202 1434 1433; 1391 1202 1203 1435 1434;
1392 1203 1204 1436 1435; 1393 1217 1197 1429 1437; 1394 1316 1217 1437 1438;
1395 1317 1316 1438 1439; 1396 1349 1317 1439 1440; 1397 1204 1362 1441 1436;
1398 1375 1349 1440 1442; 1403 1447 1448 1175 1174; 1404 1448 1449 1176 1175;
1405 1449 1450 1177 1176; 1406 1450 1451 1178 1177; 1407 1451 1452 1179 1178;
1408 1452 1453 1180 1179; 1409 1453 1454 1181 1180; 1410 1454 1455 1182 1181;
1411 1456 1447 1174 1195; 1412 1457 1456 1195 1314; 1413 1458 1457 1314 1315;
1414 1459 1458 1315 1348; 1415 1455 1460 1361 1182; 1416 1461 1459 1348 1374;
1417 1429 1428 1462 1463; 1418 1428 1430 1464 1462; 1419 1430 1431 1465 1464;
1420 1431 1432 1466 1465; 1421 1432 1433 1467 1466; 1422 1433 1434 1468 1467;
1423 1434 1435 1469 1468; 1424 1435 1436 1470 1469; 1425 1437 1429 1463 1471;
1426 1438 1437 1471 1472; 1427 1439 1438 1472 1473; 1428 1440 1439 1473 1474;
1429 1436 1441 1475 1470; 1430 1442 1440 1474 1476; 1431 1463 1462 1477 1478;
1432 1462 1464 1479 1477; 1433 1464 1465 1480 1479; 1434 1465 1466 1481 1480;
1435 1466 1467 1482 1481; 1436 1467 1468 1483 1482; 1437 1468 1469 1484 1483;
1438 1469 1470 1485 1484; 1439 1471 1463 1478 1486; 1440 1472 1471 1486 1487;
1441 1473 1472 1487 1488; 1442 1474 1473 1488 1489; 1443 1470 1475 1490 1485;
1444 1476 1474 1489 1491; 1445 1492 1493 1448 1447; 1446 1493 1494 1449 1448;
1447 1494 1495 1450 1449; 1448 1495 1496 1451 1450; 1449 1496 1497 1452 1451;
1450 1497 1498 1453 1452; 1451 1498 1499 1454 1453; 1452 1499 1500 1455 1454;
1453 1501 1492 1447 1456; 1454 1502 1501 1456 1457; 1455 1503 1502 1457 1458;
1456 1504 1503 1458 1459; 1457 1500 1505 1460 1455; 1458 1506 1504 1459 1461;
1459 1507 1508 1493 1492; 1460 1508 1509 1494 1493; 1461 1509 1510 1495 1494;
1462 1510 1511 1496 1495; 1463 1511 1512 1497 1496; 1464 1512 1513 1498 1497;
1465 1513 1514 1499 1498; 1466 1514 1515 1500 1499; 1467 1516 1507 1492 1501;
1468 1517 1516 1501 1502; 1469 1518 1517 1502 1503; 1470 1519 1518 1503 1504;
1471 1515 1520 1505 1500; 1472 1521 1519 1504 1506;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 -
384 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472
_RAFT_MOM 327 333 334 337 338 344 347 TO 350 354 TO 360 364 TO 370 -
374 TO 380 384 TO 390 394 TO 400 404 TO 410 414 417 TO 420 424 427 TO 430 -
433 TO 442 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 -
1386 TO 1398 1403 TO 1472
_RAFT_SHEAR 338 350 360 370 374 TO 380 384 TO 390 400 410 420 430 433 TO 442 -
1141 1142 1147 TO 1154 1167 TO 1175 1188 1263 1264 1269 TO 1282 1301 TO 1336 -
1385 TO 1398 1403 TO 1472
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472 THICKNESS 0.75
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.75 ZD 0.75
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************************
```



```
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472 PR GY -55
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472 PR GY -20
JOINT LOAD
128 FX 1.5 FY -274.054
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -196.17
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -237.065 FZ 3.85
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 654 FZ -240
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY -654 FZ 240
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -23.01 FZ -8.1
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 23.01 FZ 8.1
PERFORM ANALYSIS
CHANGE
SUPPORTS
327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
1404 TO 1472 PLATE MAT DIRECT Y SUBGRADE 57000 COMPRESSION
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD 201 DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 DL+L
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 DL+LR
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 DL+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75
PERFORM ANALYSIS
CHANGE
LOAD 205 DL+WL1
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 DL+WL2
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
LOAD 207 DL+0.75WL1+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 4 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 DL+0.75WL2+0.75L+0.75LR
REPEAT LOAD
1 1.0 2 0.75 3 0.75 5 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 0.6DL+WL1
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 210 0.6DL+WL2
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL+0.7SL1
REPEAT LOAD
1 0.6 6 0.7
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL+0.7SL2
REPEAT LOAD
1 0.6 7 0.7
PERFORM ANALYSIS
CHANGE
********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL+1.7L+0.5LR
REPEAT LOAD
1 1.4 3 1.7 2 0.5
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL+1.6LR+1.0L
REPEAT LOAD
1 1.2 2 1.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL+1.6WL1+1.0L+0.5LR
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL+1.6WL2+1.0L+0.5LR
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL+1.0SL1+1.0L
REPEAT LOAD
1 1.2 3 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 1.2DL+1.0SL2+1.0L
REPEAT LOAD
1 1.2 3 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL+1.6WL1
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL+1.6WL2
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL+1.0SL1
```



```
LOAD 110 0.9DL+1.0SL1
REPEAT LOAD
1 0.9 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 111 0.9DL+1.0SL2
REPEAT LOAD
1 0.9 7 1.0
PERFORM ANALYSIS
CHANGE
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 111
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F5_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.75 ZD 0.75
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 1.5 FY -274.054
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE ROOF LIVE LOAD ( LR )
JOINT LOAD
128 FY -196.17
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -237.065 FZ 3.85
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 654 FZ -240
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY -654 FZ 240
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -23.01 FZ -8.1
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 23.01 FZ 8.1
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 3 1.0
LOAD COMB 203 DL+LR
1 1.0 2 1.0
LOAD COMB 204 DL+0.75L+0.75LR
1 1.0 2 0.75 3 0.75
LOAD COMB 205 DL+WL1
1 1.0 4 1.0
LOAD COMB 206 DL+WL2
1 1.0 5 1.0
LOAD COMB 207 DL+0.75WL1+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 208 DL+0.75WL2+0.75L+0.75LR
1 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 209 0.6DL+1.0WL1
```



```
LOAD COMB 209 0.6DL+1.0WL1
1 0.6 4 1.0
LOAD COMB 210 0.6DL+1.0WL2
1 0.6 5 1.0
LOAD COMB 211 0.6DL+0.7SL1
1 0.6 6 0.7
LOAD COMB 212 0.6DL+0.7SL2
1 0.6 7 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 4 STAAD INPUT – FOUNDATION F6

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F6
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175;
41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175;
302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438;
306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438;
322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438;
325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438;
333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438;
336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438;
388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438;
391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438;
413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438;
1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438;
1554 -0.75 909.9 3.92438; 1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438;
1560 -0.75 909.9 2.92438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413;
428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548;
1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556;
1507 1549 34 410 1560;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
424 TO 428 1497 1499 1501 1503 1505 1507
_RAFT_MOM 327 333 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 427 -
428 1497 1499 1501 1503 1505 1507
_RAFT_SHEAR 334 348 354 TO 358 404 TO 408 418 428 1497 1499 1501 1503 1505 1507
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
```



```
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 PR GY -52
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 PR GY -10
JOINT LOAD
128 FY -137.6
*********************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -307.229
*********************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -1.522
*********************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 1.522
*********************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
LOAD COMB 103 1.2DL+1.0SL1+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 104 1.2DL+1.0SL2+1.0L
1 1.2 2 1.0 4 1.0
LOAD COMB 105 0.9DL+1.0SL1
1 0.9 3 1.0
LOAD COMB 106 0.9DL+1.0SL2
1 0.9 4 1.0
*********************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
415 TO 418 424 TO 428
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F6
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FY -137.6
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -307.229
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -1.522
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 1.522
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 5 STAAD INPUT – FOUNDATION F6A

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F6A
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175;
41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175;
302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438;
306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438;
322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438;
325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438;
388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438;
391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438;
413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438;
1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438;
1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438; 1560 -0.75 909.9 2.92438;
1561 -0.5 909.9 2.17438; 1562 -0.25 909.9 2.17438; 1563 -0.25 909.9 1.92438;
1564 -0.5 909.9 1.92438; 1565 0 909.9 2.17438; 1566 0 909.9 1.92438;
1567 0.25 909.9 2.17438; 1568 0.25 909.9 1.92438; 1569 0.5 909.9 2.17438;
1570 0.5 909.9 1.92438; 1571 0.75 909.9 2.17438; 1572 0.75 909.9 1.92438;
1573 -0.75 909.9 2.17438; 1574 -0.75 909.9 1.92438; 1575 -0.5 909.9 1.67438;
1576 -0.25 909.9 1.67438; 1577 -0.25 909.9 1.42438; 1578 -0.5 909.9 1.42438;
1579 0 909.9 1.67438; 1580 0 909.9 1.42438; 1581 0.25 909.9 1.67438;
1582 0.25 909.9 1.42438; 1583 0.5 909.9 1.67438; 1584 0.5 909.9 1.42438;
1585 0.75 909.9 1.67438; 1586 0.75 909.9 1.42438; 1587 -0.75 909.9 1.67438;
1588 -0.75 909.9 1.42438; 1589 -0.5 909.9 1.17438; 1590 -0.25 909.9 1.17438;
1591 -0.25 909.9 0.92438; 1592 -0.5 909.9 0.92438; 1593 0 909.9 1.17438;
1594 0 909.9 0.92438; 1595 0.25 909.9 1.17438; 1596 0.25 909.9 0.92438;
1597 0.5 909.9 1.17438; 1598 0.5 909.9 0.92438; 1599 0.75 909.9 1.17438;
1600 0.75 909.9 0.92438; 1601 -0.75 909.9 1.17438; 1602 -0.75 909.9 0.92438;
1603 -0.5 909.9 0.67438; 1604 -0.25 909.9 0.67438; 1605 -0.25 909.9 0.47538;
1606 -0.5 909.9 0.47538; 1607 0 909.9 0.67438; 1608 0 909.9 0.47538;
1609 0.25 909.9 0.67438; 1610 0.25 909.9 0.47538; 1611 0.5 909.9 0.67438;
1612 0.5 909.9 0.47538; 1613 0.75 909.9 0.67438; 1614 0.75 909.9 0.47538;
1615 -0.75 909.9 0.67438; 1616 -0.75 909.9 0.47538; 1617 -0.5 909.9 0.22538;
1618 -0.25 909.9 0.22538; 1619 -0.25 909.9 -0.0246199;
1620 -0.5 909.9 -0.0246199; 1621 0 909.9 0.22538; 1622 0 909.9 -0.0246199;
1623 0.25 909.9 0.22538; 1624 0.25 909.9 -0.0246199; 1625 0.5 909.9 0.22538;
1626 0.5 909.9 -0.0246199; 1627 0.75 909.9 0.22538; 1628 0.75 909.9 -0.0246199;
1629 -0.75 909.9 0.22538; 1630 -0.75 909.9 -0.0246199; 1631 0 913.4 0.47538;
1632 1 909.9 3.42438; 1633 1 909.9 3.175; 1634 1 909.9 3.67438;
1635 1 909.9 2.67438; 1636 1 909.9 2.42438; 1637 1 909.9 2.92438;
1638 1 909.9 2.17438; 1639 1 909.9 1.92438; 1640 1 909.9 1.67438;
1641 1 909.9 1.42438; 1642 1 909.9 1.17438; 1643 1 909.9 0.92438;
1644 1 909.9 0.67438; 1645 1 909.9 0.47538; 1646 1 909.9 0.22538;
1647 1 909.9 -0.0246199; 1648 -1 909.9 3.42438; 1649 -1 909.9 3.175;
1650 -1 909.9 3.67438; 1651 -1 909.9 2.67438; 1652 -1 909.9 2.42438;
1653 -1 909.9 2.92438; 1654 -1 909.9 2.17438; 1655 -1 909.9 1.92438;
1656 -1 909.9 1.67438; 1657 -1 909.9 1.42438; 1658 -1 909.9 1.17438;
1659 -1 909.9 0.92438; 1660 -1 909.9 0.67438; 1661 -1 909.9 0.47538;
1662 -1 909.9 0.22538; 1663 -1 909.9 -0.0246199; 1664 0 913.4 3.675;
MEMBER INCIDENCES
185 2 128; 1568 1608 1631; 1599 128 1664;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 414 410 411 400 399;
415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402;
418 414 415 404 403; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 1497 1548 302 34 1549;
1499 1552 322 302 1548; 1503 1556 399 388 1557; 1505 1560 410 399 1556;
1507 1549 34 410 1560; 1508 1561 1562 1563 1564; 1509 1562 1565 1566 1563;
1510 1565 1567 1568 1566; 1511 1567 1569 1570 1568; 1512 1569 1571 1572 1570;
1513 388 389 1562 1561; 1514 389 390 1565 1562; 1515 390 391 1567 1565;
1516 391 392 1569 1567; 1517 392 393 1571 1569; 1518 1573 1561 1564 1574;
1519 1557 388 1561 1573; 1520 1575 1576 1577 1578; 1521 1576 1579 1580 1577;
1522 1579 1581 1582 1580; 1523 1581 1583 1584 1582; 1524 1583 1585 1586 1584;
```



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1522 1579 1581 1582 1580; 1523 1581 1583 1584 1582; 1524 1583 1585 1586 1584;
1525 1564 1563 1576 1575; 1526 1563 1566 1579 1576; 1527 1566 1568 1581 1579;
1528 1568 1570 1583 1581; 1529 1570 1572 1585 1583; 1530 1587 1575 1578 1588;
1531 1574 1564 1575 1587; 1532 1589 1590 1591 1592; 1533 1590 1593 1594 1591;
1534 1593 1595 1596 1594; 1535 1595 1597 1598 1596; 1536 1597 1599 1600 1598;
1537 1578 1577 1590 1589; 1538 1577 1580 1593 1590; 1539 1580 1582 1595 1593;
1540 1582 1584 1597 1595; 1541 1584 1586 1599 1597; 1542 1601 1589 1592 1602;
1543 1588 1578 1589 1601; 1544 1603 1604 1605 1606; 1545 1604 1607 1608 1605;
1546 1607 1609 1610 1608; 1547 1609 1611 1612 1610; 1548 1611 1613 1614 1612;
1549 1592 1591 1604 1603; 1550 1591 1594 1607 1604; 1551 1594 1596 1609 1607;
1552 1596 1598 1611 1609; 1553 1598 1600 1613 1611; 1554 1615 1603 1606 1616;
1555 1602 1592 1603 1615; 1556 1617 1618 1619 1620; 1557 1618 1621 1622 1619;
1558 1621 1623 1624 1622; 1559 1623 1625 1626 1624; 1560 1625 1627 1628 1626;
1561 1606 1605 1618 1617; 1562 1605 1608 1621 1618; 1563 1608 1610 1623 1621;
1564 1610 1612 1625 1623; 1565 1612 1614 1627 1625; 1566 1629 1617 1620 1630;
1567 1616 1606 1617 1629; 1569 308 1632 1633 6; 1570 327 1634 1632 308;
1571 404 1635 1636 393; 1572 415 1637 1635 404; 1573 6 1633 1637 415;
1574 1571 1638 1639 1572; 1575 393 1636 1638 1571; 1576 1585 1640 1641 1586;
1577 1572 1639 1640 1585; 1578 1599 1642 1643 1600; 1579 1586 1641 1642 1599;
1580 1613 1644 1645 1614; 1581 1600 1643 1644 1613; 1582 1627 1646 1647 1628;
1583 1614 1645 1646 1627; 1584 1648 1548 1549 1649; 1585 1650 1552 1548 1648;
1586 1651 1556 1557 1652; 1587 1653 1560 1556 1651; 1588 1649 1549 1560 1653;
1589 1654 1573 1574 1655; 1590 1652 1557 1573 1654; 1591 1656 1587 1588 1657;
1592 1655 1574 1587 1656; 1593 1658 1601 1602 1659; 1594 1657 1588 1601 1658;
1595 1660 1615 1616 1661; 1596 1659 1602 1615 1660; 1597 1662 1629 1630 1663;
1598 1661 1616 1629 1662;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 -
1499 1503 1505 1507 TO 1567 1569 TO 1598
_RAFT_MOM 327 333 334 344 TO 348 404 TO 408 414 TO 418 424 427 428 1497 1499 -
1503 1505 1507 TO 1544 1547 TO 1561 1564 TO 1567 1569 TO 1598
_RAFT_SHEAR 404 TO 408 1503 1508 TO 1543 1569 TO 1598
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
1505 1507 TO 1567 1569 TO 1598 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1568 1599 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
1505 1507 TO 1567 1569 TO 1598 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************************
MEMBER OFFSET
1599 START 0 -0.3 0
1599 END 0 -0.3 0
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
1505 1507 TO 1567 1569 TO 1598 PR GY -52
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
1505 1507 TO 1567 1569 TO 1598 PR GY -10
JOINT LOAD
1664 FY -137.6
1631 FX 4.58 FY -87.1
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
1664 FY -307.229
1631 FX 10 FY -195
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
```

```
JOINT LOAD
1664 FY -1.522
1631 FX 0.2 FY -0.48
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1664 FY 1.522
1631 FX 0.2 FY -0.48
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
LOAD COMB 103 1.2DL+1.0SL1+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 104 1.2DL+1.0SL2+1.0L
1 1.2 2 1.0 4 1.0
LOAD COMB 105 0.9DL+1.0SL1
1 0.9 3 1.0
LOAD COMB 106 0.9DL+1.0SL2
1 0.9 4 1.0
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 MEMB 185 1568
FC 35000 MEMB 185 1568
CLB 0.05 MEMB 185 1568
CLS 0.05 MEMB 185 1568
CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 -
425 TO 428 1508 TO 1517 1520 TO 1529 1532 TO 1541 1544 TO 1553 1556 TO 1565 -
1569 TO 1583
MAXMAIN 25 MEMB 185 1568
MINMAIN 25 MEMB 185 1568
MINSEC 10 MEMB 185 1568
DESIGN COLUMN 185 1568
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F6A_STABILITY
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175; 1608 0 909.9 0.47538; 1631 0 913.4 0.47538;
1664 0 913.4 3.675;
MEMBER INCIDENCES
185 2 128; 1568 1608 1631; 1599 128 1664;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1568 1599 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 1608 FIXED
*********************************************************************************************
MEMBER OFFSET
1599 START 0 -0.3 0
1599 END 0 -0.3 0
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
1664 FY -137.6
1631 FX 4.58 FY -87.1
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
1664 FY -307.229
1631 FX 10 FY -195
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
1664 FY -1.522
1631 FX 0.2 FY -0.48
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
1664 FY 1.522
1631 FX 0.2 FY -0.48
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 6 STAAD INPUT – FOUNDATION F7

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F7
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175;
41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175;
302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438;
306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438;
322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438;
325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438;
333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438;
336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438;
388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438;
391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438;
413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438;
1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438;
1554 -0.75 909.9 3.92438; 1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438;
1560 -0.75 909.9 2.92438; 1561 -0.5 909.9 4.17438; 1562 -0.25 909.9 4.17438;
1563 0 909.9 4.17438; 1564 0.25 909.9 4.17438; 1565 0.5 909.9 4.17438;
1566 0.75 909.9 4.17438; 1567 -0.5 909.9 4.42438; 1568 -0.25 909.9 4.42438;
1569 0 909.9 4.42438; 1570 0.25 909.9 4.42438; 1571 0.5 909.9 4.42438;
1572 0.75 909.9 4.42438; 1573 -0.5 909.9 4.67438; 1574 -0.25 909.9 4.67438;
1575 0 909.9 4.67438; 1576 0.25 909.9 4.67438; 1577 0.5 909.9 4.67438;
1578 0.75 909.9 4.67438; 1579 -0.75 909.9 4.17438; 1580 -0.75 909.9 4.42438;
1581 -0.75 909.9 4.67438; 1582 -0.5 909.9 1.92438; 1583 -0.25 909.9 1.92438;
1584 -0.25 909.9 1.67438; 1585 -0.5 909.9 1.67438; 1586 0 909.9 1.92438;
1587 0 909.9 1.67438; 1588 0.25 909.9 1.92438; 1589 0.25 909.9 1.67438;
1590 0.5 909.9 1.92438; 1591 0.5 909.9 1.67438; 1592 0.75 909.9 1.92438;
1593 0.75 909.9 1.67438; 1594 -0.5 909.9 2.17438; 1595 -0.25 909.9 2.17438;
1596 0 909.9 2.17438; 1597 0.25 909.9 2.17438; 1598 0.5 909.9 2.17438;
1599 0.75 909.9 2.17438; 1600 -0.75 909.9 1.92438; 1601 -0.75 909.9 1.67438;
1602 -0.75 909.9 2.17438; 1603 -0.5 909.9 4.92438; 1604 -0.25 909.9 4.92438;
1605 0 909.9 4.92438; 1606 0.25 909.9 4.92438; 1607 0.5 909.9 4.92438;
1608 0.75 909.9 4.92438; 1609 -0.75 909.9 4.92438; 1610 -0.25 909.9 1.42438;
1611 -0.5 909.9 1.42438; 1612 0 909.9 1.42438; 1613 0.25 909.9 1.42438;
1614 0.5 909.9 1.42438; 1615 0.75 909.9 1.42438; 1616 -0.75 909.9 1.42438;
1617 0.925 909.9 3.42438; 1618 0.925 909.9 3.175; 1619 0.925 909.9 3.67438;
1620 0.925 909.9 3.92438; 1621 0.925 909.9 2.67438; 1622 0.925 909.9 2.42438;
1623 0.925 909.9 2.92438; 1624 0.925 909.9 4.17438; 1625 0.925 909.9 4.42438;
1626 0.925 909.9 4.67438; 1627 0.925 909.9 1.92438; 1628 0.925 909.9 1.67438;
1629 0.925 909.9 2.17438; 1630 0.925 909.9 4.92438; 1631 0.925 909.9 1.42438;
1632 -0.925 909.9 3.42438; 1633 -0.925 909.9 3.175; 1634 -0.925 909.9 3.67438;
1635 -0.925 909.9 3.92438; 1636 -0.925 909.9 2.67438;
1637 -0.925 909.9 2.42438; 1638 -0.925 909.9 2.92438;
1639 -0.925 909.9 4.17438; 1640 -0.925 909.9 4.42438;
1641 -0.925 909.9 4.67438; 1642 -0.925 909.9 1.92438;
1643 -0.925 909.9 1.67438; 1644 -0.925 909.9 2.17438;
1645 -0.925 909.9 4.92438; 1646 -0.925 909.9 1.42438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413;
428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548;
1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556;
1507 1549 34 410 1560; 1508 1561 1562 334 333; 1509 1562 1563 335 334;
1510 1563 1564 336 335; 1511 1564 1565 337 336; 1512 1565 1566 338 337;
1513 1567 1568 1562 1561; 1514 1568 1569 1563 1562; 1515 1569 1570 1564 1563;
1516 1570 1571 1565 1564; 1517 1571 1572 1566 1565; 1518 1573 1574 1568 1567;
1519 1574 1575 1569 1568; 1520 1575 1576 1570 1569; 1521 1576 1577 1571 1570;
1522 1577 1578 1572 1571; 1523 1579 1561 333 1554; 1524 1580 1567 1561 1579;
```



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1522 1577 1578 1572 1571; 1523 1579 1561 333 1554; 1524 1580 1567 1561 1579;
1525 1581 1573 1567 1580; 1526 1582 1583 1584 1585; 1527 1583 1586 1587 1584;
1528 1586 1588 1589 1587; 1529 1588 1590 1591 1589; 1530 1590 1592 1593 1591;
1531 1594 1595 1583 1582; 1532 1595 1596 1586 1583; 1533 1596 1597 1588 1586;
1534 1597 1598 1590 1588; 1535 1598 1599 1592 1590; 1536 388 389 1595 1594;
1537 389 390 1596 1595; 1538 390 391 1597 1596; 1539 391 392 1598 1597;
1540 392 393 1599 1598; 1541 1600 1582 1585 1601; 1542 1602 1594 1582 1600;
1543 1557 388 1594 1602; 1544 1603 1604 1574 1573; 1545 1604 1605 1575 1574;
1546 1605 1606 1576 1575; 1547 1606 1607 1577 1576; 1548 1607 1608 1578 1577;
1549 1609 1603 1573 1581; 1550 1585 1584 1610 1611; 1551 1584 1587 1612 1610;
1552 1587 1589 1613 1612; 1553 1589 1591 1614 1613; 1554 1591 1593 1615 1614;
1555 1601 1585 1611 1616; 1556 308 1617 1618 6; 1557 327 1619 1617 308;
1558 338 1620 1619 327; 1559 404 1621 1622 393; 1560 415 1623 1621 404;
1561 6 1618 1623 415; 1562 1566 1624 1620 338; 1563 1572 1625 1624 1566;
1564 1578 1626 1625 1572; 1565 1592 1627 1628 1593; 1566 1599 1629 1627 1592;
1567 393 1622 1629 1599; 1568 1608 1630 1626 1578; 1569 1593 1628 1631 1615;
1570 1632 1548 1549 1633; 1571 1634 1552 1548 1632; 1572 1635 1554 1552 1634;
1573 1636 1556 1557 1637; 1574 1638 1560 1556 1636; 1575 1633 1549 1560 1638;
1576 1639 1579 1554 1635; 1577 1640 1580 1579 1639; 1578 1641 1581 1580 1640;
1579 1642 1600 1601 1643; 1580 1644 1602 1600 1642; 1581 1637 1557 1602 1644;
1582 1645 1609 1581 1641; 1583 1643 1601 1616 1646;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
424 TO 428 1497 1499 1501 1503 1505 1507 TO 1583
_RAFT_MOM 334 348 354 TO 358 404 TO 408 418 428 1497 1499 1501 1503 1505 1507 -
1508 TO 1583
_RAFT_SHEAR 1508 TO 1583
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1583 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1582 -
1583 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
**************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1583 PR GY -52
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1583 PR GY -10
JOINT LOAD
128 FY -199 FZ 5.25
**************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -476.25 FZ 12.5
**************************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -9.512
**************************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 9.512
**************************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
```

```
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
LOAD COMB 103 1.2DL+1.0SL1+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 104 1.2DL+1.0SL2+1.0L
1 1.2 2 1.0 4 1.0
LOAD COMB 105 0.9DL+1.0SL1
1 0.9 3 1.0
LOAD COMB 106 0.9DL+1.0SL2
1 0.9 4 1.0
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
415 TO 418 424 TO 428 1508 TO 1522 1526 TO 1540 1544 TO 1548 1550 TO 1554 -
1556 TO 1569
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F7
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.4 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FY -199 FZ 5.25
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -476.25 FZ 12.5
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -9.512
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 9.512
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 7 STAAD INPUT – FOUNDATION F8

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F8
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 910.9 3.175; 6 0.75 910.9 3.175; 34 -0.5 910.9 3.175; 35 -0.25 910.9 3.175;
41 0.25 910.9 3.175; 42 0.5 910.9 3.175; 128 0 913.4 3.175;
302 -0.5 910.9 3.42438; 304 -0.25 910.9 3.42438; 305 0 910.9 3.42438;
306 0.25 910.9 3.42438; 307 0.5 910.9 3.42438; 308 0.75 910.9 3.42438;
322 -0.5 910.9 3.67438; 323 -0.25 910.9 3.67438; 324 0 910.9 3.67438;
325 0.25 910.9 3.67438; 326 0.5 910.9 3.67438; 327 0.75 910.9 3.67438;
333 -0.5 910.9 3.92438; 334 -0.25 910.9 3.92438; 335 0 910.9 3.92438;
336 0.25 910.9 3.92438; 337 0.5 910.9 3.92438; 338 0.75 910.9 3.92438;
388 -0.5 910.9 2.42438; 389 -0.25 910.9 2.42438; 390 0 910.9 2.42438;
391 0.25 910.9 2.42438; 392 0.5 910.9 2.42438; 393 0.75 910.9 2.42438;
399 -0.5 910.9 2.67438; 400 -0.25 910.9 2.67438; 401 0 910.9 2.67438;
402 0.25 910.9 2.67438; 403 0.5 910.9 2.67438; 404 0.75 910.9 2.67438;
410 -0.5 910.9 2.92438; 411 -0.25 910.9 2.92438; 412 0 910.9 2.92438;
413 0.25 910.9 2.92438; 414 0.5 910.9 2.92438; 415 0.75 910.9 2.92438;
1548 -0.75 910.9 3.42438; 1549 -0.75 910.9 3.175; 1552 -0.75 910.9 3.67438;
1554 -0.75 910.9 3.92438; 1556 -0.75 910.9 2.67438; 1557 -0.75 910.9 2.42438;
1560 -0.75 910.9 2.92438; 1561 -0.5 910.9 4.17438; 1562 -0.25 910.9 4.17438;
1563 0 910.9 4.17438; 1564 0.25 910.9 4.17438; 1565 0.5 910.9 4.17438;
1566 0.75 910.9 4.17438; 1567 -0.5 910.9 4.42438; 1568 -0.25 910.9 4.42438;
1569 0 910.9 4.42438; 1570 0.25 910.9 4.42438; 1571 0.5 910.9 4.42438;
1572 0.75 910.9 4.42438; 1573 -0.5 910.9 4.67438; 1574 -0.25 910.9 4.67438;
1575 0 910.9 4.67438; 1576 0.25 910.9 4.67438; 1577 0.5 910.9 4.67438;
1578 0.75 910.9 4.67438; 1579 -0.75 910.9 4.17438; 1580 -0.75 910.9 4.42438;
1581 -0.75 910.9 4.67438; 1603 -0.5 910.9 4.92438; 1604 -0.25 910.9 4.92438;
1605 0 910.9 4.92438; 1606 0.25 910.9 4.92438; 1607 0.5 910.9 4.92438;
1608 0.75 910.9 4.92438; 1609 -0.75 910.9 4.92438; 1617 0.925 910.9 3.42438;
1618 0.925 910.9 3.175; 1619 0.925 910.9 3.67438; 1620 0.925 910.9 3.92438;
1621 0.925 910.9 2.67438; 1622 0.925 910.9 2.42438; 1623 0.925 910.9 2.92438;
1624 0.925 910.9 4.17438; 1625 0.925 910.9 4.42438; 1626 0.925 910.9 4.67438;
1630 0.925 910.9 4.92438; 1632 -0.925 910.9 3.42438; 1633 -0.925 910.9 3.175;
1634 -0.925 910.9 3.67438; 1635 -0.925 910.9 3.92438;
1636 -0.925 910.9 2.67438; 1637 -0.925 910.9 2.42438;
1638 -0.925 910.9 2.92438; 1639 -0.925 910.9 4.17438;
1640 -0.925 910.9 4.42438; 1641 -0.925 910.9 4.67438;
1645 -0.925 910.9 4.92438; 1647 0 913.4 4.67438; 1648 -0.5 910.9 5.17438;
1649 -0.25 910.9 5.17438; 1650 0 910.9 5.17438; 1651 0.25 910.9 5.17438;
1652 0.5 910.9 5.17438; 1653 0.75 910.9 5.17438; 1654 -0.75 910.9 5.17438;
1655 0.925 910.9 5.17438; 1656 -0.925 910.9 5.17438; 1657 -0.5 910.9 5.42438;
1658 -0.25 910.9 5.42438; 1659 0 910.9 5.42438; 1660 0.25 910.9 5.42438;
1661 0.5 910.9 5.42438; 1662 0.75 910.9 5.42438; 1663 -0.75 910.9 5.42438;
1664 0.925 910.9 5.42438; 1665 -0.925 910.9 5.42438; 1675 -0.25 910.9 2.17438;
1676 -0.5 910.9 2.17438; 1677 0 910.9 2.17438; 1678 0.25 910.9 2.17438;
1679 0.5 910.9 2.17438; 1680 0.75 910.9 2.17438; 1681 -0.75 910.9 2.17438;
1682 0.925 910.9 2.17438; 1683 -0.925 910.9 2.17438; 1684 -0.25 910.9 1.92438;
1685 -0.5 910.9 1.92438; 1686 0 910.9 1.92438; 1687 0.25 910.9 1.92438;
1688 0.5 910.9 1.92438; 1689 0.75 910.9 1.92438; 1690 -0.75 910.9 1.92438;
1691 0.925 910.9 1.92438; 1692 -0.925 910.9 1.92438; 1693 -0.25 910.9 1.67438;
1694 -0.5 910.9 1.67438; 1695 0 910.9 1.67438; 1696 0.25 910.9 1.67438;
1697 0.5 910.9 1.67438; 1698 0.75 910.9 1.67438; 1699 -0.75 910.9 1.67438;
1700 0.925 910.9 1.67438; 1701 -0.925 910.9 1.67438;
MEMBER INCIDENCES
185 2 128; 1584 1575 1647;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413;
428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548;
1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556;
1507 1549 34 410 1560; 1508 1561 1562 334 333; 1509 1562 1563 335 334;
1510 1563 1564 336 335; 1511 1564 1565 337 336; 1512 1565 1566 338 337;
1513 1567 1568 1562 1561; 1514 1568 1569 1563 1562; 1515 1569 1570 1564 1563;
```



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1513 1567 1568 1562 1561; 1514 1568 1569 1563 1562; 1515 1569 1570 1564 1563;
1516 1570 1571 1565 1564; 1517 1571 1572 1566 1565; 1518 1573 1574 1568 1567;
1519 1574 1575 1569 1568; 1520 1575 1576 1570 1569; 1521 1576 1577 1571 1570;
1522 1577 1578 1572 1571; 1523 1579 1561 333 1554; 1524 1580 1567 1561 1579;
1525 1581 1573 1567 1580; 1544 1603 1604 1574 1573; 1545 1604 1605 1575 1574;
1546 1605 1606 1576 1575; 1547 1606 1607 1577 1576; 1548 1607 1608 1578 1577;
1549 1609 1603 1573 1581; 1556 308 1617 1618 6; 1557 327 1619 1617 308;
1558 338 1620 1619 327; 1559 404 1621 1622 393; 1560 415 1623 1621 404;
1561 6 1618 1623 415; 1562 1566 1624 1620 338; 1563 1572 1625 1624 1566;
1564 1578 1626 1625 1572; 1568 1608 1630 1626 1578; 1570 1632 1548 1549 1633;
1571 1634 1552 1548 1632; 1572 1635 1554 1552 1634; 1573 1636 1556 1557 1637;
1574 1638 1560 1556 1636; 1575 1633 1549 1560 1638; 1576 1639 1579 1554 1635;
1577 1640 1580 1579 1639; 1578 1641 1581 1580 1640; 1582 1645 1609 1581 1641;
1585 1648 1649 1604 1603; 1586 1649 1650 1605 1604; 1587 1650 1651 1606 1605;
1588 1651 1652 1607 1606; 1589 1652 1653 1608 1607; 1590 1654 1648 1603 1609;
1591 1653 1655 1630 1608; 1592 1656 1654 1609 1645; 1593 1657 1658 1649 1648;
1594 1658 1659 1650 1649; 1595 1659 1660 1651 1650; 1596 1660 1661 1652 1651;
1597 1661 1662 1653 1652; 1598 1663 1657 1648 1654; 1599 1662 1664 1655 1653;
1600 1665 1663 1654 1656; 1609 388 389 1675 1676; 1610 389 390 1677 1675;
1611 390 391 1678 1677; 1612 391 392 1679 1678; 1613 392 393 1680 1679;
1614 1557 388 1676 1681; 1615 393 1622 1682 1680; 1616 1637 1557 1681 1683;
1617 1676 1675 1684 1685; 1618 1675 1677 1686 1684; 1619 1677 1678 1687 1686;
1620 1678 1679 1688 1687; 1621 1679 1680 1689 1688; 1622 1681 1676 1685 1690;
1623 1680 1682 1691 1689; 1624 1683 1681 1690 1692; 1625 1685 1684 1693 1694;
1626 1684 1686 1695 1693; 1627 1686 1687 1696 1695; 1628 1687 1688 1697 1696;
1629 1688 1689 1698 1697; 1630 1690 1685 1694 1699; 1631 1689 1691 1700 1698;
1632 1692 1690 1699 1701;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
424 TO 428 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 -
1568 1570 TO 1578 1582 1585 TO 1600 1609 TO 1632
_RAFT_MOM 327 333 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 427 -
428 1497 1499 1501 1503 1505 1507 TO 1518 1521 TO 1525 1544 1547 TO 1549 -
1556 TO 1564 1568 1570 TO 1578 1582 1585 TO 1600 1609 TO 1632
_RAFT_SHEAR 334 348 358 404 TO 408 418 428 1497 1499 1501 1503 1505 -
1507 TO 1512 1517 1522 TO 1525 1548 1549 1556 TO 1564 1568 1570 TO 1578 1582 -
1589 TO 1600 1609 TO 1632
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1584 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
1571 TO 1578 1582 1585 TO 1600 1609 TO 1631 -
1632 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
**************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 PR GY -40
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 PR GY -10
JOINT LOAD
128 FY -199 FZ 5.25
1647 FY -19.3 FZ 0.54
**************************************************************************************************
```

```
**************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -476.25 FZ 12.5
1647 FY -42.02 FZ 1.55
**************************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -9.512
1647 FY -0.87
**************************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 9.512
1647 FY -0.87
**************************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
**************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
LOAD COMB 103 1.2DL+1.0SL1+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 104 1.2DL+1.0SL2+1.0L
1 1.2 2 1.0 4 1.0
LOAD COMB 105 0.9DL+1.0SL1
1 0.9 3 1.0
LOAD COMB 106 0.9DL+1.0SL2
1 0.9 4 1.0
**************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 1584
CLB 0.05 MEMB 185 1584
CLS 0.05 ALL
CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
415 TO 418 424 TO 428 1508 TO 1522 1544 TO 1548 1556 TO 1564 1568 -
1585 TO 1589 1591 1593 TO 1597 1599 1609 TO 1613 1615 1617 TO 1621 1623 1625 -
1626 TO 1629 1631
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 1584
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F8
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 910.9 3.175; 128 0 913.4 3.175; 1575 0 910.9 4.67438; 1647 0 913.4 4.67438;
MEMBER INCIDENCES
185 2 128; 1584 1575 1647;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1584 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 1575 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FY -199 FZ 5.25
1647 FY -19.3 FZ 0.54
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -476.25 FZ 12.5
1647 FY -42.02 FZ 1.55
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -9.512
1647 FY -0.87
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 9.512
1647 FY -0.87
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 8 STAAD INPUT – FOUNDATION F9

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F9
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 910.6 3.175; 6 0.75 910.6 3.175; 35 -0.25 910.6 3.175; 41 0.25 910.6 3.175;
42 0.5 910.6 3.175; 128 0 913.4 3.175; 304 -0.25 910.6 3.42438;
305 0 910.6 3.42438; 306 0.25 910.6 3.42438; 307 0.5 910.6 3.42438;
308 0.75 910.6 3.42438; 389 -0.25 910.6 2.42438; 390 0 910.6 2.42438;
391 0.25 910.6 2.42438; 392 0.5 910.6 2.42438; 393 0.75 910.6 2.42438;
400 -0.25 910.6 2.67438; 401 0 910.6 2.67438; 402 0.25 910.6 2.67438;
403 0.5 910.6 2.67438; 404 0.75 910.6 2.67438; 411 -0.25 910.6 2.92438;
412 0 910.6 2.92438; 413 0.25 910.6 2.92438; 414 0.5 910.6 2.92438;
415 0.75 910.6 2.92438; 1617 1 910.6 3.42438; 1618 1 910.6 3.175;
1621 1 910.6 2.67438; 1622 1 910.6 2.42438; 1623 1 910.6 2.92438;
1624 -0.25 910.6 2.17438; 1625 0 910.6 2.17438; 1626 0 910.6 1.92438;
1627 -0.25 910.6 1.92438; 1628 0.25 910.6 2.17438; 1629 0.25 910.6 1.92438;
1630 0.5 910.6 2.17438; 1631 0.5 910.6 1.92438; 1632 0.75 910.6 2.17438;
1633 0.75 910.6 1.92438; 1634 1 910.6 2.17438; 1635 1 910.6 1.92438;
1672 -0.25 910.6 3.67438; 1673 0 910.6 3.67438; 1674 0.25 910.6 3.67438;
1675 0.5 910.6 3.67438; 1676 0.75 910.6 3.67438; 1677 1 910.6 3.67438;
1678 -0.25 910.6 3.92438; 1679 0 910.6 3.92438; 1680 0.25 910.6 3.92438;
1681 0.5 910.6 3.92438; 1682 0.75 910.6 3.92438; 1683 1 910.6 3.92438;
1720 -0.45 910.6 3.42438; 1721 -0.45 910.6 3.175; 1722 -0.45 910.6 2.67438;
1723 -0.45 910.6 2.42438; 1724 -0.45 910.6 2.92438; 1725 -0.45 910.6 2.17438;
1726 -0.45 910.6 1.92438; 1729 -0.45 910.6 3.67438; 1730 -0.45 910.6 3.92438;
1737 1.25 910.6 3.42438; 1738 1.25 910.6 3.175; 1739 1.5 910.6 3.42438;
1740 1.5 910.6 3.175; 1741 1.75 910.6 3.42438; 1742 1.75 910.6 3.175;
1743 1.25 910.6 2.67438; 1744 1.25 910.6 2.42438; 1745 1.5 910.6 2.67438;
1746 1.5 910.6 2.42438; 1747 1.75 910.6 2.67438; 1748 1.75 910.6 2.42438;
1749 1.25 910.6 2.92438; 1750 1.5 910.6 2.92438; 1751 1.75 910.6 2.92438;
1752 2 910.6 3.42438; 1753 2 910.6 3.175; 1754 2 910.6 2.67438;
1755 2 910.6 2.42438; 1756 2 910.6 2.92438; 1757 1.25 910.6 2.17438;
1758 1.25 910.6 1.92438; 1759 1.5 910.6 2.17438; 1760 1.5 910.6 1.92438;
1761 1.75 910.6 2.17438; 1762 1.75 910.6 1.92438; 1763 2 910.6 2.17438;
1764 2 910.6 1.92438; 1773 1.25 910.6 3.67438; 1774 1.5 910.6 3.67438;
1775 1.75 910.6 3.67438; 1776 2 910.6 3.67438; 1777 1.25 910.6 3.92438;
1778 1.5 910.6 3.92438; 1779 1.75 910.6 3.92438; 1780 2 910.6 3.92438;
1805 2.25 910.6 3.42438; 1806 2.25 910.6 3.175; 1807 2.25 910.6 2.67438;
1808 2.25 910.6 2.42438; 1809 2.25 910.6 2.92438; 1810 2.5 910.6 3.42438;
1811 2.5 910.6 3.175; 1812 2.5 910.6 2.67438; 1813 2.5 910.6 2.42438;
1814 2.5 910.6 2.92438; 1815 2.25 910.6 2.17438; 1816 2.25 910.6 1.92438;
1817 2.5 910.6 2.17438; 1818 2.5 910.6 1.92438; 1823 2.25 910.6 3.67438;
1824 2.5 910.6 3.67438; 1825 2.25 910.6 3.92438; 1826 2.5 910.6 3.92438;
1839 2 913.4 3.175; 1840 -0.25 910.6 4.17438; 1841 0 910.6 4.17438;
1842 0.25 910.6 4.17438; 1843 0.5 910.6 4.17438; 1844 0.75 910.6 4.17438;
1845 1 910.6 4.17438; 1846 -0.45 910.6 4.17438; 1847 1.25 910.6 4.17438;
1848 1.5 910.6 4.17438; 1849 1.75 910.6 4.17438; 1850 2 910.6 4.17438;
1851 2.25 910.6 4.17438; 1852 2.5 910.6 4.17438;
MEMBER INCIDENCES
185 2 128; 1754 1753 1839;
ELEMENT INCIDENCES SHELL
328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41; 334 307 308 6 42;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 415 411 412 401 400; 416 412 413 402 401;
417 413 414 403 402; 418 414 415 404 403; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 1556 308 1617 1618 6;
1559 404 1621 1622 393; 1560 415 1623 1621 404; 1561 6 1618 1623 415;
1562 1624 1625 1626 1627; 1563 1625 1628 1629 1626; 1564 1628 1630 1631 1629;
1565 1630 1632 1633 1631; 1566 389 390 1625 1624; 1567 390 391 1628 1625;
1568 391 392 1630 1628; 1569 392 393 1632 1630; 1570 1632 1634 1635 1633;
1571 393 1622 1634 1632; 1602 1672 1673 305 304; 1603 1673 1674 306 305;
1604 1674 1675 307 306; 1605 1675 1676 308 307; 1606 1676 1677 1617 308;
1607 1678 1679 1673 1672; 1608 1679 1680 1674 1673; 1609 1680 1681 1675 1674;
1610 1681 1682 1676 1675; 1611 1682 1683 1677 1676; 1642 1720 304 35 1721;
1643 1722 400 389 1723; 1644 1724 411 400 1722; 1645 1721 35 411 1724;
1646 1725 1624 1627 1726; 1647 1723 389 1624 1725; 1650 1729 1672 304 1720;
1651 1730 1678 1672 1729; 1658 1617 1737 1738 1618; 1659 1737 1739 1740 1738;
1660 1739 1741 1742 1740; 1661 1621 1743 1744 1622; 1662 1743 1745 1746 1744;
1663 1745 1747 1748 1746; 1664 1623 1749 1743 1621; 1665 1749 1750 1745 1743;
1666 1750 1751 1747 1745; 1667 1618 1738 1749 1623; 1668 1738 1740 1750 1749;
1669 1740 1742 1751 1750; 1670 1741 1752 1753 1742; 1671 1747 1754 1755 1748;
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1669 1740 1742 1751 1750; 1670 1741 1752 1753 1742; 1671 1747 1754 1755 1748;
1672 1751 1756 1754 1747; 1673 1742 1753 1756 1751; 1674 1634 1757 1758 1635;
1675 1757 1759 1760 1758; 1676 1759 1761 1762 1760; 1677 1622 1744 1757 1634;
1678 1744 1746 1759 1757; 1679 1746 1748 1761 1759; 1680 1761 1763 1764 1762;
1681 1748 1755 1763 1761; 1690 1677 1773 1737 1617; 1691 1773 1774 1739 1737;
1692 1774 1775 1741 1739; 1693 1775 1776 1752 1741; 1694 1683 1777 1773 1677;
1695 1777 1778 1774 1773; 1696 1778 1779 1775 1774; 1697 1779 1780 1776 1775;
1722 1752 1805 1806 1753; 1723 1754 1807 1808 1755; 1724 1756 1809 1807 1754;
1725 1753 1806 1809 1756; 1726 1805 1810 1811 1806; 1727 1807 1812 1813 1808;
1728 1809 1814 1812 1807; 1729 1806 1811 1814 1809; 1730 1763 1815 1816 1764;
1731 1755 1808 1815 1763; 1732 1815 1817 1818 1816; 1733 1808 1813 1817 1815;
1738 1776 1823 1805 1752; 1739 1823 1824 1810 1805; 1740 1780 1825 1823 1776;
1741 1825 1826 1824 1823; 1755 1840 1841 1679 1678; 1756 1841 1842 1680 1679;
1757 1842 1843 1681 1680; 1758 1843 1844 1682 1681; 1759 1844 1845 1683 1682;
1760 1846 1840 1678 1730; 1761 1845 1847 1777 1683; 1762 1847 1848 1778 1777;
1763 1848 1849 1779 1778; 1764 1849 1850 1780 1779; 1765 1850 1851 1825 1780;
1766 1851 1852 1826 1825;
START GROUP DEFINITION
ELEMENT
_RAFT 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 1602 -
1603 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
1738 TO 1741 1755 TO 1766
_RAFT_MOM 333 334 405 TO 408 415 TO 418 427 428 1556 1559 TO 1571 -
1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1669 1671 1672 1674 TO 1681 1690 -
1691 TO 1697 1723 1724 1726 TO 1733 1738 TO 1741 1755 TO 1766
_RAFT_SHEAR 334 405 TO 408 418 428 1556 1559 TO 1571 1605 1606 1610 1611 1643 -
1646 1647 1658 1659 1661 TO 1665 1667 1668 1671 1674 TO 1681 1690 1691 1694 -
1695 TO 1697 1723 1727 1730 TO 1733 1740 1741 1755 TO 1766
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
1738 TO 1741 1755 TO 1766 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1754 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
1738 TO 1741 1755 TO 1766 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
**********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
1738 TO 1741 1755 TO 1766 PR GY -46
* LOAD DUE TO SURCHARGE
328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
1738 TO 1741 1755 TO 1766 PR GY -10
JOINT LOAD
128 1839 FY -199 FZ 5.25
**********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 1839 FY -476.25 FZ 12.5
**********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 1839 FY -9.512
**********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 1839 FY 9.512
**********************************************************************************************
```

```
**********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
**********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7L
1 1.4 2 1.7
LOAD COMB 103 1.2DL+1.0SL1+1.0L
1 1.2 2 1.0 3 1.0
LOAD COMB 104 1.2DL+1.0SL2+1.0L
1 1.2 2 1.0 4 1.0
LOAD COMB 105 0.9DL+1.0SL1
1 0.9 3 1.0
LOAD COMB 106 0.9DL+1.0SL2
1 0.9 4 1.0
**********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 1754
CLB 0.05 MEMB 185 1754
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 1754
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 01/09/2020
JOB NAME FDN. MKD. F9_STABILITY
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 910.6 3.175; 128 0 913.4 3.175; 1753 2 910.6 3.175; 1839 2 913.4 3.175;
MEMBER INCIDENCES
185 2 128; 1754 1753 1839;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 1754 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 1753 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 1839 FY -199 FZ 5.25
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 1839 FY -476.25 FZ 12.5
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 1839 FY -9.512
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 1839 FY 9.512
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL
1 1.0
LOAD COMB 202 DL+L
1 1.0 2 1.0
LOAD COMB 203 DL+0.75L
1 1.0 2 0.75
LOAD COMB 204 0.6DL+0.7SL1
1 0.6 3 0.7
LOAD COMB 205 0.6DL+0.7SL2
1 0.6 4 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

C01

ATTACHMENT 9 STAAD OUTPUT – FOUNDATION F4

PROPRIETARY & CONFIDENTIAL

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*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  3, 2020                 *
*          Time=     9: 8:45                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
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    1. STAAD SPACE
INPUT FILE: F4.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F4
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 524 1.94 909.9 3.42438; 525 1.94 909.9 3.175; 526 1.94 909.9 3.67438
   17. 527 1.94 909.9 3.92438; 528 1.94 909.9 4.17438; 529 1.94 909.9 4.42438
   18. 530 1.94 909.9 2.17438; 531 1.94 909.9 1.925; 532 1.94 909.9 2.42438
   19. 533 1.94 909.9 2.67438; 534 1.94 909.9 2.92438; 588 2.25 909.9 3.42438
   20. 589 2.25 909.9 3.175; 590 2.5 909.9 3.42438; 591 2.5 909.9 3.175
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   25. 604 2.5 909.9 4.42438; 605 2.75 909.9 4.42438; 606 2.25 909.9 2.17438
   26. 607 2.25 909.9 1.925; 608 2.5 909.9 2.17438; 609 2.5 909.9 1.925
   27. 610 2.75 909.9 2.17438; 611 2.75 909.9 1.925; 612 2.25 909.9 2.42438
   28. 613 2.5 909.9 2.42438; 614 2.75 909.9 2.42438; 615 2.25 909.9 2.67438
   29. 616 2.5 909.9 2.67438; 617 2.75 909.9 2.67438; 618 2.25 909.9 2.92438
   30. 619 2.5 909.9 2.92438; 620 2.75 909.9 2.92438; 621 3 909.9 3.42438
   31. 622 3 909.9 3.175; 623 3 909.9 3.67438; 624 3 909.9 3.92438
   32. 625 3 909.9 4.17438; 626 3 909.9 4.42438; 627 3 909.9 2.17438
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110. 1538 5.54 909.9 1
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113. ELEMENT INCIDENCES SHELL
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195. 1483 1432 1433 1530 1529; 1484 1433 1434 1531 1530; 1485 1434 1435 1532 1531
196. 1486 1435 1436 1533 1532; 1487 1436 1437 1534 1533; 1488 1437 1440 1535 1534
197. 1489 1440 1445 1536 1535; 1490 1445 1446 1537 1536; 1491 1446 1448 1538 1537
198. START GROUP DEFINITION
199. ELEMENT
200. _RAFT 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 -
201. 1157 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
202. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
203. 1459 TO 1491
204. _RAFT_MOM 589 TO 628 637 TO 646 661 TO 666 783 TO 788 1095 TO 1100 -
205. 1119 TO 1124 1127 TO 1136 1157 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 -
206. 1282 TO 1317 1364 TO 1373 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 -
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207. 1443 TO 1452 1454 1455 1458 TO 1491
208. _RAFT_SHEAR 589 TO 618 623 624 641 642 663 664 785 786 1097 1098 1121 1122 -
209. 1131 1132 1157 TO 1166 1178 TO 1187 1235 1236 1251 1253 1282 TO 1317 1364 -
210. 1365 TO 1373 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 -
211. 1454 1455 1458 TO 1491
212. MEMBER
213. END GROUP DEFINITION
214. ELEMENT PROPERTY
215. 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
216. 1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
217. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
218. 1459 TO 1491 THICKNESS 0.75
219. DEFINE MATERIAL START
220. ISOTROPIC CONCRETE
221. E 2.17185E+007
222. POISSON 0.17
223. DENSITY 25
224. ALPHA 1E-005
225. DAMP 0.05
226. TYPE CONCRETE
227. STRENGTH FCU 27579
228. END DEFINE MATERIAL
229. MEMBER PROPERTY AMERICAN
230. 1092 PRIS YD 0.75 ZD 0.75
231. CONSTANTS
232. MATERIAL CONCRETE ALL
233. SUPPORTS
234. 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
235. 1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
236. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
237. 1459 TO 1491 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
524	0.038702
525	0.038750
526	0.038750
527	0.038750
528	0.038750
529	0.038750
530	0.038702
531	0.038702
532	0.038750
533	0.038750
534	0.038798
588	0.069913
589	0.070000
590	0.062422
591	0.062500
592	0.062422
593	0.062500
594	0.070000
595	0.062500
596	0.062500
597	0.070000

598	0.062500
599	0.062500
600	0.070000
601	0.062500
602	0.062500
603	0.070000
604	0.062500
605	0.062500
606	0.069913
607	0.069913
608	0.062422
609	0.062422
610	0.062422
611	0.062422
612	0.070000
613	0.062500
614	0.062500
615	0.070000
616	0.062500
617	0.062500
618	0.070087
619	0.062577
620	0.062577
621	0.062423
622	0.062500
623	0.062500
624	0.062500
625	0.062500
626	0.062500
627	0.062423
628	0.062423
629	0.062500
630	0.062500
631	0.062578
640	0.062422
641	0.062500
642	0.062500
643	0.062500
644	0.062500
645	0.062500
646	0.062422
647	0.062422
648	0.062500
649	0.062500
650	0.062577
663	0.061174
664	0.061250
665	0.061250
666	0.061250
667	0.061250
668	0.061250
669	0.061174
670	0.061174
671	0.061250
672	0.061250
673	0.061326

792	0.061174
793	0.061250
794	0.061250
795	0.061250
796	0.061250
797	0.061250
798	0.061174
799	0.061174
800	0.061250
801	0.061250
802	0.061326
1116	0.062423
1117	0.062500
1118	0.062500
1119	0.062500
1120	0.062500
1121	0.062500
1122	0.062423
1123	0.062423
1124	0.062500
1125	0.062500
1126	0.062578
1141	0.062423
1142	0.062500
1143	0.062500
1144	0.062500
1145	0.062500
1146	0.062500
1147	0.062423
1148	0.062423
1149	0.062500
1150	0.062500
1151	0.062578
1152	0.062423
1153	0.062500
1154	0.062500
1155	0.062500
1156	0.062500
1157	0.062500
1158	0.062423
1159	0.062423
1160	0.062500
1161	0.062500
1162	0.062578
1184	0.038750
1185	0.070000
1186	0.062500
1187	0.062500
1188	0.062500
1189	0.062500
1190	0.061250
1191	0.061250
1192	0.062500
1193	0.062500
1194	0.062500
1206	0.038750

1207	0.070000
1208	0.062500
1209	0.062500
1210	0.062500
1211	0.062500
1212	0.061250
1213	0.061250
1214	0.062500
1215	0.062500
1216	0.062500
1262	0.062423
1263	0.062500
1264	0.062500
1265	0.062500
1266	0.062500
1267	0.062500
1268	0.062423
1269	0.062423
1270	0.062500
1271	0.062500
1272	0.062578
1284	0.062500
1286	0.062500
1317	0.062422
1318	0.062500
1319	0.062500
1320	0.062500
1321	0.062500
1322	0.062500
1323	0.062422
1324	0.062422
1325	0.062500
1326	0.062500
1327	0.062577
1328	0.068665
1329	0.068750
1330	0.068750
1331	0.068750
1332	0.068750
1333	0.068750
1334	0.068665
1335	0.068665
1336	0.068750
1337	0.068750
1338	0.068835
1339	0.062500
1340	0.068750
1341	0.062500
1342	0.068750
1343	0.037454
1344	0.037500
1345	0.037500
1346	0.037500
1347	0.037500
1348	0.037500
1349	0.037454

1350	0.037454
1351	0.037500
1352	0.037500
1353	0.037547
1354	0.037500
1355	0.037500
1405	0.038750
1406	0.070000
1407	0.062500
1408	0.062500
1409	0.062500
1410	0.062500
1411	0.061250
1412	0.061250
1413	0.062500
1414	0.062500
1415	0.062500
1427	0.032938
1428	0.059500
1429	0.053125
1430	0.053125
1431	0.053125
1432	0.053125
1433	0.052062
1434	0.052062
1435	0.053125
1436	0.053125
1437	0.053125
1439	0.062500
1440	0.053125
1443	0.062500
1444	0.068750
1445	0.053125
1446	0.058438
1447	0.037500
1448	0.031875
1465	0.038750
1466	0.070000
1467	0.062500
1468	0.062500
1469	0.062500
1470	0.062500
1471	0.061250
1472	0.061250
1473	0.062500
1474	0.062500
1475	0.062500
1487	0.032938
1488	0.059500
1489	0.053125
1490	0.053125
1491	0.053125
1492	0.053125
1493	0.052062
1494	0.052062
1495	0.053125

1496	0.053125
1497	0.053125
1499	0.062500
1500	0.053125
1503	0.062500
1504	0.068750
1505	0.053125
1506	0.058438
1507	0.037500
1508	0.031875
1509	0.013563
1510	0.024500
1511	0.021875
1512	0.021875
1513	0.021875
1514	0.021875
1515	0.021438
1516	0.021438
1517	0.021875
1518	0.021875
1519	0.021875
1520	0.021875
1521	0.021875
1522	0.024063
1523	0.013125
1524	0.024500
1525	0.013563
1526	0.021875
1527	0.021875
1528	0.021875
1529	0.021875
1530	0.021437
1531	0.021437
1532	0.021875
1533	0.021875
1534	0.021875
1535	0.021875
1536	0.021875
1537	0.024063
1538	0.013125
TOTAL	15.657784

```
238. ********************************************************************************
239. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
240. SELFWEIGHT Y -1
241. ELEMENT LOAD
242. * LOAD DUE TO BACKFILL
243. 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
244. 1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
245. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
246. 1459 TO 1491 PR GY -50
247. * LOAD DUE TO SURCHARGE
248. 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
249. 1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
250. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
251. 1459 TO 1491 PR GY -20
252. JOINT LOAD
```

```
253. 1115 FY -100.45
254. ********************************************************************************
255. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
256. JOINT LOAD
257. 1115 FY -0.0001
258. ********************************************************************************
259. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
260. JOINT LOAD
261. 1115 FY -0.0001
262. ********************************************************************************
263. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
264. JOINT LOAD
265. 1115 FZ -320
266. ********************************************************************************
267. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
268. JOINT LOAD
269. 1115 FZ 320
270. ********************************************************************************
271. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
272. JOINT LOAD
273. 1115 FY -0.0001
274. ********************************************************************************
275. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
276. JOINT LOAD
277. 1115 FY 0.0001
278. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 286 NUMBER OF MEMBERS 1
NUMBER OF PLATES 252 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 285

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 127/ 25/ 78 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 861
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 68 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.1/ 143461.2 MB

```
279. CHANGE
280. SUPPORTS
281. 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1093 TO 1102 1117 TO 1136 1157 -
282. 1158 TO 1166 1178 TO 1187 1231 TO 1240 1251 1253 1282 TO 1317 1364 TO 1373 -
283. 1385 TO 1394 1396 1397 1400 TO 1405 1422 TO 1431 1443 TO 1452 1454 1455 1458 -
284. 1459 TO 1491 PLATE MAT DIRECT Y SUBGRADE 57000 COMPRESSION
```



```
285. ********************************************************************************
286. * UNFACTORED LOAD COMBINATIONS
287. LOAD 201 DL
288. REPEAT LOAD
289. 1 1.0
290. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201


291. CHANGE
292. LOAD 202 DL+L
293. REPEAT LOAD
294. 1 1.0 3 1.0
295. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202


296. CHANGE
297. LOAD 203 DL+LR
298. REPEAT LOAD
299. 1 1.0 2 1.0
300. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203


301. CHANGE
302. LOAD 204 DL+0.75L+0.75LR
303. REPEAT LOAD
304. 1 1.0 2 0.75 3 0.75
305. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204


306. CHANGE
307. LOAD 205 DL+WL1
308. REPEAT LOAD
```

```
309. 1 1.0 4 1.0
310. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  205



311. CHANGE
312. LOAD 206 DL+WL2
313. REPEAT LOAD
314. 1 1.0 5 1.0
315. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206



316. CHANGE
317. LOAD 207 DL+0.75WL1+0.75L+0.75LR
318. REPEAT LOAD
319. 1 1.0 2 0.75 3 0.75 4 0.75
320. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



321. CHANGE
322. LOAD 208 DL+0.75WL2+0.75L+0.75LR
323. REPEAT LOAD
324. 1 1.0 2 0.75 3 0.75 5 0.75
325. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



326. CHANGE
327. LOAD 209 0.6DL+WL1
328. REPEAT LOAD
329. 1 0.6 4 1.0
330. PERFORM ANALYSIS
```



```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  209



331. CHANGE
332. LOAD 210 0.6DL+WL2
333. REPEAT LOAD
334. 1 0.6 5 1.0
335. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  210



336. CHANGE
337. LOAD 211 0.6DL+0.7SL1
338. REPEAT LOAD
339. 1 0.6 6 0.7
340. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



341. CHANGE
342. LOAD 212 0.6DL+0.7SL2
343. REPEAT LOAD
344. 1 0.6 7 0.7
345. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



346. CHANGE
347. ********************************************************************************
348. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
```

```
 349. LOAD 101 1.4DL
 350. REPEAT LOAD
 351. 1 1.4
 352. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101



 353. CHANGE
 354. LOAD 102 1.4DL+1.7L+0.5LR
 355. REPEAT LOAD
 356. 1 1.4 3 1.7 2 0.5
 357. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102



 358. CHANGE
 359. LOAD 103 1.2DL+1.6LR+1.0L
 360. REPEAT LOAD
 361. 1 1.2 2 1.6 3 1.0
 362. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 363. CHANGE
 364. LOAD 104 1.2DL+1.6WL1+1.0L+0.5LR
 365. REPEAT LOAD
 366. 1 1.2 4 1.6 2 0.5 3 1.0
 367. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  104



 368. CHANGE
 369. LOAD 105 1.2DL+1.6WL2+1.0L+0.5LR
 370. REPEAT LOAD
```



```
 371. 1 1.2 5 1.6 2 0.5 3 1.0
 372. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  105



 373. CHANGE
 374. LOAD 106 1.2DL+1.0SL1+1.0L
 375. REPEAT LOAD
 376. 1 1.2 3 1.0 6 1.0
 377. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



 378. CHANGE
 379. LOAD 107 1.2DL+1.0SL2+1.0L
 380. REPEAT LOAD
 381. 1 1.2 3 1.0 7 1.0
 382. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  107



 383. CHANGE
 384. LOAD 108 0.9DL+1.6WL1
 385. REPEAT LOAD
 386. 1 0.9 4 1.6
 387. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  108



 388. CHANGE
 389. LOAD 109 0.9DL+1.6WL2
```

```
 390. REPEAT LOAD
 391. 1 0.9 5 1.6
 392. PERFORM ANALYSIS
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  109



 393. CHANGE
 394. LOAD 110 0.9DL+1.0SL1
 395. REPEAT LOAD
 396. 1 0.9 6 1.0
 397. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110



 398. CHANGE
 399. LOAD 111 0.9DL+1.0SL2
 400. REPEAT LOAD
 401. 1 0.9 7 1.0
 402. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  111



 403. CHANGE
 404. ******************************************************************************
 405. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=    3



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

***WARNING -  INSTABILITY AT JOINT  1538   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  4.6723142E+06 L-MATRIX DIAG=  1.1483207E-06 EQN NO     860
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          7

***WARNING -  INSTABILITY AT JOINT  1538   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  4.6723142E+06 L-MATRIX DIAG=  1.1455268E-06 EQN NO     860
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.          8

***WARNING -  INSTABILITY AT JOINT  1538   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  4.6723142E+06 L-MATRIX DIAG=  1.1492521E-06 EQN NO     860
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.
```

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 1.1473894E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 1.1492521E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.2506249E+06 L-MATRIX DIAG= -1.9557774E-08 EQN NO 861
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 1.1483207E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 1.1445954E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 1.1445954E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 13 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 3.3946708E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 3.5120174E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 4.1229650E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 3.4477562E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 5.5786222E-07 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 8.7264925E-07 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1538 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723142E+06 L-MATRIX DIAG= 4.1397288E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 12 iterations, Case= 5

**NOTE-Tension/Compression converged after 1 iterations, Case= 6

**NOTE-Tension/Compression converged after 1 iterations, Case= 7

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 209

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 104

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 108

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

406. LOAD LIST 101 TO 111

```
408. CODE ACI
409. TRACK 2 ALL
410. FC 35000 MEMB 1092
411. CLB 0.05 MEMB 1092
412. CLS 0.05 ALL
413. CLT 0.05 MEMB 589 TO 628 637 TO 646 659 TO 668 781 TO 790 1157 TO 1163 1178 -
414. 1179 TO 1184 1364 TO 1370 1385 TO 1391 1422 TO 1428 1443 TO 1449 1464 TO 1470 -
415. 1478 TO 1484
416. MAXMAIN 32 ALL
417. MINMAIN 32 ALL
418. MINSEC 10 ALL
419. DESIGN COLUMN 1092
```



```
================================================================

     COLUMN  NO.  1092  DESIGN PER ACI 318-08 - AXIAL + BENDING


*** NO PROPER BAR ARRANGEMENT IS POSSIBLE.

        AREA OF STEEL REQUIRED = 22623.8  SQ. MM


********************END OF COLUMN DESIGN RESULTS********************


420. END CONCRETE DESIGN
421. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= OCT  3,2020   TIME=  9:10:50 ****
```

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2014 Bentley Systems, Inc.
     *                http://www.bentley.com                    *
     ************************************************************
```

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    OCT  3, 2020                *
            *           Time=     9:12:47                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F4 _STABILITY CHECK.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 01/09/2020
     4. JOB NAME FDN. MKD. F4_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 793 3.74 909.9 3.175; 1115 3.74 913.4 3.175
    14. MEMBER INCIDENCES
    15. 1092 793 1115
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 1092 PRIS YD 0.75 ZD 0.75
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 793 FIXED
    32. ******************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 1115 FY -100.45
    37. ******************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
```



```
    39. JOINT LOAD
    40. 1115 FY -0.0001
    41. ******************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    43. JOINT LOAD
    44. 1115 FY -0.0001
    45. ******************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
    47. JOINT LOAD
    48. 1115 FZ -320
    49. ******************************************************************************
    50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
    51. JOINT LOAD
    52. 1115 FZ 320
    53. ******************************************************************************
    54. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
    55. JOINT LOAD
    56. 1115 FY -0.0001
    57. ******************************************************************************
    58. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
    59. JOINT LOAD
    60. 1115 FY 0.0001
    61. ******************************************************************************
    62. * UNFACTORED LOAD COMBINATIONS
    63. LOAD COMB 201 DL
    64. 1 1.0
    65. LOAD COMB 202 DL+L
    66. 1 1.0 3 1.0
    67. LOAD COMB 203 DL+LR
    68. 1 1.0 2 1.0
    69. LOAD COMB 204 DL+0.75L+0.75LR
    70. 1 1.0 2 0.75 3 0.75
    71. LOAD COMB 205 DL+WL1
    72. 1 1.0 4 1.0
    73. LOAD COMB 206 DL+WL2
    74. 1 1.0 5 1.0
    75. LOAD COMB 207 DL+0.75WL1+0.75L+0.75LR
    76. 1 1.0 2 0.75 3 0.75 4 0.75
    77. LOAD COMB 208 DL+0.75WL2+0.75L+0.75LR
    78. 1 1.0 2 0.75 3 0.75 5 0.75
    79. LOAD COMB 209 0.6DL+1.0WL1
    80. 1 0.6 4 1.0
    81. LOAD COMB 210 0.6DL+1.0WL2
    82. 1 0.6 5 1.0
    83. LOAD COMB 211 0.6DL+0.7SL1
    84. 1 0.6 6 0.7
    85. LOAD COMB 212 0.6DL+0.7SL2
    86. 1 0.6 7 0.7
    87. ******************************************************************************
    88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 12 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 143457.2 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:12:51 ****


```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*               http://www.bentley.com                      *
*************************************************************
```

C01

ATTACHMENT 10 STAAD OUTPUT – FOUNDATION F5

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  3, 2020                 *
*          Time=     9:17: 9                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F5.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F5
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175
   17. 41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175
   18. 302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438
   19. 306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438
   20. 309 1 909.9 3.175; 311 1.25 909.9 3.175; 313 1 909.9 3.42438
   21. 314 1.25 909.9 3.42438; 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438
   22. 324 0 909.9 3.67438; 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438
   23. 327 0.75 909.9 3.67438; 328 1 909.9 3.67438; 329 1.25 909.9 3.67438
   24. 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438
   25. 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438
   26. 339 1 909.9 3.92438; 340 1.25 909.9 3.92438; 344 -0.5 909.9 4.17438
   27. 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438; 347 0.25 909.9 4.17438
   28. 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438; 350 1 909.9 4.17438
   29. 351 1.25 909.9 4.17438; 355 -0.5 909.9 4.42438; 356 -0.25 909.9 4.42438
   30. 357 0 909.9 4.42438; 358 0.25 909.9 4.42438; 359 0.5 909.9 4.42438
   31. 360 0.75 909.9 4.42438; 361 1 909.9 4.42438; 362 1.25 909.9 4.42438
   32. 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925; 369 -0.25 909.9 2.17438
   33. 370 -0.25 909.9 1.925; 371 0 909.9 2.17438; 372 0 909.9 1.925
   34. 373 0.25 909.9 2.17438; 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438
   35. 376 0.5 909.9 1.925; 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925
   36. 379 1 909.9 2.17438; 380 1 909.9 1.925; 381 1.25 909.9 2.17438
   37. 382 1.25 909.9 1.925; 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438
   38. 390 0 909.9 2.42438; 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438
```



```
   39. 393 0.75 909.9 2.42438; 394 1 909.9 2.42438; 395 1.25 909.9 2.42438
   40. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   41. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   42. 405 1 909.9 2.67438; 406 1.25 909.9 2.67438; 410 -0.5 909.9 2.92438
   43. 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438
   44. 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438
   45. 417 1.25 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   46. 422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438
   47. 425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925
   48. 428 1.5 909.9 2.42438; 429 1.5 909.9 2.67438; 430 1.5 909.9 2.92438
   49. 1163 -0.75 909.9 3.42438; 1164 -0.75 909.9 3.175; 1165 -0.75 909.9 3.67438
   50. 1166 -0.75 909.9 3.92438; 1167 -0.75 909.9 4.17438; 1168 -0.75 909.9 4.42438
   51. 1169 -0.75 909.9 2.17438; 1170 -0.75 909.9 1.925; 1171 -0.75 909.9 2.42438
   52. 1172 -0.75 909.9 2.67438; 1173 -0.75 909.9 2.92438; 1174 -0.5 909.9 4.67438
   53. 1175 -0.25 909.9 4.67438; 1176 0 909.9 4.67438; 1177 0.25 909.9 4.67438
   54. 1178 0.5 909.9 4.67438; 1179 0.75 909.9 4.67438; 1180 1 909.9 4.67438
   55. 1181 1.25 909.9 4.67438; 1182 1.5 909.9 4.67438; 1195 -0.75 909.9 4.67438
   56. 1196 -0.25 909.9 1.675; 1197 -0.5 909.9 1.675; 1198 0 909.9 1.675
   57. 1199 0.25 909.9 1.675; 1200 0.5 909.9 1.675; 1201 0.75 909.9 1.675
   58. 1202 1 909.9 1.675; 1203 1.25 909.9 1.675; 1204 1.5 909.9 1.675
   59. 1217 -0.75 909.9 1.675; 1292 -1 909.9 3.42438; 1293 -1 909.9 3.175
   60. 1294 -1 909.9 3.67438; 1295 -1 909.9 3.92438; 1296 -1 909.9 4.17438
   61. 1297 -1 909.9 4.42438; 1298 -1 909.9 2.17438; 1299 -1 909.9 1.925
   62. 1300 -1 909.9 2.42438; 1301 -1 909.9 2.67438; 1302 -1 909.9 2.92438
   63. 1303 -1.25 909.9 3.42438; 1304 -1.25 909.9 3.175; 1305 -1.25 909.9 3.67438
   64. 1306 -1.25 909.9 3.92438; 1307 -1.25 909.9 4.17438; 1308 -1.25 909.9 4.42438
   65. 1309 -1.25 909.9 2.17438; 1310 -1.25 909.9 1.925; 1311 -1.25 909.9 2.42438
   66. 1312 -1.25 909.9 2.67438; 1313 -1.25 909.9 2.92438; 1314 -1 909.9 4.67438
   67. 1315 -1.25 909.9 4.67438; 1316 -1 909.9 1.675; 1317 -1.25 909.9 1.675
   68. 1337 -1.5 909.9 3.42438; 1338 -1.5 909.9 3.175; 1339 -1.5 909.9 3.67438
   69. 1340 -1.5 909.9 3.92438; 1341 -1.5 909.9 4.17438; 1342 -1.5 909.9 4.42438
   70. 1343 -1.5 909.9 2.17438; 1344 -1.5 909.9 1.925; 1345 -1.5 909.9 2.42438
   71. 1346 -1.5 909.9 2.67438; 1347 -1.5 909.9 2.92438; 1348 -1.5 909.9 4.67438
   72. 1349 -1.5 909.9 1.675; 1350 1.8 909.9 3.42438; 1351 1.8 909.9 3.175
   73. 1352 1.8 909.9 3.67438; 1353 1.8 909.9 3.92438; 1354 1.8 909.9 4.17438
   74. 1355 1.8 909.9 4.42438; 1356 1.8 909.9 2.17438; 1357 1.8 909.9 1.925
   75. 1358 1.8 909.9 2.42438; 1359 1.8 909.9 2.67438; 1360 1.8 909.9 2.92438
   76. 1361 1.8 909.9 4.67438; 1362 1.8 909.9 1.675; 1363 -1.8 909.9 3.42438
   77. 1364 -1.8 909.9 3.175; 1365 -1.8 909.9 3.67438; 1366 -1.8 909.9 3.92438
   78. 1367 -1.8 909.9 4.17438; 1368 -1.8 909.9 4.42438; 1369 -1.8 909.9 2.17438
   79. 1370 -1.8 909.9 1.925; 1371 -1.8 909.9 2.42438; 1372 -1.8 909.9 2.67438
   80. 1373 -1.8 909.9 2.92438; 1374 -1.8 909.9 4.67438; 1375 -1.8 909.9 1.675
   81. 1428 -0.25 909.9 1.425; 1429 -0.5 909.9 1.425; 1430 0 909.9 1.425
   82. 1431 0.25 909.9 1.425; 1432 0.5 909.9 1.425; 1433 0.75 909.9 1.425
   83. 1434 1 909.9 1.425; 1435 1.25 909.9 1.425; 1436 1.5 909.9 1.425
   84. 1437 -0.75 909.9 1.425; 1438 -1 909.9 1.425; 1439 -1.25 909.9 1.425
   85. 1440 -1.5 909.9 1.425; 1441 1.8 909.9 1.425; 1442 -1.8 909.9 1.425
   86. 1447 -0.5 909.9 4.92438; 1448 -0.25 909.9 4.92438; 1449 0 909.9 4.92438
   87. 1450 0.25 909.9 4.92438; 1451 0.5 909.9 4.92438; 1452 0.75 909.9 4.92438
   88. 1453 1 909.9 4.92438; 1454 1.25 909.9 4.92438; 1455 1.5 909.9 4.92438
   89. 1456 -0.75 909.9 4.92438; 1457 -1 909.9 4.92438; 1458 -1.25 909.9 4.92438
   90. 1459 -1.5 909.9 4.92438; 1460 1.8 909.9 4.92438; 1461 -1.8 909.9 4.92438
   91. 1462 -0.25 909.9 1.175; 1463 -0.5 909.9 1.175; 1464 0 909.9 1.175
   92. 1465 0.25 909.9 1.175; 1466 0.5 909.9 1.175; 1467 0.75 909.9 1.175
   93. 1468 1 909.9 1.175; 1469 1.25 909.9 1.175; 1470 1.5 909.9 1.175
   94. 1471 -0.75 909.9 1.175; 1472 -1 909.9 1.175; 1473 -1.25 909.9 1.175
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 95. 1474 -1.5 909.9 1.175; 1475 1.8 909.9 1.175; 1476 -1.8 909.9 1.175
 96. 1477 -0.25 909.9 1; 1478 -0.5 909.9 1; 1479 0 909.9 1; 1480 0.25 909.9 1
 97. 1481 0.5 909.9 1; 1482 0.75 909.9 1; 1483 1 909.9 1; 1484 1.25 909.9 1
 98. 1485 1.5 909.9 1; 1486 -0.75 909.9 1; 1487 -1 909.9 1; 1488 -1.25 909.9 1
 99. 1489 -1.5 909.9 1; 1490 1.8 909.9 1; 1491 -1.8 909.9 1
100. 1492 -0.5 909.9 5.17438; 1493 -0.25 909.9 5.17438; 1494 0 909.9 5.17438
101. 1495 0.25 909.9 5.17438; 1496 0.5 909.9 5.17438; 1497 0.75 909.9 5.17438
102. 1498 1 909.9 5.17438; 1499 1.25 909.9 5.17438; 1500 1.5 909.9 5.17438
103. 1501 -0.75 909.9 5.17438; 1502 -1 909.9 5.17438; 1503 -1.25 909.9 5.17438
104. 1504 -1.5 909.9 5.17438; 1505 1.8 909.9 5.17438; 1506 -1.8 909.9 5.17438
105. 1507 -0.5 909.9 5.34938; 1508 -0.25 909.9 5.34938; 1509 0 909.9 5.34938
106. 1510 0.25 909.9 5.34938; 1511 0.5 909.9 5.34938; 1512 0.75 909.9 5.34938
107. 1513 1 909.9 5.34938; 1514 1.25 909.9 5.34938; 1515 1.5 909.9 5.34938
108. 1516 -0.75 909.9 5.34938; 1517 -1 909.9 5.34938; 1518 -1.25 909.9 5.34938
109. 1519 -1.5 909.9 5.34938; 1520 1.8 909.9 5.34938; 1521 -1.8 909.9 5.34938
110. MEMBER INCIDENCES
111. 185 2 128
112. ELEMENT INCIDENCES SHELL
113. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
114. 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309; 344 322 323 304 302
115. 345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306
116. 348 326 327 308 307; 349 327 328 313 308; 350 328 329 314 313
117. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
118. 357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327
119. 360 339 340 329 328; 364 344 345 334 333; 365 345 346 335 334
120. 366 346 347 336 335; 367 347 348 337 336; 368 348 349 338 337
121. 369 349 350 339 338; 370 350 351 340 339; 374 355 356 345 344
122. 375 356 357 346 345; 376 357 358 347 346; 377 358 359 348 347
123. 378 359 360 349 348; 379 360 361 350 349; 380 361 362 351 350
124. 384 366 369 370 367; 385 369 371 372 370; 386 371 373 374 372
125. 387 373 375 376 374; 388 375 377 378 376; 389 377 379 380 378
126. 390 379 381 382 380; 394 388 389 369 366; 395 389 390 371 369
127. 396 390 391 373 371; 397 391 392 375 373; 398 392 393 377 375
128. 399 393 394 379 377; 400 394 395 381 379; 404 399 400 389 388
129. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
130. 408 403 404 393 392; 409 404 405 394 393; 410 405 406 395 394
131. 414 410 411 400 399; 415 411 412 401 400; 416 412 413 402 401
132. 417 413 414 403 402; 418 414 415 404 403; 419 415 416 405 404
133. 420 416 417 406 405; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
134. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 430 309 311 417 416
135. 433 314 420 421 311; 434 329 422 420 314; 435 340 423 422 329
136. 436 351 424 423 340; 437 362 425 424 351; 438 381 426 427 382
137. 439 395 428 426 381; 440 406 429 428 395; 441 417 430 429 406
138. 442 311 421 430 417; 1137 1163 302 34 1164; 1138 1165 322 302 1163
139. 1139 1166 333 322 1165; 1140 1167 344 333 1166; 1141 1168 355 344 1167
140. 1142 1169 366 367 1170; 1143 1171 388 366 1169; 1144 1172 399 388 1171
141. 1145 1173 410 399 1172; 1146 1164 34 410 1173; 1147 1174 1175 356 355
142. 1148 1175 1176 357 356; 1149 1176 1177 358 357; 1150 1177 1178 359 358
143. 1151 1178 1179 360 359; 1152 1179 1180 361 360; 1153 1180 1181 362 361
144. 1154 1181 1182 425 362; 1167 1195 1174 355 1168; 1168 367 370 1196 1197
145. 1169 370 372 1198 1196; 1170 372 374 1199 1198; 1171 374 376 1200 1199
146. 1172 376 378 1201 1200; 1173 378 380 1202 1201; 1174 380 382 1203 1202
147. 1175 382 427 1204 1203; 1188 1170 367 1197 1217; 1259 1292 1163 1164 1293
148. 1260 1294 1165 1163 1292; 1261 1295 1166 1165 1294; 1262 1296 1167 1166 1295
149. 1263 1297 1168 1167 1296; 1264 1298 1169 1170 1299; 1265 1300 1171 1169 1298
150. 1266 1301 1172 1171 1300; 1267 1302 1173 1172 1301; 1268 1293 1164 1173 1302
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151. 1269 1303 1292 1293 1304; 1270 1305 1294 1292 1303; 1271 1306 1295 1294 1305
152. 1272 1307 1296 1295 1306; 1273 1308 1297 1296 1307; 1274 1309 1298 1299 1310
153. 1275 1311 1300 1298 1309; 1276 1312 1301 1300 1311; 1277 1313 1302 1301 1312
154. 1278 1304 1293 1302 1313; 1279 1314 1195 1168 1297; 1280 1315 1314 1297 1308
155. 1281 1299 1170 1217 1316; 1282 1310 1299 1316 1317; 1301 1337 1303 1304 1338
156. 1302 1339 1305 1303 1337; 1303 1340 1306 1305 1339; 1304 1341 1307 1306 1340
157. 1305 1342 1308 1307 1341; 1306 1343 1309 1310 1344; 1307 1345 1311 1309 1343
158. 1308 1346 1312 1311 1345; 1309 1347 1313 1312 1346; 1310 1338 1304 1313 1347
159. 1311 1348 1315 1308 1342; 1312 1344 1310 1317 1349; 1313 420 1350 1351 421
160. 1314 422 1352 1350 420; 1315 423 1353 1352 422; 1316 424 1354 1353 423
161. 1317 425 1355 1354 424; 1318 426 1356 1357 427; 1319 428 1358 1356 426
162. 1320 429 1359 1358 428; 1321 430 1360 1359 429; 1322 421 1351 1360 430
163. 1323 1182 1361 1355 425; 1324 427 1357 1362 1204; 1325 1363 1337 1338 1364
164. 1326 1365 1339 1337 1363; 1327 1366 1340 1339 1365; 1328 1367 1341 1340 1366
165. 1329 1368 1342 1341 1367; 1330 1369 1343 1344 1370; 1331 1371 1345 1343 1369
166. 1332 1372 1346 1345 1371; 1333 1373 1347 1346 1372; 1334 1364 1338 1347 1373
167. 1335 1374 1348 1342 1368; 1336 1370 1344 1349 1375; 1385 1197 1196 1428 1429
168. 1386 1196 1198 1430 1428; 1387 1198 1199 1431 1430; 1388 1199 1200 1432 1431
169. 1389 1200 1201 1433 1432; 1390 1201 1202 1434 1433; 1391 1202 1203 1435 1434
170. 1392 1203 1204 1436 1435; 1393 1217 1197 1429 1437; 1394 1316 1217 1437 1438
171. 1395 1317 1316 1438 1439; 1396 1349 1317 1439 1440; 1397 1204 1362 1441 1436
172. 1398 1375 1349 1440 1442; 1403 1447 1448 1175 1174; 1404 1448 1449 1176 1175
173. 1405 1449 1450 1177 1176; 1406 1450 1451 1178 1177; 1407 1451 1452 1179 1178
174. 1408 1452 1453 1180 1179; 1409 1453 1454 1181 1180; 1410 1454 1455 1182 1181
175. 1411 1456 1447 1174 1195; 1412 1457 1456 1195 1314; 1413 1458 1457 1314 1315
176. 1414 1459 1458 1315 1348; 1415 1455 1460 1361 1182; 1416 1461 1459 1348 1374
177. 1417 1429 1428 1462 1463; 1418 1428 1430 1464 1462; 1419 1430 1431 1465 1464
178. 1420 1431 1432 1466 1465; 1421 1432 1433 1467 1466; 1422 1433 1434 1468 1467
179. 1423 1434 1435 1469 1468; 1424 1435 1436 1470 1469; 1425 1437 1429 1463 1471
180. 1426 1438 1437 1471 1472; 1427 1439 1438 1472 1473; 1428 1440 1439 1473 1474
181. 1429 1436 1441 1475 1470; 1430 1442 1440 1474 1476; 1431 1463 1462 1477 1478
182. 1432 1462 1464 1479 1477; 1433 1464 1465 1480 1479; 1434 1465 1466 1481 1480
183. 1435 1466 1467 1482 1481; 1436 1467 1468 1483 1482; 1437 1468 1469 1484 1483
184. 1438 1469 1470 1485 1484; 1439 1471 1463 1478 1486; 1440 1472 1471 1486 1487
185. 1441 1473 1472 1487 1488; 1442 1474 1473 1488 1489; 1443 1470 1475 1490 1485
186. 1444 1476 1474 1489 1491; 1445 1492 1493 1448 1447; 1446 1493 1494 1449 1448
187. 1447 1494 1495 1450 1449; 1448 1495 1496 1451 1450; 1449 1496 1497 1452 1451
188. 1450 1497 1498 1453 1452; 1451 1498 1499 1454 1453; 1452 1499 1500 1455 1454
189. 1453 1501 1492 1447 1456; 1454 1502 1501 1456 1457; 1455 1503 1502 1457 1458
190. 1456 1504 1503 1458 1459; 1457 1500 1505 1460 1455; 1458 1506 1504 1459 1461
191. 1459 1507 1508 1493 1492; 1460 1508 1509 1494 1493; 1461 1509 1510 1495 1494
192. 1462 1510 1511 1496 1495; 1463 1511 1512 1497 1496; 1464 1512 1513 1498 1497
193. 1465 1513 1514 1499 1498; 1466 1514 1515 1500 1499; 1467 1516 1507 1492 1501
194. 1468 1517 1516 1501 1502; 1469 1518 1517 1502 1503; 1470 1519 1518 1503 1504
195. 1471 1515 1520 1505 1500; 1472 1521 1519 1504 1506
196. START GROUP DEFINITION
197. ELEMENT
198. _RAFT 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 -
199. 384 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
200. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
201. 1404 TO 1472
202. _RAFT_MOM 327 333 334 337 338 344 347 TO 350 354 TO 360 364 TO 370 -
203. 374 TO 380 384 TO 390 394 TO 400 404 TO 410 414 417 TO 420 424 427 TO 430 -
204. 433 TO 442 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 -
205. 1386 TO 1398 1403 TO 1472
206. _RAFT_SHEAR 338 350 360 370 374 TO 380 384 TO 390 400 410 420 430 433 TO 442 -
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207. 1141 1142 1147 TO 1154 1167 TO 1175 1188 1263 1264 1269 TO 1282 1301 TO 1336 -
208. 1385 TO 1398 1403 TO 1472
209. MEMBER
210. END GROUP DEFINITION
211. ELEMENT PROPERTY
212. 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
213. 385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
214. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
215. 1404 TO 1472 THICKNESS 0.75
216. DEFINE MATERIAL START
217. ISOTROPIC CONCRETE
218. E 2.17185E+007
219. POISSON 0.17
220. DENSITY 25
221. ALPHA 1E-005
222. DAMP 0.05
223. TYPE CONCRETE
224. STRENGTH FCU 27579
225. END DEFINE MATERIAL
226. MEMBER PROPERTY AMERICAN
227. 185 PRIS YD 0.75 ZD 0.75
228. CONSTANTS
229. MATERIAL CONCRETE ALL
230. SUPPORTS
231. 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
232. 385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
233. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
234. 1404 TO 1472 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.062500
311	0.062500
313	0.062423
314	0.062422
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.062500
328	0.062500
329	0.062500

333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.062500
339	0.062500
340	0.062500
344	0.062500
345	0.062500
346	0.062500
347	0.062500
348	0.062500
349	0.062500
350	0.062500
351	0.062500
355	0.062500
356	0.062500
357	0.062500
358	0.062500
359	0.062500
360	0.062500
361	0.062500
362	0.062500
366	0.062422
367	0.062422
369	0.062422
370	0.062422
371	0.062422
372	0.062422
373	0.062422
374	0.062422
375	0.062422
376	0.062422
377	0.062422
378	0.062422
379	0.062423
380	0.062423
381	0.062422
382	0.062422
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.062500
395	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.062500
406	0.062500

410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.062578
417	0.062578
420	0.068665
421	0.068750
422	0.068750
423	0.068750
424	0.068750
425	0.068750
426	0.068665
427	0.068665
428	0.068750
429	0.068750
430	0.068835
1163	0.062422
1164	0.062500
1165	0.062500
1166	0.062500
1167	0.062500
1168	0.062500
1169	0.062422
1170	0.062422
1171	0.062500
1172	0.062500
1173	0.062578
1174	0.062500
1175	0.062500
1176	0.062500
1177	0.062500
1178	0.062500
1179	0.062500
1180	0.062500
1181	0.062500
1182	0.068750
1195	0.062500
1196	0.062500
1197	0.062500
1198	0.062500
1199	0.062500
1200	0.062500
1201	0.062500
1202	0.062500
1203	0.062500
1204	0.068750
1217	0.062500
1292	0.062423
1293	0.062500
1294	0.062500
1295	0.062500
1296	0.062500
1297	0.062500

1298	0.062423
1299	0.062423
1300	0.062500
1301	0.062500
1302	0.062578
1303	0.062422
1304	0.062500
1305	0.062500
1306	0.062500
1307	0.062500
1308	0.062500
1309	0.062422
1310	0.062422
1311	0.062500
1312	0.062500
1313	0.062578
1314	0.062500
1315	0.062500
1316	0.062500
1317	0.062500
1337	0.068665
1338	0.068750
1339	0.068750
1340	0.068750
1341	0.068750
1342	0.068750
1343	0.068665
1344	0.068665
1345	0.068750
1346	0.068750
1347	0.068835
1348	0.068750
1349	0.068750
1350	0.037453
1351	0.037500
1352	0.037500
1353	0.037500
1354	0.037500
1355	0.037500
1356	0.037453
1357	0.037453
1358	0.037500
1359	0.037500
1360	0.037547
1361	0.037500
1362	0.037500
1363	0.037453
1364	0.037500
1365	0.037500
1366	0.037500
1367	0.037500
1368	0.037500
1369	0.037453
1370	0.037453
1371	0.037500
1372	0.037500

1373	0.037547
1374	0.037500
1375	0.037500
1428	0.062500
1429	0.062500
1430	0.062500
1431	0.062500
1432	0.062500
1433	0.062500
1434	0.062500
1435	0.062500
1436	0.068750
1437	0.062500
1438	0.062500
1439	0.062500
1440	0.068750
1441	0.037500
1442	0.037500
1447	0.062500
1448	0.062500
1449	0.062500
1450	0.062500
1451	0.062500
1452	0.062500
1453	0.062500
1454	0.062500
1455	0.068750
1456	0.062500
1457	0.062500
1458	0.062500
1459	0.068750
1460	0.037500
1461	0.037500
1462	0.053125
1463	0.053125
1464	0.053125
1465	0.053125
1466	0.053125
1467	0.053125
1468	0.053125
1469	0.053125
1470	0.058437
1471	0.053125
1472	0.053125
1473	0.053125
1474	0.058437
1475	0.031875
1476	0.031875
1477	0.021875
1478	0.021875
1479	0.021875
1480	0.021875
1481	0.021875
1482	0.021875
1483	0.021875
1484	0.021875

1485	0.024062
1486	0.021875
1487	0.021875
1488	0.021875
1489	0.024062
1490	0.013125
1491	0.013125
1492	0.053125
1493	0.053125
1494	0.053125
1495	0.053125
1496	0.053125
1497	0.053125
1498	0.053125
1499	0.053125
1500	0.058437
1501	0.053125
1502	0.053125
1503	0.053125
1504	0.058437
1505	0.031875
1506	0.031875
1507	0.021875
1508	0.021875
1509	0.021875
1510	0.021875
1511	0.021875
1512	0.021875
1513	0.021875
1514	0.021875
1515	0.024063
1516	0.021875
1517	0.021875
1518	0.021875
1519	0.024063
1520	0.013125
1521	0.013125
TOTAL	15.657798

```
235. ********************************************************************************
236. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
237. SELFWEIGHT Y -1
238. ELEMENT LOAD
239. * LOAD DUE TO BACKFILL
240. 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
241. 385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
242. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
243. 1404 TO 1472 PR GY -55
244. * LOAD DUE TO SURCHARGE
245. 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
246. 385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
247. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
248. 1404 TO 1472 PR GY -20
249. JOINT LOAD
250. 128 FX 1.5 FY -274.054
251. ********************************************************************************
252. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
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253. JOINT LOAD
254. 128 FY -196.17
255. ********************************************************************************
256. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
257. JOINT LOAD
258. 128 FY -237.065 FZ 3.85
259. ********************************************************************************
260. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
261. JOINT LOAD
262. 128 FY 654 FZ -240
263. ********************************************************************************
264. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
265. JOINT LOAD
266. 128 FY -654 FZ 240
267. ********************************************************************************
268. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
269. JOINT LOAD
270. 128 FY -23.01 FZ -8.1
271. ********************************************************************************
272. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
273. JOINT LOAD
274. 128 FY 23.01 FZ 8.1
275. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 286 NUMBER OF MEMBERS 1
NUMBER OF PLATES 252 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 285

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 109/ 25/ 78 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 861
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 68 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.1/ 143456.9 MB

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276. CHANGE
277. SUPPORTS
278. 327 328 332 TO 334 337 338 344 TO 350 354 TO 360 364 TO 370 374 TO 380 384 -
279. 385 TO 390 394 TO 400 404 TO 410 414 TO 420 424 TO 430 433 TO 442 -
280. 1137 TO 1154 1167 TO 1175 1188 1259 TO 1282 1301 TO 1336 1385 TO 1398 1403 -
281. 1404 TO 1472 PLATE MAT DIRECT Y SUBGRADE 57000 COMPRESSION
282. ********************************************************************************
283. * UNFACTORED LOAD COMBINATIONS
284. LOAD 201 DL
```



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285. REPEAT LOAD
286. 1 1.0
287. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201


288. CHANGE
289. LOAD 202 DL+L
290. REPEAT LOAD
291. 1 1.0 3 1.0
292. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  202


293. CHANGE
294. LOAD 203 DL+LR
295. REPEAT LOAD
296. 1 1.0 2 1.0
297. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  203


298. CHANGE
299. LOAD 204 DL+0.75L+0.75LR
300. REPEAT LOAD
301. 1 1.0 2 0.75 3 0.75
302. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  204


303. CHANGE
304. LOAD 205 DL+WL1
305. REPEAT LOAD
306. 1 1.0 4 1.0
307. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

308. CHANGE
309. LOAD 206 DL+WL2
310. REPEAT LOAD
311. 1 1.0 5 1.0
312. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

313. CHANGE
314. LOAD 207 DL+0.75WL1+0.75L+0.75LR
315. REPEAT LOAD
316. 1 1.0 2 0.75 3 0.75 4 0.75
317. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

318. CHANGE
319. LOAD 208 DL+0.75WL2+0.75L+0.75LR
320. REPEAT LOAD
321. 1 1.0 2 0.75 3 0.75 5 0.75
322. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

323. CHANGE
324. LOAD 209 0.6DL+WL1
325. REPEAT LOAD
326. 1 0.6 4 1.0
327. PERFORM ANALYSIS

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 209

328. CHANGE
329. LOAD 210 0.6DL+WL2
330. REPEAT LOAD
331. 1 0.6 5 1.0
332. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

333. CHANGE
334. LOAD 211 0.6DL+0.7SL1
335. REPEAT LOAD
336. 1 0.6 6 0.7
337. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

338. CHANGE
339. LOAD 212 0.6DL+0.7SL2
340. REPEAT LOAD
341. 1 0.6 7 0.7
342. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

343. CHANGE
344. **
345. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
346. LOAD 101 1.4DL
347. REPEAT LOAD
348. 1 1.4
349. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

350. CHANGE
351. LOAD 102 1.4DL+1.7L+0.5LR
352. REPEAT LOAD
353. 1 1.4 3 1.7 2 0.5
354. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

355. CHANGE
356. LOAD 103 1.2DL+1.6LR+1.0L
357. REPEAT LOAD
358. 1 1.2 2 1.6 3 1.0
359. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

360. CHANGE
361. LOAD 104 1.2DL+1.6WL1+1.0L+0.5LR
362. REPEAT LOAD
363. 1 1.2 4 1.6 2 0.5 3 1.0
364. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 104

365. CHANGE
366. LOAD 105 1.2DL+1.6WL2+1.0L+0.5LR
367. REPEAT LOAD
368. 1 1.2 5 1.6 2 0.5 3 1.0
369. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

370. CHANGE
371. LOAD 106 1.2DL+1.0SL1+1.0L
372. REPEAT LOAD
373. 1 1.2 3 1.0 6 1.0
374. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

375. CHANGE
376. LOAD 107 1.2DL+1.0SL2+1.0L
377. REPEAT LOAD
378. 1 1.2 3 1.0 7 1.0
379. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

380. CHANGE
381. LOAD 108 0.9DL+1.6WL1
382. REPEAT LOAD
383. 1 0.9 4 1.6
384. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723205E+06 L-MATRIX DIAG= 7.4766576E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 7 iterations, Case= 108

385. CHANGE
386. LOAD 109 0.9DL+1.6WL2
387. REPEAT LOAD
388. 1 0.9 5 1.6
389. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

390. CHANGE
391. LOAD 110 0.9DL+1.0SL1
392. REPEAT LOAD
393. 1 0.9 6 1.0
394. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

395. CHANGE
396. LOAD 111 0.9DL+1.0SL2
397. REPEAT LOAD
398. 1 0.9 7 1.0
399. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

400. CHANGE
401. **
402. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723205E+06 L-MATRIX DIAG= 7.4766576E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9731959E+04 L-MATRIX DIAG= 2.1827873E-10 EQN NO 859
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723205E+06 L-MATRIX DIAG= 7.4720010E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.2506263E+06 L-MATRIX DIAG= 2.7138740E-06 EQN NO 861
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 6 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 6

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723205E+06 L-MATRIX DIAG= -2.9522926E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9731959E+04 L-MATRIX DIAG= 2.1827873E-10 EQN NO 859
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 4.6723205E+06 L-MATRIX DIAG= 7.4720010E-06 EQN NO 860
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1490 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.2506263E+06 L-MATRIX DIAG= 2.7138740E-06 EQN NO 861
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 7

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

```
 **START ITERATION NO.          2
 **START ITERATION NO.          3
 **START ITERATION NO.          4
 **START ITERATION NO.          5
 **START ITERATION NO.          6
 **START ITERATION NO.          7

 **NOTE-Tension/Compression converged after  7 iterations, Case=  209



 **NOTE-Tension/Compression converged after  1 iterations, Case=  210



 **NOTE-Tension/Compression converged after  1 iterations, Case=  211



 **NOTE-Tension/Compression converged after  1 iterations, Case=  212



 **NOTE-Tension/Compression converged after  1 iterations, Case=  101



 **NOTE-Tension/Compression converged after  1 iterations, Case=  102



 **NOTE-Tension/Compression converged after  1 iterations, Case=  103


 **START ITERATION NO.          2
 **START ITERATION NO.          3

 **NOTE-Tension/Compression converged after  3 iterations, Case=  104



 **NOTE-Tension/Compression converged after  1 iterations, Case=  105



 **NOTE-Tension/Compression converged after  1 iterations, Case=  106



 **NOTE-Tension/Compression converged after  1 iterations, Case=  107


 **START ITERATION NO.          2
 **START ITERATION NO.          3
 **START ITERATION NO.          4
```



```
 **START ITERATION NO.          6
 **START ITERATION NO.          7

***WARNING -  INSTABILITY AT JOINT  1490   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  4.6723205E+06 L-MATRIX DIAG=  7.4766576E-06 EQN NO     860
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


 **NOTE-Tension/Compression converged after  7 iterations, Case=  108



 **NOTE-Tension/Compression converged after  1 iterations, Case=  109



 **NOTE-Tension/Compression converged after  1 iterations, Case=  110



 **NOTE-Tension/Compression converged after  1 iterations, Case=  111


 403. LOAD LIST 101 TO 111
 404. START CONCRETE DESIGN
```

```
405. CODE ACI
406. TRACK 2 ALL
407. FC 35000 MEMB 185
408. CLB 0.05 MEMB 185
409. CLS 0.05 ALL
410. CLT 0.05 ALL
411. MAXMAIN 25 ALL
412. MINMAIN 25 ALL
413. MINSEC 10 ALL
414. DESIGN COLUMN 185
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 750.0 X 750.0 MMS, TIED
        AREA OF STEEL REQUIRED = 13848.8  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

 32 - 25 MM               2.793      108      STA     0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
22765.17  18212.14   7303.10   2874.61     393.6
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1981.83  -6498.10   -839.39      5.25   -0.00179
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
22765.17  18212.14   7303.10   2874.61     393.6
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1981.83  -6498.10   -839.39   1493.31   -0.50830
 ------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |             16811.20  1652.11  8405.60  2782.32
       P0 |*            15410.27  1918.51  7004.67  2870.69
          | *           14009.34  2151.63  5603.73  2825.47
   Pn,max|__*          12608.40  2344.09  4202.80  2716.83
          |   *         11207.47  2509.22  2801.87  2543.87
    Pn    |    *         9806.54  2653.85  1400.93  2302.13
  NOMINAL |     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL |      *       16811.20  1652.11  8405.60  2782.32
COMPRESSION|       *     15410.27  1918.51  7004.67  2870.69
       Pb|-------*Mb     14009.34  2151.63  5603.73  2825.47
          |      *       12608.40  2344.09  4202.80  2716.83
__________|____*_______11207.47  2509.22  2801.87  2543.87
          |  * M0   Mn, 9806.54  2653.85  1400.93  2302.13
          | *   BENDING
    P-tens|*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************


415. END CONCRETE DESIGN
```

416. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:19:10 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

```
            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    OCT  3, 2020                   *
            *           Time=     9:19:54                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F5_STABILITY CHECK.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 01/09/2020
     4. JOB NAME FDN. MKD. F5_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 913.4 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 0.75 ZD 0.75
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 2 FIXED
    32. ******************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 128 FX 1.5 FY -274.054
    37. ******************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE ROOF LIVE LOAD ( LR )
```



```
    39. JOINT LOAD
    40. 128 FY -196.17
    41. ******************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    43. JOINT LOAD
    44. 128 FY -237.065 FZ 3.85
    45. ******************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
    47. JOINT LOAD
    48. 128 FY 654 FZ -240
    49. ******************************************************************************
    50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
    51. JOINT LOAD
    52. 128 FY -654 FZ 240
    53. ******************************************************************************
    54. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
    55. JOINT LOAD
    56. 128 FY -23.01 FZ -8.1
    57. ******************************************************************************
    58. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
    59. JOINT LOAD
    60. 128 FY 23.01 FZ 8.1
    61. ******************************************************************************
    62. * UNFACTORED LOAD COMBINATIONS
    63. LOAD COMB 201 DL
    64. 1 1.0
    65. LOAD COMB 202 DL+L
    66. 1 1.0 3 1.0
    67. LOAD COMB 203 DL+LR
    68. 1 1.0 2 1.0
    69. LOAD COMB 204 DL+0.75L+0.75LR
    70. 1 1.0 2 0.75 3 0.75
    71. LOAD COMB 205 DL+WL1
    72. 1 1.0 4 1.0
    73. LOAD COMB 206 DL+WL2
    74. 1 1.0 5 1.0
    75. LOAD COMB 207 DL+0.75WL1+0.75L+0.75LR
    76. 1 1.0 2 0.75 3 0.75 4 0.75
    77. LOAD COMB 208 DL+0.75WL2+0.75L+0.75LR
    78. 1 1.0 2 0.75 3 0.75 5 0.75
    79. LOAD COMB 209 0.6DL+1.0WL1
    80. 1 0.6 4 1.0
    81. LOAD COMB 210 0.6DL+1.0WL2
    82. 1 0.6 5 1.0
    83. LOAD COMB 211 0.6DL+0.7SL1
    84. 1 0.6 6 0.7
    85. LOAD COMB 212 0.6DL+0.7SL2
    86. 1 0.6 7 0.7
    87. ******************************************************************************
    88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 12 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 143453.1 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:19:59 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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****

ATTACHMENT 11 STAAD OUTPUT – FOUNDATION F6

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 30, 2020                 *
*          Time=    14:16:38                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F6.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F6
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175
   14. 41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175
   15. 302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438
   16. 306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438
   17. 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438
   18. 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438
   19. 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438
   20. 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438
   21. 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438
   22. 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438
   23. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   24. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   25. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   26. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   27. 1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438
   28. 1554 -0.75 909.9 3.92438; 1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438
   29. 1560 -0.75 909.9 2.92438
   30. MEMBER INCIDENCES
   31. 185 2 128
   32. ELEMENT INCIDENCES SHELL
   33. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   34. 334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304
   35. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   36. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
   37. 357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388
   38. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
```



```
   39. 408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400
   40. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   41. 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413
   42. 428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548
   43. 1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556
   44. 1507 1549 34 410 1560
   45. START GROUP DEFINITION
   46. ELEMENT
   47. _RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
   48. 424 TO 428 1497 1499 1501 1503 1505 1507
   49. _RAFT_MOM 327 333 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 427 -
   50. 428 1497 1499 1501 1503 1505 1507
   51. _RAFT_SHEAR 334 348 354 TO 358 404 TO 408 418 428 1497 1499 1501 1503 1505 1507
   52. MEMBER
   53. END GROUP DEFINITION
   54. ELEMENT PROPERTY
   55. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
   56. 1497 1499 1501 1503 1505 1507 THICKNESS 0.4
   57. DEFINE MATERIAL START
   58. ISOTROPIC CONCRETE
   59. E 2.17185E+007
   60. POISSON 0.17
   61. DENSITY 25
   62. ALPHA 1E-005
   63. DAMP 0.05
   64. TYPE CONCRETE
   65. STRENGTH FCU 27579
   66. END DEFINE MATERIAL
   67. MEMBER PROPERTY AMERICAN
   68. 185 PRIS YD 0.6 ZD 0.6
   69. CONSTANTS
   70. MATERIAL CONCRETE ALL
   71. SUPPORTS
   72. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
   73. 1497 1499 1501 1503 1505 1507 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.031250
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.031211
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500

```
       327          0.031250
       333          0.031250
       334          0.031250
       335          0.031250
       336          0.031250
       337          0.031250
       338          0.015625
       388          0.031250
       389          0.031250
       390          0.031250
       391          0.031250
       392          0.031250
       393          0.015625
       399          0.062500
       400          0.062500
       401          0.062500
       402          0.062500
       403          0.062500
       404          0.031250
       410          0.062578
       411          0.062578
       412          0.062578
       413          0.062578
       414          0.062578
       415          0.031289
      1548          0.031211
      1549          0.031250
      1552          0.031250
      1554          0.015625
      1556          0.031250
      1557          0.015625
      1560          0.031289
     TOTAL          2.250000
 74. ********************************************************************************
 75. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
 76. SELFWEIGHT Y -1
 77. ELEMENT LOAD
 78. * LOAD DUE TO BACKFILL
 79. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
 80. 1497 1499 1501 1503 1505 1507 PR GY -52
 81. * LOAD DUE TO SURCHARGE
 82. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
 83. 1497 1499 1501 1503 1505 1507 PR GY -10
 84. JOINT LOAD
 85. 128 FY -137.6
 86. ********************************************************************************
 87. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
 88. JOINT LOAD
 89. 128 FY -307.229
 90. ********************************************************************************
 91. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
 92. JOINT LOAD
 93. 128 FY -1.522
 94. ********************************************************************************
 95. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
 96. JOINT LOAD
```



```
 97. 128 FY 1.522
 98. ********************************************************************************
 99. * UNFACTORED LOAD COMBINATIONS
100. LOAD COMB 201 DL
101. 1 1.0
102. LOAD COMB 202 DL+L
103. 1 1.0 2 1.0
104. LOAD COMB 203 DL+0.75L
105. 1 1.0 2 0.75
106. LOAD COMB 204 0.6DL+0.7SL1
107. 1 0.6 3 0.7
108. LOAD COMB 205 0.6DL+0.7SL2
109. 1 0.6 4 0.7
110. ********************************************************************************
111. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
112. LOAD COMB 101 1.4DL
113. 1 1.4
114. LOAD COMB 102 1.4DL+1.7L
115. 1 1.4 2 1.7
116. LOAD COMB 103 1.2DL+1.0SL1+1.0L
117. 1 1.2 2 1.0 3 1.0
118. LOAD COMB 104 1.2DL+1.0SL2+1.0L
119. 1 1.2 2 1.0 4 1.0
120. LOAD COMB 105 0.9DL+1.0SL1
121. 1 0.9 3 1.0
122. LOAD COMB 106 0.9DL+1.0SL2
123. 1 0.9 4 1.0
124. ********************************************************************************
125. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS         50  NUMBER OF MEMBERS       1
  NUMBER OF PLATES         36  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS     49


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=    47/    13/     45 DOF
TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =     153
TOTAL LOAD COMBINATION  CASES =    11  SO FAR.
  SIZE OF STIFFNESS MATRIX =        7 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.3/ 143560.5 MB



126. LOAD LIST 101 TO 106
127. START CONCRETE DESIGN
```

```
128. CODE ACI
129. TRACK 2 ALL
130. FC 35000 MEMB 185
131. CLB 0.05 MEMB 185
132. CLS 0.05 ALL
133. CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
134. 415 TO 418 424 TO 428
135. MAXMAIN 25 ALL
136. MINMAIN 25 ALL
137. MINSEC 10 ALL
138. DESIGN COLUMN 185
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 25 MM               1.091      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    296.37      0.00    0.00000
 ------------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53    296.37      0.00    0.00000
 ------------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              9022.30   709.00  4511.15  1061.99
       P0 |*             8270.44   812.98  3759.29  1019.91
          | *            7518.58   897.94  3007.43   956.53
   Pn,max|__*           6766.73   962.48  2255.58   875.37
          |   *          6014.87  1009.20  1503.72   752.23
    Pn    |    *         5263.01  1041.19   751.86   595.66
  NOMINAL |     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL |      *       9022.30   709.00  4511.15  1061.99
COMPRESSION|       *     8270.44   812.98  3759.29  1019.91
        Pb|-------*Mb    7518.58   897.94  3007.43   956.53
          |      *       6766.73   962.48  2255.58   875.37
__________|____*_______  6014.87  1009.20  1503.72   752.23
          |  * M0   Mn,  5263.01  1041.19   751.86   595.66
          | *   BENDING
    P-tens|*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************
```

```
139. END CONCRETE DESIGN
140. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= SEP 30,2020   TIME= 14:16:43 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
```

```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  3, 2020                 *
*          Time=     9:21:28                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F6 - STABILITY.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F6
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 913.4 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 PRIS YD 0.6 ZD 0.6
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 FIXED
   32. **************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FY -137.6
   37. **************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
```



```
   39. JOINT LOAD
   40. 128 FY -307.229
   41. **************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
   43. JOINT LOAD
   44. 128 FY -1.522
   45. **************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
   47. JOINT LOAD
   48. 128 FY 1.522
   49. **************************************************************************
   50. * UNFACTORED LOAD COMBINATIONS
   51. LOAD COMB 201 DL
   52. 1 1.0
   53. LOAD COMB 202 DL+L
   54. 1 1.0 2 1.0
   55. LOAD COMB 203 DL+0.75L
   56. 1 1.0 2 0.75
   57. LOAD COMB 204 0.6DL+0.7SL1
   58. 1 0.6 3 0.7
   59. LOAD COMB 205 0.6DL+0.7SL2
   60. 1 0.6 4 0.7
   61. **************************************************************************
   62. PERFORM ANALYSIS
```

```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          2  NUMBER OF MEMBERS       1
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      1


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
 TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =       6
 TOTAL LOAD COMBINATION  CASES =     5  SO FAR.
   SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.0/ 143452.9 MB
```

```
   63. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:21:32 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
****

C01

ATTACHMENT 12 STAAD OUTPUT – FOUNDATION F6A

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 30, 2020                 *
*          Time=    14:13:48                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F6A.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F6A
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175
   14. 41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175
   15. 302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438
   16. 306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438
   17. 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438
   18. 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438
   19. 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438
   20. 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438
   21. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   22. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   23. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   24. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   25. 1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438
   26. 1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438; 1560 -0.75 909.9 2.92438
   27. 1561 -0.5 909.9 2.17438; 1562 -0.25 909.9 2.17438; 1563 -0.25 909.9 1.92438
   28. 1564 -0.5 909.9 1.92438; 1565 0 909.9 2.17438; 1566 0 909.9 1.92438
   29. 1567 0.25 909.9 2.17438; 1568 0.25 909.9 1.92438; 1569 0.5 909.9 2.17438
   30. 1570 0.5 909.9 1.92438; 1571 0.75 909.9 2.17438; 1572 0.75 909.9 1.92438
   31. 1573 -0.75 909.9 2.17438; 1574 -0.75 909.9 1.92438; 1575 -0.5 909.9 1.67438
   32. 1576 -0.25 909.9 1.67438; 1577 -0.25 909.9 1.42438; 1578 -0.5 909.9 1.42438
   33. 1579 0 909.9 1.67438; 1580 0 909.9 1.42438; 1581 0.25 909.9 1.67438
   34. 1582 0.25 909.9 1.42438; 1583 0.5 909.9 1.67438; 1584 0.5 909.9 1.42438
   35. 1585 0.75 909.9 1.67438; 1586 0.75 909.9 1.42438; 1587 -0.75 909.9 1.67438
   36. 1588 -0.75 909.9 1.42438; 1589 -0.5 909.9 1.17438; 1590 -0.25 909.9 1.17438
   37. 1591 -0.25 909.9 0.92438; 1592 -0.5 909.9 0.92438; 1593 0 909.9 1.17438
   38. 1594 0 909.9 0.92438; 1595 0.25 909.9 1.17438; 1596 0.25 909.9 0.92438
```



```
   39. 1597 0.5 909.9 1.17438; 1598 0.5 909.9 0.92438; 1599 0.75 909.9 1.17438
   40. 1600 0.75 909.9 0.92438; 1601 -0.75 909.9 1.17438; 1602 -0.75 909.9 0.92438
   41. 1603 -0.5 909.9 0.67438; 1604 -0.25 909.9 0.67438; 1605 -0.25 909.9 0.47538
   42. 1606 -0.5 909.9 0.47538; 1607 0 909.9 0.67438; 1608 0 909.9 0.47538
   43. 1609 0.25 909.9 0.67438; 1610 0.25 909.9 0.47538; 1611 0.5 909.9 0.67438
   44. 1612 0.5 909.9 0.47538; 1613 0.75 909.9 0.67438; 1614 0.75 909.9 0.47538
   45. 1615 -0.75 909.9 0.67438; 1616 -0.75 909.9 0.47538; 1617 -0.5 909.9 0.22538
   46. 1618 -0.25 909.9 0.22538; 1619 -0.25 909.9 -0.0246199
   47. 1620 -0.5 909.9 -0.0246199; 1621 0 909.9 0.22538; 1622 0 909.9 -0.0246199
   48. 1623 0.25 909.9 0.22538; 1624 0.25 909.9 -0.0246199; 1625 0.5 909.9 0.22538
   49. 1626 0.5 909.9 -0.0246199; 1627 0.75 909.9 0.22538; 1628 0.75 909.9 -0.0246199
   50. 1629 -0.75 909.9 0.22538; 1630 -0.75 909.9 -0.0246199; 1631 0 913.4 0.47538
   51. 1632 1 909.9 3.42438; 1633 1 909.9 3.175; 1634 1 909.9 3.67438
   52. 1635 1 909.9 2.67438; 1636 1 909.9 2.42438; 1637 1 909.9 2.92438
   53. 1638 1 909.9 2.17438; 1639 1 909.9 1.92438; 1640 1 909.9 1.67438
   54. 1641 1 909.9 1.42438; 1642 1 909.9 1.17438; 1643 1 909.9 0.92438
   55. 1644 1 909.9 0.67438; 1645 1 909.9 0.47538; 1646 1 909.9 0.22538
   56. 1647 1 909.9 -0.0246199; 1648 -1 909.9 3.42438; 1649 -1 909.9 3.175
   57. 1650 -1 909.9 3.67438; 1651 -1 909.9 2.67438; 1652 -1 909.9 2.42438
   58. 1653 -1 909.9 2.92438; 1654 -1 909.9 2.17438; 1655 -1 909.9 1.92438
   59. 1656 -1 909.9 1.67438; 1657 -1 909.9 1.42438; 1658 -1 909.9 1.17438
   60. 1659 -1 909.9 0.92438; 1660 -1 909.9 0.67438; 1661 -1 909.9 0.47538
   61. 1662 -1 909.9 0.22538; 1663 -1 909.9 -0.0246199; 1664 0 913.4 3.675
   62. MEMBER INCIDENCES
   63. 185 2 128; 1568 1608 1631; 1599 128 1664
   64. ELEMENT INCIDENCES SHELL
   65. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   66. 334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304
   67. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   68. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
   69. 407 402 403 392 391; 408 403 404 393 392; 414 410 411 400 399
   70. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
   71. 418 414 415 404 403; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
   72. 427 41 42 414 413; 428 42 6 415 414; 1497 1548 302 34 1549
   73. 1499 1552 322 302 1548; 1503 1556 399 388 1557; 1505 1560 410 399 1556
   74. 1507 1549 34 410 1560; 1508 1561 1562 1563 1564; 1509 1562 1565 1566 1563
   75. 1510 1565 1567 1568 1566; 1511 1567 1569 1570 1568; 1512 1569 1571 1572 1570
   76. 1513 388 389 1562 1561; 1514 389 390 1565 1562; 1515 390 391 1567 1565
   77. 1516 391 392 1569 1567; 1517 392 393 1571 1569; 1518 1573 1561 1564 1574
   78. 1519 1557 388 1561 1573; 1520 1575 1576 1577 1578; 1521 1576 1579 1580 1577
   79. 1522 1579 1581 1582 1580; 1523 1581 1583 1584 1582; 1524 1583 1585 1586 1584
   80. 1525 1564 1563 1576 1575; 1526 1563 1566 1579 1576; 1527 1566 1568 1581 1579
   81. 1528 1568 1570 1583 1581; 1529 1570 1572 1585 1583; 1530 1587 1575 1578 1588
   82. 1531 1574 1564 1575 1587; 1532 1589 1590 1591 1592; 1533 1590 1593 1594 1591
   83. 1534 1593 1595 1596 1594; 1535 1595 1597 1598 1596; 1536 1597 1599 1600 1598
   84. 1537 1578 1577 1590 1589; 1538 1577 1580 1593 1590; 1539 1580 1582 1595 1593
   85. 1540 1582 1584 1597 1595; 1541 1584 1586 1599 1597; 1542 1601 1589 1592 1602
   86. 1543 1588 1578 1589 1601; 1544 1603 1604 1605 1606; 1545 1604 1607 1608 1605
   87. 1546 1607 1609 1610 1608; 1547 1609 1611 1612 1610; 1548 1611 1613 1614 1612
   88. 1549 1592 1591 1604 1603; 1550 1591 1594 1607 1604; 1551 1594 1596 1609 1607
   89. 1552 1596 1598 1611 1609; 1553 1598 1600 1613 1611; 1554 1615 1603 1606 1616
   90. 1555 1602 1592 1603 1615; 1556 1617 1618 1619 1620; 1557 1618 1621 1622 1619
   91. 1558 1621 1623 1624 1622; 1559 1623 1625 1626 1624; 1560 1625 1627 1628 1626
   92. 1561 1606 1605 1618 1617; 1562 1605 1608 1621 1618; 1563 1608 1610 1623 1621
   93. 1564 1610 1612 1625 1623; 1565 1612 1614 1627 1625; 1566 1629 1617 1620 1630
   94. 1567 1616 1606 1617 1629; 1569 308 1632 1633 6; 1570 327 1634 1632 308
```

```
 95. 1571 404 1635 1636 393; 1572 415 1637 1635 404; 1573 6 1633 1637 415
 96. 1574 1571 1638 1639 1572; 1575 393 1636 1638 1571; 1576 1585 1640 1641 1586
 97. 1577 1572 1639 1640 1585; 1578 1599 1642 1643 1600; 1579 1586 1641 1642 1599
 98. 1580 1613 1644 1645 1614; 1581 1600 1643 1644 1613; 1582 1627 1646 1647 1628
 99. 1583 1614 1645 1646 1627; 1584 1648 1548 1549 1649; 1585 1650 1552 1548 1648
100. 1586 1651 1556 1557 1652; 1587 1653 1560 1556 1651; 1588 1649 1549 1560 1653
101. 1589 1654 1573 1574 1655; 1590 1652 1557 1573 1654; 1591 1656 1587 1588 1657
102. 1592 1655 1574 1587 1656; 1593 1658 1601 1602 1659; 1594 1657 1588 1601 1658
103. 1595 1660 1615 1616 1661; 1596 1659 1602 1615 1660; 1597 1662 1629 1630 1663
104. 1598 1661 1616 1629 1662
105. START GROUP DEFINITION
106. ELEMENT
107. _RAFT 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 -
108. 1499 1503 1505 1507 TO 1567 1569 TO 1598
109. _RAFT_MOM 327 333 334 344 TO 348 404 TO 408 414 TO 418 424 427 428 1497 1499 -
110. 1503 1505 1507 TO 1544 1547 TO 1561 1564 TO 1567 1569 TO 1598
111. _RAFT_SHEAR 404 TO 408 1503 1508 TO 1543 1569 TO 1598
112. MEMBER
113. END GROUP DEFINITION
114. ELEMENT PROPERTY
115. 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
116. 1505 1507 TO 1567 1569 TO 1598 THICKNESS 0.4
117. DEFINE MATERIAL START
118. ISOTROPIC CONCRETE
119. E 2.17185E+007
120. POISSON 0.17
121. DENSITY 25
122. ALPHA 1E-005
123. DAMP 0.05
124. TYPE CONCRETE
125. STRENGTH FCU 27579
126. END DEFINE MATERIAL
127. MEMBER PROPERTY AMERICAN
128. 185 1568 1599 PRIS YD 0.6 ZD 0.6
129. CONSTANTS
130. MATERIAL CONCRETE ALL
131. SUPPORTS
132. 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
133. 1505 1507 TO 1567 1569 TO 1598 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
322	0.031250

323	0.031250
324	0.031250
325	0.031250
326	0.031250
327	0.031250
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
1548	0.062422
1549	0.062500
1552	0.031250
1556	0.062500
1557	0.062500
1560	0.062578
1561	0.062500
1562	0.062500
1563	0.062500
1564	0.062500
1565	0.062500
1566	0.062500
1567	0.062500
1568	0.062500
1569	0.062500
1570	0.062500
1571	0.062500
1572	0.062500
1573	0.062500
1574	0.062500
1575	0.062500
1576	0.062500
1577	0.062500
1578	0.062500
1579	0.062500
1580	0.062500
1581	0.062500
1582	0.062500
1583	0.062500
1584	0.062500
1585	0.062500
1586	0.062500
1587	0.062500

1588	0.062500
1589	0.062500
1590	0.062500
1591	0.062500
1592	0.062500
1593	0.062500
1594	0.062500
1595	0.062500
1596	0.062500
1597	0.062500
1598	0.062500
1599	0.062500
1600	0.062500
1601	0.062500
1602	0.062500
1603	0.056125
1604	0.056125
1605	0.056125
1606	0.056125
1607	0.056125
1608	0.056125
1609	0.056125
1610	0.056125
1611	0.056125
1612	0.056125
1613	0.056125
1614	0.056125
1615	0.056125
1616	0.056125
1617	0.062500
1618	0.062500
1619	0.031250
1620	0.031250
1621	0.062500
1622	0.031250
1623	0.062500
1624	0.031250
1625	0.062500
1626	0.031250
1627	0.062500
1628	0.031250
1629	0.062500
1630	0.031250
1632	0.031211
1633	0.031250
1634	0.015625
1635	0.031250
1636	0.031250
1637	0.031289
1638	0.031250
1639	0.031250
1640	0.031250
1641	0.031250
1642	0.031250
1643	0.031250
1644	0.028063

1645	0.028063
1646	0.031250
1647	0.015625
1648	0.031211
1649	0.031250
1650	0.015625
1651	0.031250
1652	0.031250
1653	0.031289
1654	0.031250
1655	0.031250
1656	0.031250
1657	0.031250
1658	0.031250
1659	0.031250
1660	0.028063
1661	0.028063
1662	0.031250
1663	0.015625
TOTAL	7.398002

```
134. ********************************************************************************
135. MEMBER OFFSET
136. 1599 START 0 -0.3 0
137. 1599 END 0 -0.3 0
138. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
139. SELFWEIGHT Y -1
140. ELEMENT LOAD
141. * LOAD DUE TO BACKFILL
142. 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
143. 1505 1507 TO 1567 1569 TO 1598 PR GY -52
144. * LOAD DUE TO SURCHARGE
145. 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 TO 428 1497 1499 1503 -
146. 1505 1507 TO 1567 1569 TO 1598 PR GY -10
147. JOINT LOAD
148. 1664 FY -137.6
149. 1631 FX 4.58 FY -87.1
150. ********************************************************************************
151. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
152. JOINT LOAD
153. 1664 FY -307.229
154. 1631 FX 10 FY -195
155. ********************************************************************************
156. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
157. JOINT LOAD
158. 1664 FY -1.522
159. 1631 FX 0.2 FY -0.48
160. ********************************************************************************
161. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
162. JOINT LOAD
163. 1664 FY 1.522
164. 1631 FX 0.2 FY -0.48
165. ********************************************************************************
166. * UNFACTORED LOAD COMBINATIONS
167. LOAD COMB 201 DL
168. 1 1.0
169. LOAD COMB 202 DL+L
```

```
170. 1 1.0 2 1.0
171. LOAD COMB 203 DL+0.75L
172. 1 1.0 2 0.75
173. LOAD COMB 204 0.6DL+0.7SL1
174. 1 0.6 3 0.7
175. LOAD COMB 205 0.6DL+0.7SL2
176. 1 0.6 4 0.7
177. ********************************************************************************
178. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
179. LOAD COMB 101 1.4DL
180. 1 1.4
181. LOAD COMB 102 1.4DL+1.7L
182. 1 1.4 2 1.7
183. LOAD COMB 103 1.2DL+1.0SL1+1.0L
184. 1 1.2 2 1.0 3 1.0
185. LOAD COMB 104 1.2DL+1.0SL2+1.0L
186. 1 1.2 2 1.0 4 1.0
187. LOAD COMB 105 0.9DL+1.0SL1
188. 1 0.9 3 1.0
189. LOAD COMB 106 0.9DL+1.0SL2
190. 1 0.9 4 1.0
191. ********************************************************************************
192. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**147**	**NUMBER OF MEMBERS**	**3**
NUMBER OF PLATES	**120**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**144**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 140/ 16/ 51 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 450
TOTAL LOAD COMBINATION CASES = 11 SO FAR.
SIZE OF STIFFNESS MATRIX = 23 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 143563.1 MB

```
193. LOAD LIST 101 TO 106
194. START CONCRETE DESIGN
```



```
195. CODE ACI
196. TRACK 2 MEMB 185 1568
197. FC 35000 MEMB 185 1568
198. CLB 0.05 MEMB 185 1568
199. CLS 0.05 MEMB 185 1568
200. CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 404 TO 408 414 TO 418 424 -
201. 425 TO 428 1508 TO 1517 1520 TO 1529 1532 TO 1541 1544 TO 1553 1556 TO 1565 -
202. 1569 TO 1583
203. MAXMAIN 25 MEMB 185 1568
204. MINMAIN 25 MEMB 185 1568
205. MINSEC 10 MEMB 185 1568
206. DESIGN COLUMN 185 1568
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	306.06	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	306.06	150.61	0.14060

	Pn	Mn	Pn	Mn	(@ Z)
	9022.30	709.00	4511.15	1061.99	
P0	8270.44	812.98	3759.29	1019.91	
	7518.58	897.94	3007.43	956.53	
Pn,max	6766.73	962.48	2255.58	875.37	
	6014.87	1009.20	1503.72	752.23	
Pn	5263.01	1041.19	751.86	595.66	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	9022.30	709.00	4511.15	1061.99	
COMPRESSION	8270.44	812.98	3759.29	1019.91	
Pb	7518.58	897.94	3007.43	956.53	
	6766.73	962.48	2255.58	875.37	
	6014.87	1009.20	1503.72	752.23	
	5263.01	1041.19	751.86	595.66	
	M0 Mn,				
	BENDING				
P-tens	MOMENT				

COLUMN NO. 1568 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	187.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	187.60	0.00	0.00000

	Pn	Mn	Pn	Mn	(@ Z)
	9022.30	709.00	4511.15	1061.99	
P0	8270.44	812.98	3759.29	1019.91	
	7518.58	897.94	3007.43	956.53	
Pn,max	6766.73	962.48	2255.58	875.37	
	6014.87	1009.20	1503.72	752.23	
Pn	5263.01	1041.19	751.86	595.66	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	9022.30	709.00	4511.15	1061.99	
COMPRESSION	8270.44	812.98	3759.29	1019.91	
Pb	7518.58	897.94	3007.43	956.53	
	6766.73	962.48	2255.58	875.37	
	6014.87	1009.20	1503.72	752.23	
	5263.01	1041.19	751.86	595.66	
	M0 Mn,				
	BENDING				
P-tens	MOMENT				

**********END OF COLUMN DESIGN RESULTS**********

```
207. END CONCRETE DESIGN
208. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= SEP 30,2020   TIME= 14:13:54 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
```

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    OCT  3, 2020                *
            *           Time=     9:24:24                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F6A - STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 01/09/2020
     4. JOB NAME FDN. MKD. F6A_STABILITY
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 913.4 3.175; 1608 0 909.9 0.47538; 1631 0 913.4 0.47538
    14. 1664 0 913.4 3.675
    15. MEMBER INCIDENCES
    16. 185 2 128; 1568 1608 1631; 1599 128 1664
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 25
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 185 1568 1599 PRIS YD 0.6 ZD 0.6
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 1608 FIXED
    33. ******************************************************************************
    34. MEMBER OFFSET
    35. 1599 START 0 -0.3 0
    36. 1599 END 0 -0.3 0
    37. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
```



```
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    38. SELFWEIGHT Y -1
    39. JOINT LOAD
    40. 1664 FY -137.6
    41. 1631 FX 4.58 FY -87.1
    42. ******************************************************************************
    43. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    44. JOINT LOAD
    45. 1664 FY -307.229
    46. 1631 FX 10 FY -195
    47. ******************************************************************************
    48. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
    49. JOINT LOAD
    50. 1664 FY -1.522
    51. 1631 FX 0.2 FY -0.48
    52. ******************************************************************************
    53. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
    54. JOINT LOAD
    55. 1664 FY 1.522
    56. 1631 FX 0.2 FY -0.48
    57. ******************************************************************************
    58. * UNFACTORED LOAD COMBINATIONS
    59. LOAD COMB 201 DL
    60. 1 1.0
    61. LOAD COMB 202 DL+L
    62. 1 1.0 2 1.0
    63. LOAD COMB 203 DL+0.75L
    64. 1 1.0 2 0.75
    65. LOAD COMB 204 0.6DL+0.7SL1
    66. 1 0.6 3 0.7
    67. LOAD COMB 205 0.6DL+0.7SL2
    68. 1 0.6 4 0.7
    69. ******************************************************************************
    70. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	5	NUMBER OF MEMBERS	3
NUMBER OF PLATES	0	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
 ORIGINAL/FINAL BAND-WIDTH=     3/     3/     18 DOF
TOTAL      PRIMARY LOAD CASES =      4, TOTAL DEGREES OF FREEDOM =      18
TOTAL LOAD COMBINATION  CASES =      5  SO FAR.
  SIZE OF STIFFNESS MATRIX =         1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/ 143452.7 MB
```

71. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:24:28 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*              http://www.bentley.com                      *
************************************************************
```

C01

ATTACHMENT 13 STAAD OUTPUT – FOUNDATION F7

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 30, 2020                 *
*          Time=    14:16: 8                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F7.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F7
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175
   14. 41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 913.4 3.175
   15. 302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438
   16. 306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438
   17. 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438
   18. 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438
   19. 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438; 335 0 909.9 3.92438
   20. 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438; 338 0.75 909.9 3.92438
   21. 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438
   22. 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438
   23. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   24. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   25. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   26. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   27. 1548 -0.75 909.9 3.42438; 1549 -0.75 909.9 3.175; 1552 -0.75 909.9 3.67438
   28. 1554 -0.75 909.9 3.92438; 1556 -0.75 909.9 2.67438; 1557 -0.75 909.9 2.42438
   29. 1560 -0.75 909.9 2.92438; 1561 -0.5 909.9 4.17438; 1562 -0.25 909.9 4.17438
   30. 1563 0 909.9 4.17438; 1564 0.25 909.9 4.17438; 1565 0.5 909.9 4.17438
   31. 1566 0.75 909.9 4.17438; 1567 -0.5 909.9 4.42438; 1568 -0.25 909.9 4.42438
   32. 1569 0 909.9 4.42438; 1570 0.25 909.9 4.42438; 1571 0.5 909.9 4.42438
   33. 1572 0.75 909.9 4.42438; 1573 -0.5 909.9 4.67438; 1574 -0.25 909.9 4.67438
   34. 1575 0 909.9 4.67438; 1576 0.25 909.9 4.67438; 1577 0.5 909.9 4.67438
   35. 1578 0.75 909.9 4.67438; 1579 -0.75 909.9 4.17438; 1580 -0.75 909.9 4.42438
   36. 1581 -0.75 909.9 4.67438; 1582 -0.5 909.9 1.92438; 1583 -0.25 909.9 1.92438
   37. 1584 -0.25 909.9 1.67438; 1585 -0.5 909.9 1.67438; 1586 0 909.9 1.92438
   38. 1587 0 909.9 1.67438; 1588 0.25 909.9 1.92438; 1589 0.25 909.9 1.67438
```



```
   39. 1590 0.5 909.9 1.92438; 1591 0.5 909.9 1.67438; 1592 0.75 909.9 1.92438
   40. 1593 0.75 909.9 1.67438; 1594 -0.5 909.9 2.17438; 1595 -0.25 909.9 2.17438
   41. 1596 0 909.9 2.17438; 1597 0.25 909.9 2.17438; 1598 0.5 909.9 2.17438
   42. 1599 0.75 909.9 2.17438; 1600 -0.75 909.9 1.92438; 1601 -0.75 909.9 1.67438
   43. 1602 -0.75 909.9 2.17438; 1603 -0.5 909.9 4.92438; 1604 -0.25 909.9 4.92438
   44. 1605 0 909.9 4.92438; 1606 0.25 909.9 4.92438; 1607 0.5 909.9 4.92438
   45. 1608 0.75 909.9 4.92438; 1609 -0.75 909.9 4.92438; 1610 -0.25 909.9 1.42438
   46. 1611 -0.5 909.9 1.42438; 1612 0 909.9 1.42438; 1613 0.25 909.9 1.42438
   47. 1614 0.5 909.9 1.42438; 1615 0.75 909.9 1.42438; 1616 -0.75 909.9 1.42438
   48. 1617 0.925 909.9 3.42438; 1618 0.925 909.9 3.175; 1619 0.925 909.9 3.67438
   49. 1620 0.925 909.9 3.92438; 1621 0.925 909.9 2.67438; 1622 0.925 909.9 2.42438
   50. 1623 0.925 909.9 2.92438; 1624 0.925 909.9 4.17438; 1625 0.925 909.9 4.42438
   51. 1626 0.925 909.9 4.67438; 1627 0.925 909.9 1.92438; 1628 0.925 909.9 1.67438
   52. 1629 0.925 909.9 2.17438; 1630 0.925 909.9 4.92438; 1631 0.925 909.9 1.42438
   53. 1632 -0.925 909.9 3.42438; 1633 -0.925 909.9 3.175; 1634 -0.925 909.9 3.67438
   54. 1635 -0.925 909.9 3.92438; 1636 -0.925 909.9 2.67438
   55. 1637 -0.925 909.9 2.42438; 1638 -0.925 909.9 2.92438
   56. 1639 -0.925 909.9 4.17438; 1640 -0.925 909.9 4.42438
   57. 1641 -0.925 909.9 4.67438; 1642 -0.925 909.9 1.92438
   58. 1643 -0.925 909.9 1.67438; 1644 -0.925 909.9 2.17438
   59. 1645 -0.925 909.9 4.92438; 1646 -0.925 909.9 1.42438
   60. MEMBER INCIDENCES
   61. 185 2 128
   62. ELEMENT INCIDENCES SHELL
   63. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   64. 334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304
   65. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   66. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
   67. 357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388
   68. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
   69. 408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400
   70. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   71. 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413
   72. 428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548
   73. 1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556
   74. 1507 1549 34 410 1560; 1508 1561 1562 334 333; 1509 1562 1563 335 334
   75. 1510 1563 1564 336 335; 1511 1564 1565 337 336; 1512 1565 1566 338 337
   76. 1513 1567 1568 1562 1561; 1514 1568 1569 1563 1562; 1515 1569 1570 1564 1563
   77. 1516 1570 1571 1565 1564; 1517 1571 1572 1566 1565; 1518 1573 1574 1568 1567
   78. 1519 1574 1575 1569 1568; 1520 1575 1576 1570 1569; 1521 1576 1577 1571 1570
   79. 1522 1577 1578 1572 1571; 1523 1579 1561 333 1554; 1524 1580 1567 1561 1579
   80. 1525 1581 1573 1567 1580; 1526 1582 1583 1584 1585; 1527 1583 1586 1587 1584
   81. 1528 1586 1588 1589 1587; 1529 1588 1590 1591 1589; 1530 1590 1592 1593 1591
   82. 1531 1594 1595 1583 1582; 1532 1595 1596 1586 1583; 1533 1596 1597 1588 1586
   83. 1534 1597 1598 1590 1588; 1535 1598 1599 1592 1590; 1536 388 389 1595 1594
   84. 1537 389 390 1596 1595; 1538 390 391 1597 1596; 1539 391 392 1598 1597
   85. 1540 392 393 1599 1598; 1541 1600 1582 1585 1601; 1542 1602 1594 1582 1600
   86. 1543 1557 388 1594 1602; 1544 1603 1604 1574 1573; 1545 1604 1605 1575 1574
   87. 1546 1605 1606 1576 1575; 1547 1606 1607 1577 1576; 1548 1607 1608 1578 1577
   88. 1549 1609 1603 1573 1581; 1550 1585 1584 1610 1611; 1551 1584 1587 1612 1610
   89. 1552 1587 1589 1613 1612; 1553 1589 1591 1614 1613; 1554 1591 1593 1615 1614
   90. 1555 1601 1585 1611 1616; 1556 308 1617 1618 6; 1557 327 1619 1617 308
   91. 1558 338 1620 1619 327; 1559 404 1621 1622 393; 1560 415 1623 1621 404
   92. 1561 6 1618 1623 415; 1562 1566 1624 1620 338; 1563 1572 1625 1624 1566
   93. 1564 1578 1626 1625 1572; 1565 1592 1627 1628 1593; 1566 1599 1629 1627 1592
   94. 1567 393 1622 1629 1599; 1568 1608 1630 1626 1578; 1569 1593 1628 1631 1615
```

```
 95. 1570 1632 1548 1549 1633; 1571 1634 1552 1548 1632; 1572 1635 1554 1552 1634
 96. 1573 1636 1556 1557 1637; 1574 1638 1560 1556 1636; 1575 1633 1549 1560 1638
 97. 1576 1639 1579 1554 1635; 1577 1640 1580 1579 1639; 1578 1641 1581 1580 1640
 98. 1579 1642 1600 1601 1643; 1580 1644 1602 1600 1642; 1581 1637 1557 1602 1644
 99. 1582 1645 1609 1581 1641; 1583 1643 1601 1616 1646
100. START GROUP DEFINITION
101. ELEMENT
102. _RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
103. 424 TO 428 1497 1499 1501 1503 1505 1507 TO 1583
104. _RAFT_MOM 334 348 354 TO 358 404 TO 408 418 428 1497 1499 1501 1503 1505 1507 -
105. 1508 TO 1583
106. _RAFT_SHEAR 1508 TO 1583
107. MEMBER
108. END GROUP DEFINITION
109. ELEMENT PROPERTY
110. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
111. 1497 1499 1501 1503 1505 1507 TO 1583 THICKNESS 0.4
112. DEFINE MATERIAL START
113. ISOTROPIC CONCRETE
114. E 2.17185E+007
115. POISSON 0.17
116. DENSITY 25
117. ALPHA 1E-005
118. DAMP 0.05
119. TYPE CONCRETE
120. STRENGTH FCU 27579
121. END DEFINE MATERIAL
122. MEMBER PROPERTY AMERICAN
123. 185 PRIS YD 0.6 ZD 0.6
124. CONSTANTS
125. MATERIAL CONCRETE ALL
126. SUPPORTS
127. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
128. 1497 1499 1501 1503 1505 1507 TO 1582 -
129. 1583 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.053125
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.053059
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500

327	0.053125
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.053125
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.053125
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.053125
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.053191
1548	0.053059
1549	0.053125
1552	0.053125
1554	0.053125
1556	0.053125
1557	0.053125
1560	0.053191
1561	0.062500
1562	0.062500
1563	0.062500
1564	0.062500
1565	0.062500
1566	0.053125
1567	0.062500
1568	0.062500
1569	0.062500
1570	0.062500
1571	0.062500
1572	0.053125
1573	0.062500
1574	0.062500
1575	0.062500
1576	0.062500
1577	0.062500
1578	0.053125
1579	0.053125
1580	0.053125
1581	0.053125
1582	0.062500
1583	0.062500
1584	0.062500

1585	0.062500
1586	0.062500
1587	0.062500
1588	0.062500
1589	0.062500
1590	0.062500
1591	0.062500
1592	0.053125
1593	0.053125
1594	0.062500
1595	0.062500
1596	0.062500
1597	0.062500
1598	0.062500
1599	0.053125
1600	0.053125
1601	0.053125
1602	0.053125
1603	0.031250
1604	0.031250
1605	0.031250
1606	0.031250
1607	0.031250
1608	0.026562
1609	0.026562
1610	0.031250
1611	0.031250
1612	0.031250
1613	0.031250
1614	0.031250
1615	0.026562
1616	0.026562
1617	0.021848
1618	0.021875
1619	0.021875
1620	0.021875
1621	0.021875
1622	0.021875
1623	0.021902
1624	0.021875
1625	0.021875
1626	0.021875
1627	0.021875
1628	0.021875
1629	0.021875
1630	0.010937
1631	0.010937
1632	0.021848
1633	0.021875
1634	0.021875
1635	0.021875
1636	0.021875
1637	0.021875
1638	0.021902
1639	0.021875
1640	0.021875

1641	0.021875
1642	0.021875
1643	0.021875
1644	0.021875
1645	0.010937
1646	0.010937
TOTAL	6.474998

```
130. ********************************************************************************
131. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
132. SELFWEIGHT Y -1
133. ELEMENT LOAD
134. * LOAD DUE TO BACKFILL
135. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
136. 1497 1499 1501 1503 1505 1507 TO 1583 PR GY -52
137. * LOAD DUE TO SURCHARGE
138. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
139. 1497 1499 1501 1503 1505 1507 TO 1583 PR GY -10
140. JOINT LOAD
141. 128 FY -199 FZ 5.25
142. ********************************************************************************
143. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
144. JOINT LOAD
145. 128 FY -476.25 FZ 12.5
146. ********************************************************************************
147. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
148. JOINT LOAD
149. 128 FY -9.512
150. ********************************************************************************
151. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
152. JOINT LOAD
153. 128 FY 9.512
154. ********************************************************************************
155. * UNFACTORED LOAD COMBINATIONS
156. LOAD COMB 201 DL
157. 1 1.0
158. LOAD COMB 202 DL+L
159. 1 1.0 2 1.0
160. LOAD COMB 203 DL+0.75L
161. 1 1.0 2 0.75
162. LOAD COMB 204 0.6DL+0.7SL1
163. 1 0.6 3 0.7
164. LOAD COMB 205 0.6DL+0.7SL2
165. 1 0.6 4 0.7
166. ********************************************************************************
167. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
168. LOAD COMB 101 1.4DL
169. 1 1.4
170. LOAD COMB 102 1.4DL+1.7L
171. 1 1.4 2 1.7
172. LOAD COMB 103 1.2DL+1.0SL1+1.0L
173. 1 1.2 2 1.0 3 1.0
174. LOAD COMB 104 1.2DL+1.0SL2+1.0L
175. 1 1.2 2 1.0 4 1.0
176. LOAD COMB 105 0.9DL+1.0SL1
177. 1 0.9 3 1.0
178. LOAD COMB 106 0.9DL+1.0SL2
```

```
179. 1 0.9 4 1.0
180. ********************************************************************************
181. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 136 NUMBER OF MEMBERS 1
NUMBER OF PLATES 112 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 135

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 111/ 16/ 51 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 411
TOTAL LOAD COMBINATION CASES = 11 SO FAR.
SIZE OF STIFFNESS MATRIX = 21 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.8/ 143561.4 MB

```
182. LOAD LIST 101 TO 106
183. START CONCRETE DESIGN
```



```
184. CODE ACI
185. TRACK 2 ALL
186. FC 35000 MEMB 185
187. CLB 0.05 MEMB 185
188. CLS 0.05 ALL
189. CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
190. 415 TO 418 424 TO 428 1508 TO 1522 1526 TO 1540 1544 TO 1548 1550 TO 1554 -
191. 1556 TO 1569
192. MAXMAIN 25 ALL
193. MINMAIN 25 ALL
194. MINSEC 10 ALL
195. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

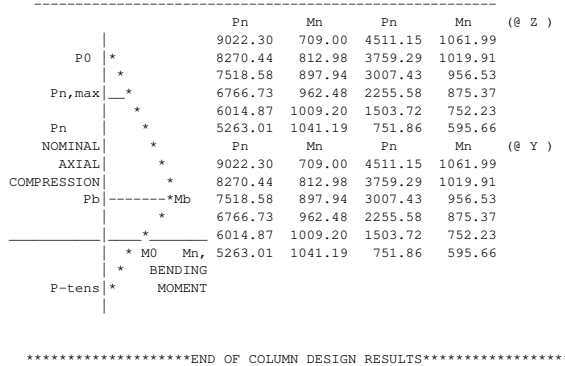
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

196. END CONCRETE DESIGN
197. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 30,2020 TIME= 14:16:14 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
****

```
            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    OCT  3, 2020                   *
            *           Time=     9:26:41                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F7 - STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 01/09/2020
     4. JOB NAME FDN. MKD. F7
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 913.4 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 0.6 ZD 0.6
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 2 FIXED
    32. **************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 128 FY -199 FZ 5.25
    37. **************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
```



```
    39. JOINT LOAD
    40. 128 FY -476.25 FZ 12.5
    41. **************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
    43. JOINT LOAD
    44. 128 FY -9.512
    45. **************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
    47. JOINT LOAD
    48. 128 FY 9.512
    49. **************************************************************************
    50. * UNFACTORED LOAD COMBINATIONS
    51. LOAD COMB 201 DL
    52. 1 1.0
    53. LOAD COMB 202 DL+L
    54. 1 1.0 2 1.0
    55. LOAD COMB 203 DL+0.75L
    56. 1 1.0 2 0.75
    57. LOAD COMB 204 0.6DL+0.7SL1
    58. 1 0.6 3 0.7
    59. LOAD COMB 205 0.6DL+0.7SL2
    60. 1 0.6 4 0.7
    61. **************************************************************************
    62. PERFORM ANALYSIS
```

```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          2  NUMBER OF MEMBERS       1
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      1

       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
 TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =       6
 TOTAL LOAD COMBINATION  CASES =     5  SO FAR.
   SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.0/ 143452.5 MB
```

```
    63. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:26:45 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
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****

ATTACHMENT 14 STAAD OUTPUT – FOUNDATION F8

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  3, 2020                 *
*          Time=     9:29:53                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F8.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F8
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 910.9 3.175; 6 0.75 910.9 3.175; 34 -0.5 910.9 3.175; 35 -0.25 910.9 3.175
   14. 41 0.25 910.9 3.175; 42 0.5 910.9 3.175; 128 0 913.4 3.175
   15. 302 -0.5 910.9 3.42438; 304 -0.25 910.9 3.42438; 305 0 910.9 3.42438
   16. 306 0.25 910.9 3.42438; 307 0.5 910.9 3.42438; 308 0.75 910.9 3.42438
   17. 322 -0.5 910.9 3.67438; 323 -0.25 910.9 3.67438; 324 0 910.9 3.67438
   18. 325 0.25 910.9 3.67438; 326 0.5 910.9 3.67438; 327 0.75 910.9 3.67438
   19. 333 -0.5 910.9 3.92438; 334 -0.25 910.9 3.92438; 335 0 910.9 3.92438
   20. 336 0.25 910.9 3.92438; 337 0.5 910.9 3.92438; 338 0.75 910.9 3.92438
   21. 388 -0.5 910.9 2.42438; 389 -0.25 910.9 2.42438; 390 0 910.9 2.42438
   22. 391 0.25 910.9 2.42438; 392 0.5 910.9 2.42438; 393 0.75 910.9 2.42438
   23. 399 -0.5 910.9 2.67438; 400 -0.25 910.9 2.67438; 401 0 910.9 2.67438
   24. 402 0.25 910.9 2.67438; 403 0.5 910.9 2.67438; 404 0.75 910.9 2.67438
   25. 410 -0.5 910.9 2.92438; 411 -0.25 910.9 2.92438; 412 0 910.9 2.92438
   26. 413 0.25 910.9 2.92438; 414 0.5 910.9 2.92438; 415 0.75 910.9 2.92438
   27. 1548 -0.75 910.9 3.42438; 1549 -0.75 910.9 3.175; 1552 -0.75 910.9 3.67438
   28. 1554 -0.75 910.9 3.92438; 1556 -0.75 910.9 2.67438; 1557 -0.75 910.9 2.42438
   29. 1560 -0.75 910.9 2.92438; 1561 -0.5 910.9 4.17438; 1562 -0.25 910.9 4.17438
   30. 1563 0 910.9 4.17438; 1564 0.25 910.9 4.17438; 1565 0.5 910.9 4.17438
   31. 1566 0.75 910.9 4.17438; 1567 -0.5 910.9 4.42438; 1568 -0.25 910.9 4.42438
   32. 1569 0 910.9 4.42438; 1570 0.25 910.9 4.42438; 1571 0.5 910.9 4.42438
   33. 1572 0.75 910.9 4.42438; 1573 -0.5 910.9 4.67438; 1574 -0.25 910.9 4.67438
   34. 1575 0 910.9 4.67438; 1576 0.25 910.9 4.67438; 1577 0.5 910.9 4.67438
   35. 1578 0.75 910.9 4.67438; 1579 -0.75 910.9 4.17438; 1580 -0.75 910.9 4.42438
   36. 1581 -0.75 910.9 4.67438; 1603 -0.5 910.9 4.92438; 1604 -0.25 910.9 4.92438
   37. 1605 0 910.9 4.92438; 1606 0.25 910.9 4.92438; 1607 0.5 910.9 4.92438
   38. 1608 0.75 910.9 4.92438; 1609 -0.75 910.9 4.92438; 1617 0.925 910.9 3.42438
```



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   39. 1618 0.925 910.9 3.175; 1619 0.925 910.9 3.67438; 1620 0.925 910.9 3.92438
   40. 1621 0.925 910.9 2.67438; 1622 0.925 910.9 2.42438; 1623 0.925 910.9 2.92438
   41. 1624 0.925 910.9 4.17438; 1625 0.925 910.9 4.42438; 1626 0.925 910.9 4.67438
   42. 1630 0.925 910.9 4.92438; 1632 -0.925 910.9 3.42438; 1633 -0.925 910.9 3.175
   43. 1634 -0.925 910.9 3.67438; 1635 -0.925 910.9 3.92438
   44. 1636 -0.925 910.9 2.67438; 1637 -0.925 910.9 2.42438
   45. 1638 -0.925 910.9 2.92438; 1639 -0.925 910.9 4.17438
   46. 1640 -0.925 910.9 4.42438; 1641 -0.925 910.9 4.67438
   47. 1645 -0.925 910.9 4.92438; 1647 0 913.4 4.67438; 1648 -0.5 910.9 5.17438
   48. 1649 -0.25 910.9 5.17438; 1650 0 910.9 5.17438; 1651 0.25 910.9 5.17438
   49. 1652 0.5 910.9 5.17438; 1653 0.75 910.9 5.17438; 1654 -0.75 910.9 5.17438
   50. 1655 0.925 910.9 5.17438; 1656 -0.925 910.9 5.17438; 1657 -0.5 910.9 5.42438
   51. 1658 -0.25 910.9 5.42438; 1659 0 910.9 5.42438; 1660 0.25 910.9 5.42438
   52. 1661 0.5 910.9 5.42438; 1662 0.75 910.9 5.42438; 1663 -0.75 910.9 5.42438
   53. 1664 0.925 910.9 5.42438; 1665 -0.925 910.9 5.42438; 1675 -0.25 910.9 2.17438
   54. 1676 -0.5 910.9 2.17438; 1677 0 910.9 2.17438; 1678 0.25 910.9 2.17438
   55. 1679 0.5 910.9 2.17438; 1680 0.75 910.9 2.17438; 1681 -0.75 910.9 2.17438
   56. 1682 0.925 910.9 2.17438; 1683 -0.925 910.9 2.17438; 1684 -0.25 910.9 1.92438
   57. 1685 -0.5 910.9 1.92438; 1686 0 910.9 1.92438; 1687 0.25 910.9 1.92438
   58. 1688 0.5 910.9 1.92438; 1689 0.75 910.9 1.92438; 1690 -0.75 910.9 1.92438
   59. 1691 0.925 910.9 1.92438; 1692 -0.925 910.9 1.92438; 1693 -0.25 910.9 1.67438
   60. 1694 -0.5 910.9 1.67438; 1695 0 910.9 1.67438; 1696 0.25 910.9 1.67438
   61. 1697 0.5 910.9 1.67438; 1698 0.75 910.9 1.67438; 1699 -0.75 910.9 1.67438
   62. 1700 0.925 910.9 1.67438; 1701 -0.925 910.9 1.67438
   63. MEMBER INCIDENCES
   64. 185 2 128; 1584 1575 1647
   65. ELEMENT INCIDENCES SHELL
   66. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   67. 334 307 308 6 42; 344 322 323 304 302; 345 323 324 305 304
   68. 346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307
   69. 354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324
   70. 357 336 337 326 325; 358 337 338 327 326; 404 399 400 389 388
   71. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
   72. 408 403 404 393 392; 414 410 411 400 399; 415 411 412 401 400
   73. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   74. 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412; 427 41 42 414 413
   75. 428 42 6 415 414; 1497 1548 302 34 1549; 1499 1552 322 302 1548
   76. 1501 1554 333 322 1552; 1503 1556 399 388 1557; 1505 1560 410 399 1556
   77. 1507 1549 34 410 1560; 1508 1561 1562 334 333; 1509 1562 1563 335 334
   78. 1510 1563 1564 336 335; 1511 1564 1565 337 336; 1512 1565 1566 338 337
   79. 1513 1567 1568 1562 1561; 1514 1568 1569 1563 1562; 1515 1569 1570 1564 1563
   80. 1516 1570 1571 1565 1564; 1517 1571 1572 1566 1565; 1518 1573 1574 1568 1567
   81. 1519 1574 1575 1569 1568; 1520 1575 1576 1570 1569; 1521 1576 1577 1571 1570
   82. 1522 1577 1578 1572 1571; 1523 1579 1561 333 1554; 1524 1580 1567 1561 1579
   83. 1525 1581 1573 1567 1580; 1544 1603 1604 1574 1573; 1545 1604 1605 1575 1574
   84. 1546 1605 1606 1576 1575; 1547 1606 1607 1577 1576; 1548 1607 1608 1578 1577
   85. 1549 1609 1603 1573 1581; 1556 308 1617 1618 6; 1557 327 1619 1617 308
   86. 1558 338 1620 1619 327; 1559 404 1621 1622 393; 1560 415 1623 1621 404
   87. 1561 6 1618 1623 415; 1562 1566 1624 1620 338; 1563 1572 1625 1624 1566
   88. 1564 1578 1626 1625 1572; 1568 1608 1630 1626 1578; 1570 1632 1548 1549 1633
   89. 1571 1634 1552 1548 1632; 1572 1635 1554 1552 1634; 1573 1636 1556 1557 1637
   90. 1574 1638 1560 1556 1636; 1575 1633 1549 1560 1638; 1576 1639 1579 1554 1635
   91. 1577 1640 1580 1579 1639; 1578 1641 1581 1580 1640; 1582 1645 1609 1581 1641
   92. 1585 1648 1649 1604 1603; 1586 1649 1650 1605 1604; 1587 1650 1651 1606 1605
   93. 1588 1651 1652 1607 1606; 1589 1652 1653 1608 1607; 1590 1654 1648 1603 1609
   94. 1591 1653 1655 1630 1608; 1592 1656 1654 1609 1645; 1593 1657 1658 1649 1648
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 95. 1594 1658 1659 1650 1649; 1595 1659 1660 1651 1650; 1596 1660 1661 1652 1651
 96. 1597 1661 1662 1653 1652; 1598 1663 1657 1648 1654; 1599 1662 1664 1655 1653
 97. 1600 1665 1663 1654 1656; 1609 388 389 1675 1676; 1610 389 390 1677 1675
 98. 1611 390 391 1678 1677; 1612 391 392 1679 1678; 1613 392 393 1680 1679
 99. 1614 1557 388 1676 1681; 1615 393 1622 1682 1680; 1616 1637 1557 1681 1683
100. 1617 1676 1675 1684 1685; 1618 1675 1677 1686 1684; 1619 1677 1678 1687 1686
101. 1620 1678 1679 1688 1687; 1621 1679 1680 1689 1688; 1622 1681 1676 1685 1690
102. 1623 1680 1682 1691 1689; 1624 1683 1681 1690 1692; 1625 1685 1684 1693 1694
103. 1626 1684 1686 1695 1693; 1627 1686 1687 1696 1695; 1628 1687 1688 1697 1696
104. 1629 1688 1689 1698 1697; 1630 1690 1685 1694 1699; 1631 1689 1691 1700 1698
105. 1632 1692 1690 1699 1701
106. START GROUP DEFINITION
107. ELEMENT
108. _RAFT 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 -
109. 424 TO 428 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 -
110. 1568 1570 TO 1578 1582 1585 TO 1600 1609 TO 1632
111. _RAFT_MOM 327 333 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 427 -
112. 428 1497 1499 1501 1503 1505 1507 TO 1518 1521 TO 1525 1544 1547 TO 1549 -
113. 1556 TO 1564 1568 1570 TO 1578 1582 1585 TO 1600 1609 TO 1632
114. _RAFT_SHEAR 334 348 358 404 TO 408 418 428 1497 1499 1501 1503 1505 -
115. 1507 TO 1512 1517 1522 TO 1525 1548 1549 1556 TO 1564 1568 1570 TO 1578 1582 -
116. 1589 TO 1600 1609 TO 1632
117. MEMBER
118. END GROUP DEFINITION
119. ELEMENT PROPERTY
120. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
121. 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
122. 1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 THICKNESS 0.4
123. DEFINE MATERIAL START
124. ISOTROPIC CONCRETE
125. E 2.17185E+007
126. POISSON 0.17
127. DENSITY 25
128. ALPHA 1E-005
129. DAMP 0.05
130. TYPE CONCRETE
131. STRENGTH FCU 27579
132. END DEFINE MATERIAL
133. MEMBER PROPERTY AMERICAN
134. 185 1584 PRIS YD 0.6 ZD 0.6
135. CONSTANTS
136. MATERIAL CONCRETE ALL
137. SUPPORTS
138. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
139. 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
140. 1571 TO 1578 1582 1585 TO 1600 1609 TO 1631 -
141. 1632 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.053125
34	0.062500
35	0.062500
41	0.062500

42	0.062500
302	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.053059
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.053125
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.053125
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.053125
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.053125
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.053191
1548	0.053059
1549	0.053125
1552	0.053125
1554	0.053125
1556	0.053125
1557	0.053125
1560	0.053191
1561	0.062500
1562	0.062500
1563	0.062500
1564	0.062500
1565	0.062500
1566	0.053125
1567	0.062500
1568	0.062500
1569	0.062500
1570	0.062500
1571	0.062500
1572	0.053125

1573	0.062500
1574	0.062500
1575	0.062500
1576	0.062500
1577	0.062500
1578	0.053125
1579	0.053125
1580	0.053125
1581	0.053125
1603	0.062500
1604	0.062500
1605	0.062500
1606	0.062500
1607	0.062500
1608	0.053125
1609	0.053125
1617	0.021848
1618	0.021875
1619	0.021875
1620	0.021875
1621	0.021875
1622	0.021875
1623	0.021902
1624	0.021875
1625	0.021875
1626	0.021875
1630	0.021875
1632	0.021848
1633	0.021875
1634	0.021875
1635	0.021875
1636	0.021875
1637	0.021875
1638	0.021902
1639	0.021875
1640	0.021875
1641	0.021875
1645	0.021875
1648	0.062500
1649	0.062500
1650	0.062500
1651	0.062500
1652	0.062500
1653	0.053125
1654	0.053125
1655	0.021875
1656	0.021875
1657	0.031250
1658	0.031250
1659	0.031250
1660	0.031250
1661	0.031250
1662	0.026563
1663	0.026563
1664	0.010938
1665	0.010938

1675	0.062500
1676	0.062500
1677	0.062500
1678	0.062500
1679	0.062500
1680	0.053125
1681	0.053125
1682	0.021875
1683	0.021875
1684	0.062500
1685	0.062500
1686	0.062500
1687	0.062500
1688	0.062500
1689	0.053125
1690	0.053125
1691	0.021875
1692	0.021875
1693	0.031250
1694	0.031250
1695	0.031250
1696	0.031250
1697	0.031250
1698	0.026563
1699	0.026563
1700	0.010938
1701	0.010938
TOTAL	6.937497

```
142. ********************************************************************************
143. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
144. SELFWEIGHT Y -1
145. ELEMENT LOAD
146. * LOAD DUE TO BACKFILL
147. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
148. 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
149. 1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 PR GY -40
150. * LOAD DUE TO SURCHARGE
151. 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 TO 418 424 TO 428 -
152. 1497 1499 1501 1503 1505 1507 TO 1525 1544 TO 1549 1556 TO 1564 1568 1570 -
153. 1571 TO 1578 1582 1585 TO 1600 1609 TO 1632 PR GY -10
154. JOINT LOAD
155. 128 FY -199 FZ 5.25
156. 1647 FY -19.3 FZ 0.54
157. ********************************************************************************
158. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
159. JOINT LOAD
160. 128 FY -476.25 FZ 12.5
161. 1647 FY -42.02 FZ 1.55
162. ********************************************************************************
163. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
164. JOINT LOAD
165. 128 FY -9.512
166. 1647 FY -0.87
167. ********************************************************************************
168. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
169. JOINT LOAD
```

```
170. 128 FY 9.512
171. 1647 FY -0.87
172. ********************************************************************************
173. * UNFACTORED LOAD COMBINATIONS
174. LOAD COMB 201 DL
175. 1 1.0
176. LOAD COMB 202 DL+L
177. 1 1.0 2 1.0
178. LOAD COMB 203 DL+0.75L
179. 1 1.0 2 0.75
180. LOAD COMB 204 0.6DL+0.7SL1
181. 1 0.6 3 0.7
182. LOAD COMB 205 0.6DL+0.7SL2
183. 1 0.6 4 0.7
184. ********************************************************************************
185. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
186. LOAD COMB 101 1.4DL
187. 1 1.4
188. LOAD COMB 102 1.4DL+1.7L
189. 1 1.4 2 1.7
190. LOAD COMB 103 1.2DL+1.0SL1+1.0L
191. 1 1.2 2 1.0 3 1.0
192. LOAD COMB 104 1.2DL+1.0SL2+1.0L
193. 1 1.2 2 1.0 4 1.0
194. LOAD COMB 105 0.9DL+1.0SL1
195. 1 0.9 3 1.0
196. LOAD COMB 106 0.9DL+1.0SL2
197. 1 0.9 4 1.0
198. ********************************************************************************
199. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS        146  NUMBER OF MEMBERS       2
  NUMBER OF PLATES        120  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    144


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=   100/    16/     51 DOF
TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =     444
TOTAL LOAD COMBINATION  CASES =    11  SO FAR.
  SIZE OF STIFFNESS MATRIX =       23 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.9/ 143452.2 MB



200. LOAD LIST 101 TO 106
201. START CONCRETE DESIGN
```



```
202. CODE ACI
203. TRACK 2 ALL
204. FC 35000 MEMB 185 1584
205. CLB 0.05 MEMB 185 1584
206. CLS 0.05 ALL
207. CLT 0.05 MEMB 185 327 328 332 TO 334 344 TO 348 354 TO 358 404 TO 408 414 -
208. 415 TO 418 424 TO 428 1508 TO 1522 1544 TO 1548 1556 TO 1564 1568 -
209. 1585 TO 1589 1591 1593 TO 1597 1599 1609 TO 1613 1615 1617 TO 1621 1623 1625 -
210. 1626 TO 1629 1631
211. MAXMAIN 25 ALL
212. MINMAIN 25 ALL
213. MINSEC 10 ALL
214. DESIGN COLUMN 185 1584
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

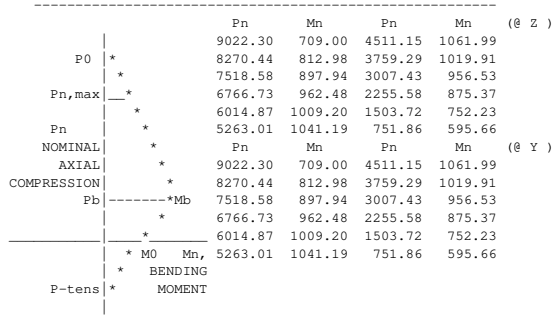
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000



==========

COLUMN NO. 1584 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	41.57	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	41.57	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

```
215. END CONCRETE DESIGN
216. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= OCT  3,2020   TIME=  9:29:59 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
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**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT  3, 2020                 *
*          Time=     9:30:18                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F8 - STABILITY.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F8
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 910.9 3.175; 128 0 913.4 3.175; 1575 0 910.9 4.67438; 1647 0 913.4 4.67438
   14. MEMBER INCIDENCES
   15. 185 2 128; 1584 1575 1647
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 1584 PRIS YD 0.6 ZD 0.6
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 1575 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
            MASTER/SLAVE OR IF UNCONNECTED JOINTS.
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FY -199 FZ 5.25
```



```
   37. 1647 FY -19.3 FZ 0.54
   38. ******************************************************************************
   39. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
   40. JOINT LOAD
   41. 128 FY -476.25 FZ 12.5
   42. 1647 FY -42.02 FZ 1.55
   43. ******************************************************************************
   44. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
   45. JOINT LOAD
   46. 128 FY -9.512
   47. 1647 FY -0.87
   48. ******************************************************************************
   49. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
   50. JOINT LOAD
   51. 128 FY 9.512
   52. 1647 FY -0.87
   53. ******************************************************************************
   54. * UNFACTORED LOAD COMBINATIONS
   55. LOAD COMB 201 DL
   56. 1 1.0
   57. LOAD COMB 202 DL+L
   58. 1 1.0 2 1.0
   59. LOAD COMB 203 DL+0.75L
   60. 1 1.0 2 0.75
   61. LOAD COMB 204 0.6DL+0.7SL1
   62. 1 0.6 3 0.7
   63. LOAD COMB 205 0.6DL+0.7SL2
   64. 1 0.6 4 0.7
   65. ******************************************************************************
   66. PERFORM ANALYSIS
```

```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

 NUMBER OF JOINTS          4  NUMBER OF MEMBERS       2
 NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
 NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      2

      SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =      12
TOTAL LOAD COMBINATION  CASES =     5  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/ 143451.2 MB
```

```
   67. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:30:22 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

C01

ATTACHMENT 15 STAAD OUTPUT – FOUNDATION F9

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 30, 2020                 *
*          Time=    14:15:33                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F9.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F9
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 910.6 3.175; 6 0.75 910.6 3.175; 35 -0.25 910.6 3.175; 41 0.25 910.6 3.175
   14. 42 0.5 910.6 3.175; 128 0 913.4 3.175; 304 -0.25 910.6 3.42438
   15. 305 0 910.6 3.42438; 306 0.25 910.6 3.42438; 307 0.5 910.6 3.42438
   16. 308 0.75 910.6 3.42438; 389 -0.25 910.6 2.42438; 390 0 910.6 2.42438
   17. 391 0.25 910.6 2.42438; 392 0.5 910.6 2.42438; 393 0.75 910.6 2.42438
   18. 400 -0.25 910.6 2.67438; 401 0 910.6 2.67438; 402 0.25 910.6 2.67438
   19. 403 0.5 910.6 2.67438; 404 0.75 910.6 2.67438; 411 -0.25 910.6 2.92438
   20. 412 0 910.6 2.92438; 413 0.25 910.6 2.92438; 414 0.5 910.6 2.92438
   21. 415 0.75 910.6 2.92438; 1617 1 910.6 3.42438; 1618 1 910.6 3.175
   22. 1621 1 910.6 2.67438; 1622 1 910.6 2.42438; 1623 1 910.6 2.92438
   23. 1624 -0.25 910.6 2.17438; 1625 0 910.6 2.17438; 1626 0 910.6 1.92438
   24. 1627 -0.25 910.6 1.92438; 1628 0.25 910.6 2.17438; 1629 0.25 910.6 1.92438
   25. 1630 0.5 910.6 2.17438; 1631 0.5 910.6 1.92438; 1632 0.75 910.6 2.17438
   26. 1633 0.75 910.6 1.92438; 1634 1 910.6 2.17438; 1635 1 910.6 1.92438
   27. 1672 -0.25 910.6 3.67438; 1673 0 910.6 3.67438; 1674 0.25 910.6 3.67438
   28. 1675 0.5 910.6 3.67438; 1676 0.75 910.6 3.67438; 1677 1 910.6 3.67438
   29. 1678 -0.25 910.6 3.92438; 1679 0 910.6 3.92438; 1680 0.25 910.6 3.92438
   30. 1681 0.5 910.6 3.92438; 1682 0.75 910.6 3.92438; 1683 1 910.6 3.92438
   31. 1720 -0.45 910.6 3.42438; 1721 -0.45 910.6 3.175; 1722 -0.45 910.6 2.67438
   32. 1723 -0.45 910.6 2.42438; 1724 -0.45 910.6 2.92438; 1725 -0.45 910.6 2.17438
   33. 1726 -0.45 910.6 1.92438; 1729 -0.45 910.6 3.67438; 1730 -0.45 910.6 3.92438
   34. 1737 1.25 910.6 3.42438; 1738 1.25 910.6 3.175; 1739 1.5 910.6 3.42438
   35. 1740 1.5 910.6 3.175; 1741 1.75 910.6 3.42438; 1742 1.75 910.6 3.175
   36. 1743 1.25 910.6 2.67438; 1744 1.25 910.6 2.42438; 1745 1.5 910.6 2.67438
   37. 1746 1.5 910.6 2.42438; 1747 1.75 910.6 2.67438; 1748 1.75 910.6 2.42438
   38. 1749 1.25 910.6 2.92438; 1750 1.5 910.6 2.92438; 1751 1.75 910.6 2.92438
```



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   39. 1752 2 910.6 3.42438; 1753 2 910.6 3.175; 1754 2 910.6 2.67438
   40. 1755 2 910.6 2.42438; 1756 2 910.6 2.92438; 1757 1.25 910.6 2.17438
   41. 1758 1.25 910.6 1.92438; 1759 1.5 910.6 2.17438; 1760 1.5 910.6 1.92438
   42. 1761 1.75 910.6 2.17438; 1762 1.75 910.6 1.92438; 1763 2 910.6 2.17438
   43. 1764 2 910.6 1.92438; 1773 1.25 910.6 3.67438; 1774 1.5 910.6 3.67438
   44. 1775 1.75 910.6 3.67438; 1776 2 910.6 3.67438; 1777 1.25 910.6 3.92438
   45. 1778 1.5 910.6 3.92438; 1779 1.75 910.6 3.92438; 1780 2 910.6 3.92438
   46. 1805 2.25 910.6 3.42438; 1806 2.25 910.6 3.175; 1807 2.25 910.6 2.67438
   47. 1808 2.25 910.6 2.42438; 1809 2.25 910.6 2.92438; 1810 2.5 910.6 3.42438
   48. 1811 2.5 910.6 3.175; 1812 2.5 910.6 2.67438; 1813 2.5 910.6 2.42438
   49. 1814 2.5 910.6 2.92438; 1815 2.25 910.6 2.17438; 1816 2.25 910.6 1.92438
   50. 1817 2.5 910.6 2.17438; 1818 2.5 910.6 1.92438; 1823 2.25 910.6 3.67438
   51. 1824 2.5 910.6 3.67438; 1825 2.25 910.6 3.92438; 1826 2.5 910.6 3.92438
   52. 1839 2 913.4 3.175; 1840 -0.25 910.6 4.17438; 1841 0 910.6 4.17438
   53. 1842 0.25 910.6 4.17438; 1843 0.5 910.6 4.17438; 1844 0.75 910.6 4.17438
   54. 1845 1 910.6 4.17438; 1846 -0.45 910.6 4.17438; 1847 1.25 910.6 4.17438
   55. 1848 1.5 910.6 4.17438; 1849 1.75 910.6 4.17438; 1850 2 910.6 4.17438
   56. 1851 2.25 910.6 4.17438; 1852 2.5 910.6 4.17438
   57. MEMBER INCIDENCES
   58. 185 2 128; 1754 1753 1839
   59. ELEMENT INCIDENCES SHELL
   60. 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41; 334 307 308 6 42
   61. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
   62. 408 403 404 393 392; 415 411 412 401 400; 416 412 413 402 401
   63. 417 413 414 403 402; 418 414 415 404 403; 425 35 2 412 411; 426 2 41 413 412
   64. 427 41 42 414 413; 428 42 6 415 414; 1556 308 1617 1618 6
   65. 1559 404 1621 1622 393; 1560 415 1623 1621 404; 1561 6 1618 1623 415
   66. 1562 1624 1625 1626 1627; 1563 1625 1628 1629 1626; 1564 1628 1630 1631 1629
   67. 1565 1630 1632 1633 1631; 1566 389 390 1625 1624; 1567 390 391 1628 1625
   68. 1568 391 392 1630 1628; 1569 392 393 1632 1630; 1570 1632 1634 1635 1633
   69. 1571 393 1622 1634 1632; 1602 1672 1673 305 304; 1603 1673 1674 306 305
   70. 1604 1674 1675 307 306; 1605 1675 1676 308 307; 1606 1676 1677 1617 308
   71. 1607 1678 1679 1673 1672; 1608 1679 1680 1674 1673; 1609 1680 1681 1675 1674
   72. 1610 1681 1682 1676 1675; 1611 1682 1683 1677 1676; 1642 1720 304 35 1721
   73. 1643 1722 400 389 1723; 1644 1724 411 400 1722; 1645 1721 35 411 1724
   74. 1646 1725 1624 1627 1726; 1647 1723 389 1624 1725; 1650 1729 1672 304 1720
   75. 1651 1730 1678 1672 1729; 1658 1617 1737 1738 1618; 1659 1737 1739 1740 1738
   76. 1660 1739 1741 1742 1740; 1661 1621 1743 1744 1622; 1662 1743 1745 1746 1744
   77. 1663 1745 1747 1748 1746; 1664 1623 1749 1743 1621; 1665 1749 1750 1745 1743
   78. 1666 1750 1751 1747 1745; 1667 1618 1738 1749 1623; 1668 1738 1740 1750 1749
   79. 1669 1740 1742 1751 1750; 1670 1741 1752 1753 1742; 1671 1747 1754 1755 1748
   80. 1672 1751 1756 1754 1747; 1673 1742 1753 1756 1751; 1674 1634 1757 1758 1635
   81. 1675 1757 1759 1760 1758; 1676 1759 1761 1762 1760; 1677 1622 1744 1757 1634
   82. 1678 1744 1746 1759 1757; 1679 1746 1748 1761 1759; 1680 1761 1763 1764 1762
   83. 1681 1748 1755 1763 1761; 1690 1677 1773 1737 1617; 1691 1773 1774 1739 1737
   84. 1692 1774 1775 1741 1739; 1693 1775 1776 1752 1741; 1694 1683 1777 1773 1677
   85. 1695 1777 1778 1774 1773; 1696 1778 1779 1775 1774; 1697 1779 1780 1776 1775
   86. 1722 1752 1805 1806 1753; 1723 1754 1807 1808 1755; 1724 1756 1809 1807 1754
   87. 1725 1753 1806 1809 1756; 1726 1805 1810 1811 1806; 1727 1807 1812 1813 1808
   88. 1728 1809 1814 1812 1807; 1729 1806 1811 1814 1809; 1730 1763 1815 1816 1764
   89. 1731 1755 1808 1815 1763; 1732 1815 1817 1818 1816; 1733 1808 1813 1817 1815
   90. 1738 1776 1823 1805 1752; 1739 1823 1824 1810 1805; 1740 1780 1825 1823 1776
   91. 1741 1825 1826 1824 1823; 1755 1840 1841 1679 1678; 1756 1841 1842 1680 1679
   92. 1757 1842 1843 1681 1680; 1758 1843 1844 1682 1681; 1759 1844 1845 1683 1682
   93. 1760 1846 1840 1678 1730; 1761 1845 1847 1777 1683; 1762 1847 1848 1778 1777
   94. 1763 1848 1849 1779 1778; 1764 1849 1850 1780 1779; 1765 1850 1851 1825 1780
```

```
 95. 1766 1851 1852 1826 1825
 96. START GROUP DEFINITION
 97. ELEMENT
 98. _RAFT 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 1602 -
 99. 1603 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
100. 1738 TO 1741 1755 TO 1766
101. _RAFT_MOM 333 334 405 TO 408 415 TO 418 427 428 1556 1559 TO 1571 -
102. 1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1669 1671 1672 1674 TO 1681 1690 -
103. 1691 TO 1697 1723 1724 1726 TO 1733 1738 TO 1741 1755 TO 1766
104. _RAFT_SHEAR 334 405 TO 408 418 428 1556 1559 TO 1571 1605 1606 1610 1611 1643 -
105. 1646 1647 1658 1659 1661 TO 1665 1667 1668 1671 1674 TO 1681 1690 1691 1694 -
106. 1695 TO 1697 1723 1727 1730 TO 1733 1740 1741 1755 TO 1766
107. MEMBER
108. END GROUP DEFINITION
109. ELEMENT PROPERTY
110. 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
111. 1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
112. 1738 TO 1741 1755 TO 1766 THICKNESS 0.4
113. DEFINE MATERIAL START
114. ISOTROPIC CONCRETE
115. E 2.17185E+007
116. POISSON 0.17
117. DENSITY 25
118. ALPHA 1E-005
119. DAMP 0.05
120. TYPE CONCRETE
121. STRENGTH FCU 27579
122. END DEFINE MATERIAL
123. MEMBER PROPERTY AMERICAN
124. 185 1754 PRIS YD 0.6 ZD 0.6
125. CONSTANTS
126. MATERIAL CONCRETE ALL
127. SUPPORTS
128. 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
129. 1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
130. 1738 TO 1741 1755 TO 1766 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
35	0.056250
41	0.062500
42	0.062500
304	0.056180
305	0.062422
306	0.062422
307	0.062422
308	0.062422
389	0.056250
390	0.062500
391	0.062500
392	0.062500
393	0.062500
400	0.056250

401	0.062500
402	0.062500
403	0.062500
404	0.062500
411	0.056320
412	0.062578
413	0.062578
414	0.062578
415	0.062578
1617	0.062423
1618	0.062500
1621	0.062500
1622	0.062500
1623	0.062578
1624	0.056250
1625	0.062500
1626	0.031250
1627	0.028125
1628	0.062500
1629	0.031250
1630	0.062500
1631	0.031250
1632	0.062500
1633	0.031250
1634	0.062500
1635	0.031250
1672	0.056250
1673	0.062500
1674	0.062500
1675	0.062500
1676	0.062500
1677	0.062500
1678	0.056250
1679	0.062500
1680	0.062500
1681	0.062500
1682	0.062500
1683	0.062500
1720	0.024969
1721	0.025000
1722	0.025000
1723	0.025000
1724	0.025031
1725	0.025000
1726	0.012500
1729	0.025000
1730	0.025000
1737	0.062422
1738	0.062500
1739	0.062422
1740	0.062500
1741	0.062422
1742	0.062500
1743	0.062500
1744	0.062500
1745	0.062500

1746	0.062500
1747	0.062500
1748	0.062500
1749	0.062578
1750	0.062577
1751	0.062578
1752	0.062423
1753	0.062500
1754	0.062500
1755	0.062500
1756	0.062578
1757	0.062500
1758	0.031250
1759	0.062500
1760	0.031250
1761	0.062500
1762	0.031250
1763	0.062500
1764	0.031250
1773	0.062500
1774	0.062500
1775	0.062500
1776	0.062500
1777	0.062500
1778	0.062500
1779	0.062500
1780	0.062500
1805	0.062423
1806	0.062500
1807	0.062500
1808	0.062500
1809	0.062578
1810	0.031211
1811	0.031250
1812	0.031250
1813	0.031250
1814	0.031289
1815	0.062500
1816	0.031250
1817	0.031250
1818	0.015625
1823	0.062500
1824	0.031250
1825	0.062500
1826	0.031250
1840	0.028125
1841	0.031250
1842	0.031250
1843	0.031250
1844	0.031250
1845	0.031250
1846	0.012500
1847	0.031250
1848	0.031250
1849	0.031250
1850	0.031250

1851	0.031250
1852	0.015625
TOTAL	6.637501

```
131. ********************************************************************************
132. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
133. SELFWEIGHT Y -1
134. ELEMENT LOAD
135. * LOAD DUE TO BACKFILL
136. 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
137. 1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
138. 1738 TO 1741 1755 TO 1766 PR GY -46
139. * LOAD DUE TO SURCHARGE
140. 328 332 TO 334 405 TO 408 415 TO 418 425 TO 428 1556 1559 TO 1571 -
141. 1602 TO 1611 1642 TO 1647 1650 1651 1658 TO 1681 1690 TO 1697 1722 TO 1733 -
142. 1738 TO 1741 1755 TO 1766 PR GY -10
143. JOINT LOAD
144. 128 1839 FY -199 FZ 5.25
145. ********************************************************************************
146. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
147. JOINT LOAD
148. 128 1839 FY -476.25 FZ 12.5
149. ********************************************************************************
150. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
151. JOINT LOAD
152. 128 1839 FY -9.512
153. ********************************************************************************
154. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
155. JOINT LOAD
156. 128 1839 FY 9.512
157. ********************************************************************************
158. * UNFACTORED LOAD COMBINATIONS
159. LOAD COMB 201 DL
160. 1 1.0
161. LOAD COMB 202 DL+L
162. 1 1.0 2 1.0
163. LOAD COMB 203 DL+0.75L
164. 1 1.0 2 0.75
165. LOAD COMB 204 0.6DL+0.7SL1
166. 1 0.6 3 0.7
167. LOAD COMB 205 0.6DL+0.7SL2
168. 1 0.6 4 0.7
169. ********************************************************************************
170. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
171. LOAD COMB 101 1.4DL
172. 1 1.4
173. LOAD COMB 102 1.4DL+1.7L
174. 1 1.4 2 1.7
175. LOAD COMB 103 1.2DL+1.0SL1+1.0L
176. 1 1.2 2 1.0 3 1.0
177. LOAD COMB 104 1.2DL+1.0SL2+1.0L
178. 1 1.2 2 1.0 4 1.0
179. LOAD COMB 105 0.9DL+1.0SL1
180. 1 0.9 3 1.0
181. LOAD COMB 106 0.9DL+1.0SL2
182. 1 0.9 4 1.0
183. ********************************************************************************
```

184. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 132 NUMBER OF MEMBERS 2
NUMBER OF PLATES 108 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 130

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 76/ 18/ 57 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 402
TOTAL LOAD COMBINATION CASES = 11 SO FAR.
SIZE OF STIFFNESS MATRIX = 23 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.8/ 143562.2 MB

185. LOAD LIST 101 TO 106
186. START CONCRETE DESIGN

187. CODE ACI
188. TRACK 2 ALL
189. FC 35000 MEMB 185 1754
190. CLB 0.05 MEMB 185 1754
191. CLS 0.05 ALL
192. CLT 0.05 ALL
193. MAXMAIN 25 ALL
194. MINMAIN 25 ALL
195. MINSEC 10 ALL
196. DESIGN COLUMN 185 1754

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

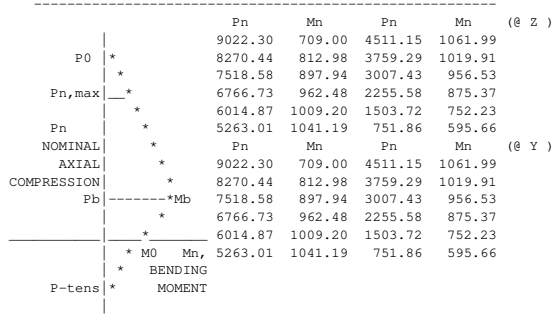
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000



COLUMN NO. 1754 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	428.62	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

197. END CONCRETE DESIGN
198. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 30,2020 TIME= 14:15:39 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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****

```
            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    OCT  3, 2020                   *
            *           Time=     9:32:25                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F9 -STABILITY.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01/09/2020
    4. JOB NAME FDN. MKD. F9_STABILITY
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 910.6 3.175; 128 0 913.4 3.175; 1753 2 910.6 3.175; 1839 2 913.4 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128; 1754 1753 1839
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 1754 PRIS YD 0.6 ZD 0.6
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 1753 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
            MASTER/SLAVE OR IF UNCONNECTED JOINTS.
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 1839 FY -199 FZ 5.25
```



```
   37. ******************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( L )
   39. JOINT LOAD
   40. 128 1839 FY -476.25 FZ 12.5
   41. ******************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
   43. JOINT LOAD
   44. 128 1839 FY -9.512
   45. ******************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
   47. JOINT LOAD
   48. 128 1839 FY 9.512
   49. ******************************************************************************
   50. * UNFACTORED LOAD COMBINATIONS
   51. LOAD COMB 201 DL
   52. 1 1.0
   53. LOAD COMB 202 DL+L
   54. 1 1.0 2 1.0
   55. LOAD COMB 203 DL+0.75L
   56. 1 1.0 2 0.75
   57. LOAD COMB 204 0.6DL+0.7SL1
   58. 1 0.6 3 0.7
   59. LOAD COMB 205 0.6DL+0.7SL2
   60. 1 0.6 4 0.7
   61. ******************************************************************************
   62. PERFORM ANALYSIS
```

```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          4  NUMBER OF MEMBERS       2
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      2


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
 TOTAL      PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =      12
 TOTAL LOAD COMBINATION  CASES =     5  SO FAR.
   SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.0/ 143451.0 MB
```

```
   63. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 3,2020 TIME= 9:32:29 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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ATTACHMENT 16 COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-005234, 01 October 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 01 October 2020	Comment by	EPC Response DATE: 03 October 2020	Status	Remarks
			General Comments (cc)				
MD-216-3420-EG-CV-CAL-2002-A01	3420	1	Load combination used in this document follows load combination indicated in SBC 301 - 2018, which is different than the code established in our current project design criteria	Bechtel	Load combinations as per SBC-301 - 2007 as per project design criteria is used in revised document.	Closed	
MD-216-3420-EG-CV-CAL-2002-A01	3420	2	This calculation follows SBC 301-2018, which is not indicated in this current document	Bechtel	SBC-301 - 2007 as per project design criteria is used in revised document.	Closed	